# Lecture notes. Lecture 11

Quantum information for newcomers
one qubit, the two-spin register, density matrices and noise

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These notes introduce the language of quantum information and quantum computing for bachelor students of physics, mathematics and computer science who want to learn quantum machine learning. They assume no course in quantum mechanics. We describe a register of $N$ qubits by a vector of $2^N$ complex amplitudes and use gates, which are evolution operators, together with the Born rule to predict the bit strings recorded in different measurement bases. For the two-spin register we then study the joint correlations of the two spins and the state of either spin alone. Density matrices and the partial trace describe these reduced states, and they also describe how measurement back-action and noise change a state. The standard textbook of the field is Ref. [1], and Ref. [11] is a free complement. The numerical values quoted below come from exact state vectors or from simulated measurement records with fixed random seeds, and no model is trained.

**Prerequisites**

Complex numbers and their modulus and phase. Vectors and matrices with complex entries, the matrix product, the conjugate transpose, eigenvalues and eigenvectors. Elementary probability, a probability distribution, the mean and the standard deviation. Some Python with NumPy for the numerical examples. Physical words such as spin, Hamiltonian, energy level or magnetic field are explained at their first appearance.

The same objects carry different names in physics, mathematics and computer science. Table 1 collects the names used in these notes.

| physics | mathematics | computer science |
|---|---|---|
| qubit state $\|\psi\rangle$ | unit vector in $\mathbb{C}^2$ | two complex amplitudes |
| register of $N$ qubits | unit vector in $\mathbb{C}^{2^N} = (\mathbb{C}^2)^{\otimes N}$ | array of $2^N$ complex numbers |
| computational basis state $\|b_0 \dots b_{N-1}\rangle$ | standard basis vector $e_i$ | bit string, binary index $i$ |
| ket $\|\psi\rangle$, bra $\langle\psi\|$ | column vector, conjugate row vector | `psi`, `psi.conj()` |
| tensor product $\otimes$ | Kronecker product | `np.kron` |
| observable | Hermitian matrix | quantity read out, with real values |
| Hamiltonian $\hat{\mathcal{H}}$ | Hermitian generator of $e^{-it\hat{\mathcal{H}}}$ | energy matrix of the device |
| gate, time evolution | unitary matrix | reversible instruction |
| circuit | product of unitary matrices | program |
| measurement, Born rule | projectors, $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$ | sampling a bit string |
| expectation value $\langle\hat{O}\rangle$ | $\mathrm{Tr}(\hat{O}\hat{\rho})$ | mean over samples |
| density matrix $\hat{\rho}$ | positive matrix of unit trace | generalised probability distribution |
| partial trace | linear map $\mathrm{Tr}_1$ on matrices | marginalisation |
| noise channel | completely positive trace-preserving map | noisy instruction |

**Table 1:** Physics, mathematics and computer-science names for the same objects.

# 1 Prepare, evolve, measure

A classical bit is a physical system with two distinguishable states, such as a voltage that is high or low or a magnetic domain that points up or down. We prepare it in one of its two states, transform it with logic gates and read it out. Quantum information keeps these three operations and replaces the two-state system by a quantum one. Examples are a spin-1/2, two energy levels of an atom or the two lowest levels of a superconducting circuit. Such a system carries one quantum bit, a *qubit*. A qubit can be prepared in a superposition of its two basis states. Its readout still returns one bit, 0 or 1, drawn at random with probabilities fixed by the state, and one run says little about the state. Quantum mechanics predicts probabilities, which we compare with records of many identically prepared runs.

A classical bit has the value 0 or 1. A *probabilistic bit*, a coin whose outcome we do not yet know, is described by the vector $(p_0, p_1)$ of its two probabilities. The entries of this stochastic vector are non-negative and sum to one. Random operations act on it with stochastic matrices, whose columns are such vectors. A qubit is described by a vector of two *complex* numbers $(a, b)$ with $|a|^2 + |b|^2 = 1$, a unit vector in $\mathbb{C}^2$, and unitary matrices change it. Reading it out samples a bit with the probabilities $|a|^2$ and $|b|^2$. Because $a$ and $b$ carry complex phases, two paths to the same outcome can cancel, which probabilities never do. Classical logic gates such as AND map two input bits to one output bit and cannot be undone. Quantum gates are unitary and therefore reversible, so a quantum computer runs reversible versions of the classical gates. A quantum program is a circuit, a sequence of gates, whose only output is a list of sampled bit strings.

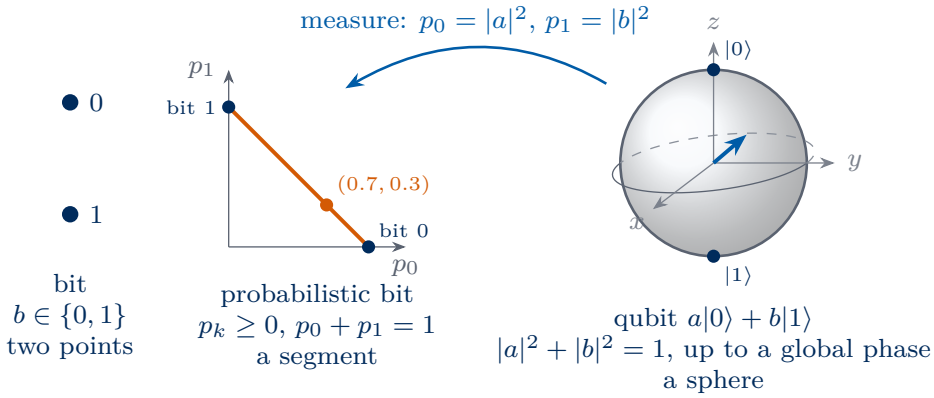



**Figure 1:** A bit takes one of two values, and a probabilistic bit is a point on the segment of probability vectors $(p_0, p_1)$. A pure qubit state is a point on a sphere, the Bloch sphere.

Every experiment described here, and every quantum algorithm built on it, consists of three operations (Figure 2). We *prepare* a known initial state, usually all qubits in the state $|0\rangle$. We let the state *evolve* under a Hamiltonian, the matrix of energies of the system, switched on for a chosen time. Such a fixed evolution is a *gate*. We *measure* every qubit and record a bit string, with the measurement basis chosen by the gates applied just before the readout. Our goal is to predict the measurement records of a register of two spins, the *two-spin register*, for any preparation, evolution and measurement basis. We also predict measurements performed on either spin alone. Each concept appears first on one qubit. We then apply it to the register of two qubits, labelled qubit 0 and qubit 1, and finally state the rule for $N$ qubits. Every quantum machine-learning method takes as input the objects defined along this route. They are Pauli strings and their expectation values estimated from bit strings, measurement settings, density matrices, Hamiltonians, time evolution and tensor contractions.

On paper every operation is linear algebra with vectors of length $2^N$ and matrices of size $2^N \times 2^N$ built with the Kronecker product. We write each operation first in this form. A computer evaluates the same index sums more efficiently as tensor contractions, written as `einsum` strings of the Python libraries NumPy and JAX and pictured as tensor-network diagrams. A short remark after each matrix form adds this view. The two forms agree number for number, and the contraction view leads to the

tensor networks used to simulate many qubits.

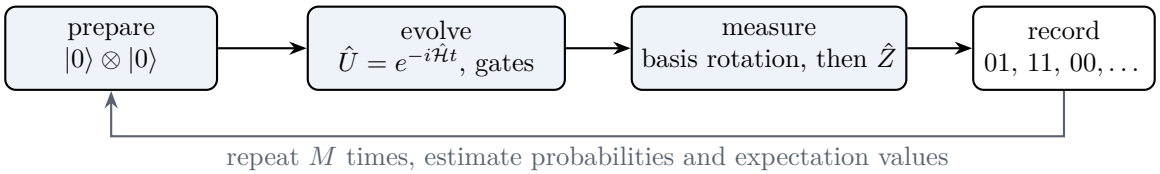



**Figure 2:** The three operations of every quantum experiment. A single run returns one bit string, so probabilities and expectation values are estimated from a record of $M$ repeated runs, the shots.

### The rules of quantum mechanics used here

Quantum mechanics enters through four rules, which we state as definitions and motivate physically where they are first used.

1. *State.* The state of $N$ qubits is a unit vector $|\psi\rangle$ in $\mathbb{C}^{2^N}$, defined up to a global phase factor. When the preparation is uncertain, it is a density matrix.
2. *Evolution.* An isolated system evolves by a unitary matrix, $|\psi\rangle \to \hat{U}|\psi\rangle$, with $\hat{U} = e^{-it\hat{\mathcal{H}}}$ for a time-independent Hamiltonian $\hat{\mathcal{H}}$ switched on for a time $t$.
3. *Measurement.* Reading out the qubits returns the bit string $b$ with the probability $|\langle b|\psi\rangle|^2$, the Born rule, and leaves the register in $|b\rangle$.
4. *Composition.* The state space of several qubits is the tensor product of their state spaces, computed with the Kronecker product.

## 2 One qubit

### 2.1 The qubit as a two-level system

A qubit is a quantum system whose states form a two-dimensional complex vector space spanned by two distinguishable basis states. In most laboratories these are two energy levels, the lower *ground state* $|g\rangle$ and the higher *excited state* $|e\rangle$. We use this implementation as the running physical example, with $|g\rangle = |0\rangle$ and $|e\rangle = |1\rangle$. Quantum mechanics allows superpositions of the two levels, although a measurement in the energy basis always finds the system in one of them. The energy difference $\Delta E = \hbar\omega_0$, with the reduced Planck constant $\hbar$, fixes the angular frequency $\omega_0$ at which the qubit oscillates and at which it absorbs or emits energy. We use units with $\hbar = 1$, in which energies are measured as angular frequencies, so $\Delta E = \omega_0$.

A single atom illustrates the three operations of Figure 2. Optical pumping prepares the atom in its ground state. A laser pulse tuned to the energy difference of the two levels drives the atom between them, and the duration and phase of the pulse decide the final superposition. A final laser makes the atom fluoresce only in one of the two levels, and a camera that detects light or no light returns one bit. The quantity to predict is the probability of each bit. Several platforms engineer two-level systems of this kind (Table 2), and photon polarisation provides a qubit whose two basis states have the same energy. In each platform, light, microwaves or voltage pulses drive transitions between the two basis states. Two-qubit gates come from an interaction between qubits, and a final measurement reads out the basis state of each qubit. A qubit is thus an addressable two-level system. A register of $N$ qubits is a set of $N$ such systems that are driven and measured individually and made to interact.

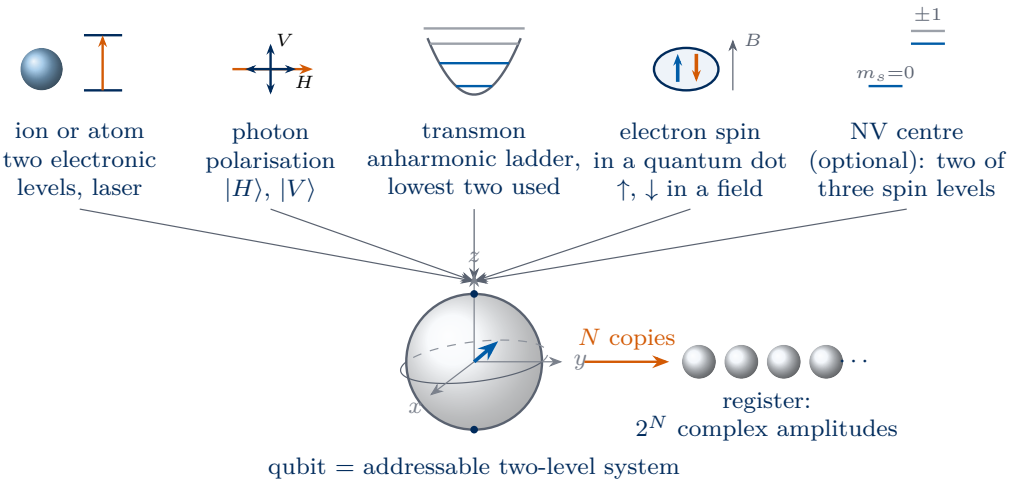



**Figure 3:** Different physical platforms provide the same abstract object, an addressable two-level system, and a register is $N$ such systems that can be driven, coupled and measured.

| platform | the two levels | driven by | coupled by | read out by |
|---|---|---|---|---|
| trapped ions | two electronic or hyperfine levels of an ion | lasers or microwaves | shared vibrations of the ion chain | laser-induced fluorescence |
| neutral atoms in optical tweezers | two levels of an atom | lasers | excitation to strongly interacting Rydberg states | fluorescence imaging |
| superconducting circuits (transmon) | two lowest levels of an anharmonic electrical oscillator | microwave pulses | capacitors or tunable couplers | microwave resonator |
| spins in semiconductor quantum dots | spin up and down of an electron | microwaves or electric fields | exchange interaction | charge sensing |
| photons | horizontal and vertical polarisation | wave plates | interference and measurement | photon detectors |

**Table 2:** Qualitative overview of physical platforms for qubits. The same mathematical description applies to all of them. Optical tweezers are tightly focused laser beams that each hold one atom. Rydberg states are highly excited atomic states with a large orbit, and a resonator is an electrical circuit or cavity that stores one oscillation mode.

## 2.2 The state vector and Dirac notation

We choose the two levels as a basis. A spin-1/2 is the intrinsic angular momentum of an electron or a nucleus. Its two basis states have the spin pointing up or down along a magnetic field in the $z$ direction. Quantum information calls the basis states of any qubit $|0\rangle$ and $|1\rangle$ and represents them by the column vectors

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.1}$$

They form the *computational basis*. Dirac notation is shorthand for column and row vectors and the matrices built from them. The symbol $|\cdot\rangle$, a *ket*, denotes a column vector. Its conjugate transpose is a row vector, the *bra* $\langle\cdot| = (|\cdot\rangle)^\dagger$, where the dagger $\dagger$ means complex conjugation followed by transposition. A bra times a ket is the *inner product*

$$\langle\phi|\psi\rangle = \sum_k \phi_k^* \psi_k, \tag{2.2}$$

a complex number. Two states are orthogonal when their inner product vanishes, as $\langle 0|1\rangle = 0$, and a state is normalised when $\langle\psi|\psi\rangle = 1$. A general state of one qubit is a superposition

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad a, b \in \mathbb{C}, \qquad |a|^2 + |b|^2 = 1. \tag{2.3}$$

The complex numbers $a$ and $b$ are the *amplitudes*. On a computer the state of one qubit is an array of two complex numbers. The Born rule turns $|a|^2$ and $|b|^2$ into probabilities, and the normalisation condition takes its meaning from there. As a running example we take

$$|\psi_{\text{ex}}\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{e^{i\pi/4}}{2}|1\rangle, \tag{2.4}$$

whose normalisation is $3/4 + 1/4 = 1$.

A ket times a bra, in the opposite order, is the *outer product* $|u\rangle\langle v|$, a $2 \times 2$ matrix with the entries $u_j v_k^*$. Two of them recur constantly,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.5}$$

Such matrices are *projectors*. Applying either one twice has the same effect as applying it once, and $|0\rangle\langle 0|\psi\rangle = a|0\rangle$ keeps only the $|0\rangle$ component of the state. Together the two projectors add up to the identity, $|0\rangle\langle 0| + |1\rangle\langle 1| = \hat{I}$. A matrix equal to its own conjugate transpose, $\hat{O}^\dagger = \hat{O}$, is *Hermitian* and has real eigenvalues. A *unitary* matrix, $\hat{U}^\dagger\hat{U} = \hat{I}$, preserves inner products and with them the normalisation.

### 2.3 Measurement in the computational basis and the Born rule

A measurement in the computational basis returns one bit, 0 or 1, and the *Born rule* gives the probabilities of the two outcomes,

$$p(0) = |\langle 0|\psi\rangle|^2 = |a|^2, \qquad p(1) = |\langle 1|\psi\rangle|^2 = |b|^2. \tag{2.6}$$

Normalisation is the statement $p(0) + p(1) = 1$. The measurement leaves the qubit in the basis state that was found, and a second measurement in the same basis repeats the result. We estimate the probabilities from many runs on identically prepared qubits, as in Figure 2. For the example state (2.4), $p(0) = |\sqrt{3}/2|^2 = 3/4$ and $p(1) = |e^{i\pi/4}/2|^2 = 1/4$. The phase factor $e^{i\pi/4}$ plays no role in this measurement.

### 2.4 Global and relative phase

Multiplying the whole state by a phase factor, $|\psi\rangle \to e^{i\alpha}|\psi\rangle$, changes no prediction of quantum mechanics. Every probability contains the amplitudes in pairs such as $a^*a$, $b^*b$ or $a^*b$, where the factor $e^{-i\alpha}e^{i\alpha} = 1$ drops out. Such a *global phase* is unobservable, and we may always choose the amplitude $a$ real and non-negative.

The phase between the two amplitudes, the *relative phase*, can be measured. The states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \tag{2.7}$$

differ only by the relative phase $e^{i\pi} = -1$ of the second amplitude. Their values of $|a|^2$ and $|b|^2$ coincide, and so do their outcome probabilities in the computational basis. Yet they are orthogonal, $\langle +|-\rangle = \frac{1}{2}(1 - 1) = 0$, and orthogonal states can be distinguished with certainty by a suitable measurement.

## 2.5 Observables and the Pauli matrices

A measurement that records $+1$ for the outcome 0 and $-1$ for the outcome 1 measures the spin along $z$ in units of $\hbar/2$. The average of the recorded values over many runs is

$$\langle \hat{Z} \rangle = (+1)\, p(0) + (-1)\, p(1) = p(0) - p(1). \tag{2.8}$$

Quantum mechanics represents a measured quantity by a Hermitian matrix, an *observable*. Its eigenvalues are the values that can be recorded, and its eigenvectors, the *eigenstates*, are the states left behind by the measurement. By the spectral theorem, every Hermitian matrix has real eigenvalues and an orthonormal basis of eigenvectors. Recorded values are therefore real, and eigenstates of different outcomes are orthogonal. For the spin along $z$ the matrix is $\hat{Z} = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$. Adding the spin components along $x$ and $y$, we obtain the three *Pauli matrices*

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.9}$$

Operators carry a hat throughout these notes. An array of numbers equated with an operator, as in equation (2.9), is the matrix of that operator in the computational basis and carries no hat. The Pauli matrices are also written $\hat{\sigma}_x \equiv \hat{X}$, $\hat{\sigma}_y \equiv \hat{Y}$, $\hat{\sigma}_z \equiv \hat{Z}$, with the vector $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$. Each Pauli matrix is Hermitian and unitary, squares to the identity, $\hat{X}^2 = \hat{Y}^2 = \hat{Z}^2 = \hat{I}$, and has the eigenvalues $\pm 1$. Two different Pauli matrices anticommute, for example $\hat{X}\hat{Z} = -\hat{Z}\hat{X}$. The eigenvectors are

$$\hat{Z}:\ |0\rangle, |1\rangle, \qquad \hat{X}:\ |\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \qquad \hat{Y}:\ |\pm i\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}, \tag{2.10}$$

with the eigenvalue $+1$ for the first state of each pair. The average of an observable $\hat{O}$ with eigenvalues $\lambda_k$ in the state $|\psi\rangle$ is the *expectation value*

$$\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle = \sum_k \lambda_k\, p(\lambda_k), \tag{2.11}$$

the eigenvalues weighted with their Born probabilities. Expanding $|\psi\rangle$ in the eigenvectors of $\hat{O}$ yields the second equality. For $\hat{Z}$ we recover $p(0) - p(1)$, and the example state has $\langle \hat{Z} \rangle = 3/4 - 1/4 = 1/2$.

## 2.6 Pauli algebra

A product of two Pauli matrices is again a Pauli matrix up to a factor. With the indices $a, b, c \in \{x, y, z\}$, the Kronecker delta $\delta_{ab}$ and the totally antisymmetric symbol $\varepsilon_{abc}$, with $\varepsilon_{xyz} = 1$, the rule reads

$$\hat{\sigma}_a \hat{\sigma}_b = \delta_{ab}\, \hat{I} + i \sum_c \varepsilon_{abc}\, \hat{\sigma}_c, \tag{2.12}$$

as multiplying the $2 \times 2$ matrices confirms. For $a = b$ it yields $\hat{\sigma}_a^2 = \hat{I}$, and for different indices

$$\hat{X}\hat{Y} = i\hat{Z}, \qquad \hat{Y}\hat{Z} = i\hat{X}, \qquad \hat{Z}\hat{X} = i\hat{Y}, \qquad \hat{Y}\hat{X} = -i\hat{Z}, \quad \hat{Z}\hat{Y} = -i\hat{X}, \quad \hat{X}\hat{Z} = -i\hat{Y}. \tag{2.13}$$

We add and subtract (2.12) with $a$ and $b$ exchanged and find the anticommutators and commutators,

$$\{\hat{\sigma}_a, \hat{\sigma}_b\} = \hat{\sigma}_a\hat{\sigma}_b + \hat{\sigma}_b\hat{\sigma}_a = 2\delta_{ab}\, \hat{I}, \qquad [\hat{\sigma}_a, \hat{\sigma}_b] = \hat{\sigma}_a\hat{\sigma}_b - \hat{\sigma}_b\hat{\sigma}_a = 2i \sum_c \varepsilon_{abc}\, \hat{\sigma}_c. \tag{2.14}$$

Since the Pauli matrices are traceless, the trace of (2.12) is $\mathrm{Tr}(\hat{\sigma}_a\hat{\sigma}_b) = 2\delta_{ab}$. The four matrices $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ are therefore orthogonal with respect to $\mathrm{Tr}(\hat{A}^\dagger \hat{B})$ and form a basis of all $2 \times 2$ matrices,

$$\hat{M} = c_0\, \hat{I} + \mathbf{c} \cdot \hat{\boldsymbol{\sigma}}, \qquad c_0 = \tfrac{1}{2}\mathrm{Tr}\hat{M}, \qquad c_a = \tfrac{1}{2}\mathrm{Tr}(\hat{\sigma}_a \hat{M}). \tag{2.15}$$

The matrix $\hat{M}$ is Hermitian if and only if $c_0$ and $\mathbf{c}$ are real. For real coefficients the anticommutators yield $(\mathbf{c} \cdot \hat{\boldsymbol{\sigma}})^2 = |\mathbf{c}|^2 \hat{I}$. The eigenvalues of $\hat{M}$ are then $c_0 \pm |\mathbf{c}|$, and every unit-vector combination $\mathbf{n} \cdot \hat{\boldsymbol{\sigma}}$ has the eigenvalues $\pm 1$.

The *ladder operators*

$$\hat\sigma^\pm = \hat\sigma_x \pm i\hat\sigma_y, \qquad \hat\sigma^+ = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = 2|0\rangle\langle 1|, \qquad \hat\sigma^- = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} = 2|1\rangle\langle 0|, \tag{2.16}$$

move between the two levels. On the basis states $\hat\sigma^+|1\rangle = 2|0\rangle$, $\hat\sigma^+|0\rangle = 0$, $\hat\sigma^-|0\rangle = 2|1\rangle$ and $\hat\sigma^-|1\rangle = 0$. With $|0\rangle = |\uparrow\rangle$ the operator $\hat\sigma^+$ raises the spin from down to up and $\hat\sigma^-$ lowers it. The inverse relations are $\hat X = \frac{1}{2}(\hat\sigma^+ + \hat\sigma^-)$ and $\hat Y = \frac{1}{2i}(\hat\sigma^+ - \hat\sigma^-)$, and from the algebra (2.12) follow

$$\hat\sigma^+\hat\sigma^- = 2(\hat I + \hat Z) = 4|0\rangle\langle 0|, \quad \hat\sigma^-\hat\sigma^+ = 2(\hat I - \hat Z), \quad [\hat\sigma^+, \hat\sigma^-] = 4\hat Z, \quad [\hat Z, \hat\sigma^\pm] = \pm 2\hat\sigma^\pm, \quad (\hat\sigma^\pm)^2 = 0. \tag{2.17}$$

The spin operators of a spin-1/2 are $\hat S_a = \hat\sigma_a/2$, and their ladder operators $\hat S^\pm = \hat S_x \pm i\hat S_y = \hat\sigma^\pm/2$ are the unit transition operators,

$$\hat S^+ = |0\rangle\langle 1|, \qquad \hat S^- = |1\rangle\langle 0|. \tag{2.18}$$

With the ground state $|g\rangle = |0\rangle$ and the excited state $|e\rangle = |1\rangle$ of Section 2.1, the two energy jumps of the qubit are

$$\text{excitation} \quad |e\rangle\langle g| = |1\rangle\langle 0| = \tfrac{1}{2}\hat\sigma^- = \hat S^-, \qquad \text{decay} \quad |g\rangle\langle e| = |0\rangle\langle 1| = \tfrac{1}{2}\hat\sigma^+ = \hat S^+. \tag{2.19}$$

The decay operator describes the loss of energy to the surroundings.

**Convention warning**

In these notes spin up is the ground state, $|g\rangle = |0\rangle = |\uparrow\rangle$, and the Hamiltonian of a free qubit is $-\frac{\omega_0}{2}\hat Z$. The operator $\hat\sigma^+$ thus raises the spin and lowers the energy, and the decay of the excited state is $\hat S^+ = |g\rangle\langle e|$. Atomic physics often uses the opposite labelling, $\hat\sigma_z^{\text{atom}} = |e\rangle\langle e| - |g\rangle\langle g| = -\hat Z$, with Pauli matrices built on the excited state as the first basis vector. In that convention $\hat\sigma^+_{\text{atom}} = |e\rangle\langle g|$ excites the atom and $\hat\sigma^-_{\text{atom}} = |g\rangle\langle e|$ describes decay, so $\hat\sigma^\pm_{\text{atom}}$ correspond to our $\hat S^\mp = \hat\sigma^\mp/2$. A text on atoms and a text on qubits can therefore use the same symbol for opposite operators.

## 2.7 The Bloch sphere

Up to the global phase, every normalised state (2.3) can be written with two angles,

$$|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle, \qquad 0 \le \vartheta \le \pi, \quad 0 \le \varphi < 2\pi. \tag{2.20}$$

The angles are the polar and azimuthal angles of a point on the unit sphere, the *Bloch sphere*. Its north pole is $|0\rangle$, its south pole $|1\rangle$, and its equator holds the equal superpositions $|\pm\rangle$ and $|\pm i\rangle$. The Cartesian coordinates of the point are the expectation values of the three Pauli matrices, obtained from (2.11) by matrix multiplication,

$$\mathbf{r} = (\langle\hat X\rangle, \langle\hat Y\rangle, \langle\hat Z\rangle) = (\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta). \tag{2.21}$$

The vector $\mathbf{r}$ is the *Bloch vector*, and every pure state has $|\mathbf{r}| = 1$. The relative phase is the azimuth $\varphi$, so it drops out of $\langle\hat Z\rangle$ and appears in $\langle\hat X\rangle$ and $\langle\hat Y\rangle$. Orthogonal states sit at antipodal points, as $|+\rangle$ and $|-\rangle$ at $\pm\hat x$. In the example state, $\cos(\vartheta/2) = \sqrt{3}/2$ fixes $\vartheta = \pi/3$ and the phase fixes $\varphi = \pi/4$, so

$$\mathbf{r}_{\text{ex}} = \left(\tfrac{\sqrt3}{2}\cos\tfrac{\pi}{4},\ \tfrac{\sqrt3}{2}\sin\tfrac{\pi}{4},\ \tfrac{1}{2}\right) \approx (0.612,\ 0.612,\ 0.500). \tag{2.22}$$

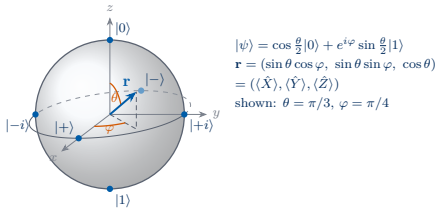



**Figure 4:** A pure state as a point on the Bloch sphere. The polar angle $\vartheta$ sets the probabilities of the bits, and the azimuth $\varphi$ is the relative phase.

# 3 Evolution and gates as rotations

## 3.1 Evolution under a Hamiltonian

Gates move the state of a qubit on the Bloch sphere, and each gate is the evolution of the qubit under a Hamiltonian switched on for a fixed time. A closed quantum system evolves according to the Schrödinger equation, in units with $\hbar = 1$,

$$i\,\frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle, \tag{3.1}$$

where the *Hamiltonian* $\hat{\mathcal{H}}$ is a Hermitian matrix, $2 \times 2$ for one qubit. Its eigenvalues are the energies of the system, the *energy levels*, and its eigenvectors are the states of definite energy. Through (3.1) the same matrix generates the time evolution. A device is controlled by switching fields or couplings on and off, which changes $\hat{\mathcal{H}}$. For a Hamiltonian that does not depend on time the equation is a linear system of differential equations with constant coefficients, solved by the *matrix exponential*. For any square matrix $\hat{A}$ we define it by the power series

$$e^{\hat{A}} = \sum_{k\geq 0} \frac{\hat{A}^k}{k!} = \hat{I} + \hat{A} + \frac{\hat{A}^2}{2} + \frac{\hat{A}^3}{6} + \dots, \tag{3.2}$$

which converges for every matrix, as the series of the ordinary exponential does. For $\hat{A} = -it\hat{\mathcal{H}}$ we differentiate each term of the series. Every term contains only powers of the one matrix $\hat{\mathcal{H}}$, so the factors commute, and

$$\frac{d}{dt}e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 1} \frac{k\,(-i\hat{\mathcal{H}})^k\,t^{k-1}}{k!} = (-i\hat{\mathcal{H}}) \sum_{k\geq 1} \frac{(-it\hat{\mathcal{H}})^{k-1}}{(k-1)!} = -i\hat{\mathcal{H}}\,e^{-it\hat{\mathcal{H}}}. \tag{3.3}$$

The state

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle, \qquad \hat{U}(t) = e^{-it\hat{\mathcal{H}}}, \tag{3.4}$$

therefore satisfies $i\frac{d}{dt}|\psi(t)\rangle = i(-i\hat{\mathcal{H}})\hat{U}(t)|\psi(0)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle$, and at $t = 0$ it starts from the right initial state, since $\hat{U}(0) = \hat{I}$. The *evolution operator* $\hat{U}(t)$ is unitary. Its conjugate transpose is $\hat{U}^\dagger = e^{+it\hat{\mathcal{H}}^\dagger} = e^{it\hat{\mathcal{H}}}$ because $\hat{\mathcal{H}}$ is Hermitian, and $\hat{U}^\dagger\hat{U} = e^{it\hat{\mathcal{H}}}e^{-it\hat{\mathcal{H}}} = \hat{I}$, since the exponents commute. Evolution therefore preserves normalisation, and the Born probabilities sum to one at every time. Because the exponents commute, two evolutions in sequence also add their times, $\hat{U}(t_2)\hat{U}(t_1) = \hat{U}(t_1 + t_2)$.

The exponential is easiest to evaluate in the eigenbasis of the Hamiltonian. We diagonalise $\hat{\mathcal{H}} = \sum_k E_k|E_k\rangle\langle E_k|$, with the energies $E_k$ and orthonormal eigenvectors $|E_k\rangle$. Every power is then $\hat{\mathcal{H}}^n = \sum_k E_k^n|E_k\rangle\langle E_k|$, and the series sums to the *spectral form*

$$\hat{U}(t) = \sum_k e^{-iE_kt}|E_k\rangle\langle E_k|. \tag{3.5}$$

An energy eigenstate $|E_k\rangle$ only acquires the phase $e^{-iE_kt}$. Its measurement predictions stay constant in time, and it is called a *stationary state*. A superposition of stationary states does change, since its components acquire different phases. On a computer $\hat{U}(t)$ is obtained either from this eigendecomposition or from a library routine for the matrix exponential, `scipy.linalg.expm` or `jax.scipy.linalg.expm`.

## 3.2 An atom in laser light as a qubit

The gates of a quantum computer are evolutions of this kind. For an atom driven by a classical laser field, the two-level description of Section 2.1 follows from the atomic Hamiltonian, as in the textbook treatments [2, 3]. Hydrogen serves as the example wherever numbers are needed. We keep $\hbar$ explicit until the end of the derivation.

### The atom and its Hamiltonian

An atom consists of a nucleus with the charge $Ze_0$ and the mass $M$ and of $N$ electrons with the charge $-e_0$ and the mass $m_e$. Here $e_0$ is the elementary charge. The nucleus has the position operator $\hat{\mathbf{R}}$ and the momentum operator $\hat{\mathbf{P}}$, and electron $i$ has $\hat{\mathbf{r}}_i$ and $\hat{\mathbf{p}}_i$. The Hamiltonian of the atom is

$$\hat{\mathcal{H}}_{\mathrm{atom}} = \frac{\hat{\mathbf{P}}^2}{2M} + \sum_{i=1}^{N} \frac{\hat{\mathbf{p}}_i^2}{2m_e} - \sum_{i=1}^{N} \frac{Ze_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{R}}|} + \sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}, \tag{3.6}$$

with the vacuum permittivity $\varepsilon_0$. The first two terms are the kinetic energies of the nucleus and of the electrons. The third is the attraction between every electron and the nucleus, and the fourth is the repulsion between pairs of electrons. Relativistic corrections, which produce the fine structure, and the coupling to the nuclear spin, which produces the hyperfine structure, are omitted at this level.

### Born–Oppenheimer approximation

The nucleus is heavier than an electron by the factor $M/m_e \approx 1840\,A$, with the mass number $A$, so it moves much more slowly than the electrons. In the *Born–Oppenheimer approximation* [4] its position is treated as a fixed parameter, here $\mathbf{R} = 0$, and the electrons obey the *electronic Hamiltonian*

$$\hat{\mathcal{H}}_{\mathrm{el}} = \sum_{i=1}^{N} \frac{\hat{\mathbf{p}}_i^2}{2m_e} - \sum_{i=1}^{N} \frac{Ze_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i|} + \sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}, \tag{3.7}$$

whose eigenvalues are the energy levels of the atom. For a single atom the motion of the centre of mass separates without approximation. For hydrogen the only correction is the replacement of $m_e$ by the reduced mass $\mu = m_e M/(m_e + M)$, which shifts the levels by about $0.05\,\%$. For several electrons a small mass-polarisation term proportional to $\sum_{i<j} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_j / M$ adds to it. The same approximation underlies molecular physics and quantum chemistry, where the nuclei of a molecule are clamped while the electronic states are computed.

### Energy eigenstates as a basis

The eigenstates of the electronic Hamiltonian form a discrete set of bound states,

$$\hat{\mathcal{H}}_{\mathrm{el}} |n\rangle = E_n |n\rangle, \qquad \langle n' | n \rangle = \delta_{nn'}, \tag{3.8}$$

where $n$ collects all quantum numbers of the state. Every bound state of the atom is a superposition

$$|\psi(t)\rangle = \sum_n c_n(t)\, |n\rangle, \qquad c_n = \langle n | \psi \rangle, \tag{3.9}$$

and $|c_n|^2$ is the probability of finding the atom in the state $|n\rangle$, with $\sum_n |c_n|^2 = 1$. Without light each amplitude only acquires a phase, $c_n(t) = c_n(0)\, e^{-iE_n t/\hbar}$. Inserting (3.9) into the Schrödinger equation $i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{\mathcal{H}} |\psi(t)\rangle$ with any Hamiltonian $\hat{\mathcal{H}}$ and projecting on $\langle n|$ gives the linear system

$$i\hbar\, \dot{c}_n = \sum_m H_{nm}\, c_m, \qquad H_{nm} = \langle n | \hat{\mathcal{H}} | m \rangle, \tag{3.10}$$

where the dot denotes the time derivative. The vector of amplitudes is the state vector of Section 2, and (3.10) is the matrix form (3.1) of the Schrödinger equation, which we write with $\hbar = 1$. Alkali atoms and singly charged ions such as $Ca^+$, $Sr^+$ and $Yb^+$, with one valence electron outside closed shells and such a discrete set of levels, serve as qubits.

### Coupling to classical light

A laser produces the electric field $\mathbf{E}(t) = E_0\,\mathbf{e}_z\cos(\omega_L t)$, which we take polarised along $z$, with the amplitude $E_0$ and the angular frequency $\omega_L$. We treat the light as a classical field. An optical wavelength of about a hundred nanometres or more is much larger than the atom, so the field is uniform across it. In this *dipole approximation* the field adds the potential energy of the electrons,

$$\hat{\mathcal{H}}_{\rm int}(t) = -\hat{\mathbf{d}}\cdot\mathbf{E}(t), \qquad \hat{\mathbf{d}} = -e_0\sum_{i=1}^{N}\hat{\mathbf{r}}_i, \tag{3.11}$$

with the *electric dipole operator* $\hat{\mathbf{d}}$ of the atom and its matrix elements $d_{nm} = \langle n|\hat{\mathbf{d}}\cdot\mathbf{e}_z|m\rangle$. The electronic Hamiltonian is unchanged when all electron positions change sign. Its eigenstates can therefore be chosen even or odd under this *parity* operation, whereas $\hat{\mathbf{d}}$ changes sign. Hence $d_{nn} = 0$, and the light couples only levels of opposite parity, a *selection rule*.

### Two levels near resonance

The Hamiltonian matrix is $H_{nm}(t) = E_n\delta_{nm} - d_{nm}E_0\cos(\omega_L t)$. The free evolution makes every amplitude rotate fast at its own frequency, and we remove this rotation by writing $c_n(t) = a_n(t)\,e^{-iE_n t/\hbar}$. Inserting this into (3.10), the terms $E_n c_n$ cancel and

$$i\hbar\,\dot{a}_n = -E_0\cos(\omega_L t)\sum_m d_{nm}\,e^{i\omega_{nm}t}\,a_m, \qquad \omega_{nm} = \frac{E_n - E_m}{\hbar}. \tag{3.12}$$

The amplitudes $a_n$ now change only because of the light. We tune the laser close to the transition between a ground state $g$ and an excited state $e$ of opposite parity, $\omega_L \approx \omega_0 = \omega_{eg}$, and write the cosine as $\frac{1}{2}(e^{i\omega_L t} + e^{-i\omega_L t})$. The coupling of $a_e$ to $a_g$ then contains the two factors

$$e^{i\omega_0 t}\cos(\omega_L t) = \tfrac{1}{2}\,e^{-i\Delta t} + \tfrac{1}{2}\,e^{i(\omega_0+\omega_L)t}, \qquad \Delta = \omega_L - \omega_0. \tag{3.13}$$

The first factor varies slowly, at the *detuning* $\Delta$. The second oscillates at nearly twice the optical frequency and averages to almost zero over the slow change of the amplitudes. The couplings to every other level average out in the same way, because their transition frequencies differ from $\omega_L$ by much more than the coupling strength. Keeping only the slow terms, which is the *rotating-wave approximation*, leaves two coupled amplitudes,

$$i\dot{a}_g = \frac{\Omega}{2}\,e^{i\Delta t}\,a_e, \qquad i\dot{a}_e = \frac{\Omega}{2}\,e^{-i\Delta t}\,a_g, \qquad \Omega = -\frac{d_{eg}E_0}{\hbar}, \tag{3.14}$$

where the phases of $|g\rangle$ and $|e\rangle$ are chosen so that $d_{eg}$ is real. The *Rabi frequency* $\Omega$ measures the strength of the coupling and is proportional to the field amplitude. For $|\Omega|, |\Delta| \ll \omega_0$, and $\Omega$ small compared with the detuning of every other allowed transition, all other levels stay nearly empty. The atom has become a two-level system with the state vector $(a_g, a_e)$, a qubit with $|g\rangle = |0\rangle$ and $|e\rangle = |1\rangle$. Since the argument uses only discrete levels and dipole couplings, it holds for every atom and ion of the kind described above.

### The qubit Hamiltonian

The factors $e^{\pm i\Delta t}$ disappear in the amplitudes $b_g = a_g e^{-i\Delta t/2}$ and $b_e = a_e e^{i\Delta t/2}$. Differentiating and using (3.14) gives $i\dot{b}_g = \frac{\Delta}{2}b_g + \frac{\Omega}{2}b_e$ and $i\dot{b}_e = \frac{\Omega}{2}b_g - \frac{\Delta}{2}b_e$, a Schrödinger equation with the constant Hamiltonian

$$\hat{\mathcal{H}}_{\rm RF} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}, \qquad \Delta = \omega_L - \omega_0, \tag{3.15}$$

in units with $\hbar = 1$. The amplitudes $b$ differ from the original amplitudes $c$ only by phases. With the zero of energy halfway between the two levels, the free two-level atom has the Hamiltonian $-\frac{\omega_0}{2}\hat{Z}$.

The ground state then has the energy $-\omega_0/2$ and the excited state $+\omega_0/2$, as in the convention box of Section 2.6. In the same units the two-level atom in the laser field has $\hat{\mathcal{H}}(t) = -\frac{\omega_0}{2}\hat{Z} + \Omega\cos(\omega_L t)\,\hat{X}$. The transformation $(b_g, b_e) = e^{-i\omega_L t\hat{Z}/2}(c_g, c_e)$ relates the two descriptions. It passes to a reference frame that rotates with the laser, the *rotating frame*, and all probabilities are the same in both frames. In this frame the laser field becomes a constant $2\times 2$ Hamiltonian, and the evolution operator (3.4) converts it into gates. A phase $\phi_L$ of the laser, in the field $E_0\,\mathbf{e}_z\cos(\omega_L t - \phi_L)$, replaces $\hat{X}$ by $\cos\phi_L\,\hat{X} + \sin\phi_L\,\hat{Y}$, so the laser phase selects the rotation axis in the $x$–$y$ plane.

**Hydrogen as an example**

Hydrogen has $Z = N = 1$, and its electronic Hamiltonian is $\hat{\mathcal{H}}_{\rm H} = \hat{\mathbf{p}}^2/2m_e - e_0^2/(4\pi\varepsilon_0|\hat{\mathbf{r}}|)$. Its eigenstates $|nlm\rangle$ carry the principal quantum number $n = 1, 2, \ldots$, the orbital angular momentum $l = 0, \ldots, n-1$ and its projection $m = -l, \ldots, l$ on the $z$ axis, and the energies

$$E_n = -\frac{\mathrm{Ry}}{n^2} \tag{3.16}$$

depend only on $n$, with the Rydberg energy $\mathrm{Ry} \approx 13.6$ eV. States with $l = 0, 1, 2$ are labelled s, p, d. In the position representation the eigenstates are $\langle\mathbf{r}|nlm\rangle = R_{nl}(r)\,Y_{lm}(\vartheta,\varphi)$, a radial function times a spherical harmonic. A spherical harmonic $Y_{lm}$ is a standard function of the two angles of the position vector that fixes the angular shape of the state. Two examples are

$$\langle\mathbf{r}|1s\rangle = \frac{2}{a_0^{3/2}}\,e^{-r/a_0}\,\frac{1}{\sqrt{4\pi}}, \qquad \langle\mathbf{r}|2p_0\rangle = \frac{1}{2\sqrt{6}\,a_0^{3/2}}\,\frac{r}{a_0}\,e^{-r/2a_0}\,\sqrt{\frac{3}{4\pi}}\cos\vartheta, \tag{3.17}$$

with the Bohr radius $a_0 \approx 0.053$ nm. For light polarised along $z$ the matrix element of $\hat{z}$ factors into a radial integral and an angular integral,

$$\langle n'l'm'|\hat{z}|nlm\rangle = \int_0^\infty R_{n'l'}R_{nl}\,r^3\,dr \int Y^*_{l'm'}\cos\vartheta\,Y_{lm}\,d\Omega_{\rm s}, \tag{3.18}$$

where $d\Omega_{\rm s}$ is the element of solid angle. The angular integral vanishes unless $m' = m$ and $l' = l \pm 1$, the selection rules for this polarisation, and for $m = 0$ it equals $(l+1)/\sqrt{(2l+1)(2l+3)}$ when $l' = l+1$. The rule $l' \neq l$ is the parity rule, since $|nlm\rangle$ has the parity $(-1)^l$. In particular 1s does not couple to 2s, although 2s has the same energy as 2p. We quote the standard result of these integrals. For the 1s–2p transition the radial integral and the angular factor $1/\sqrt{3}$ give

$$\langle 2p_0|\hat{z}|1s\rangle = \frac{128\sqrt{2}}{243}\,a_0 \approx 0.745\,a_0, \qquad \Omega = \frac{e_0 E_0}{\hbar}\,\langle 2p_0|\hat{z}|1s\rangle, \tag{3.19}$$

since $d_{eg} = -e_0\langle 2p_0|\hat{z}|1s\rangle$. The transition energy is $\hbar\omega_0 = E_2 - E_1 = \frac{3}{4}\mathrm{Ry} \approx 10.2$ eV. It is the first line of the ultraviolet Lyman series of hydrogen, called Lyman-$\alpha$, at the wavelength 121.5 nm for an infinitely heavy nucleus. The other allowed transitions, from 1s to 3p and higher p states and from 2p to 3s and 3d, lie far from this frequency. The qubit $|g\rangle = |1s\rangle = |0\rangle$, $|e\rangle = |2p_0\rangle = |1\rangle$ is therefore well isolated for a weak field.

**Limits of the two-level description**

Far-detuned states are only nearly empty. A state $r$ coupled to the ground state with the Rabi frequency $\Omega_r$ and the detuning $\Delta_r = \omega_L - \omega_{rg}$ shifts the ground-state energy by

$$\delta E_g = \frac{\Omega_r^2}{4\Delta_r}, \tag{3.20}$$

the *light shift*, which moves the resonance of the qubit transition. We quote it as a standard result of second-order perturbation theory. It also follows from the eigenvalue of the $2\times 2$ rotating-frame

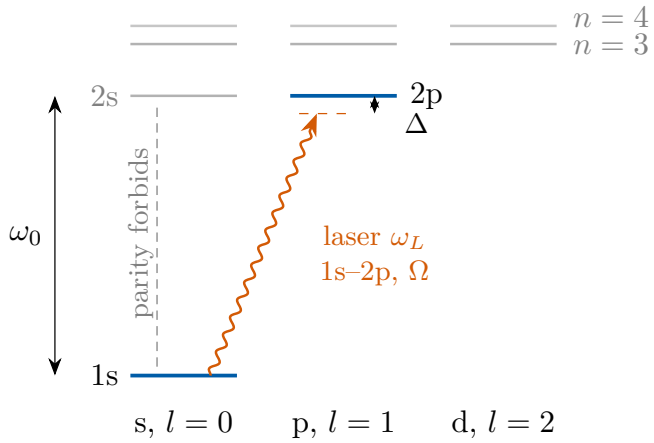



**Figure 5:** The lowest levels of hydrogen. The laser of frequency $\omega_L$ is close to the 1s–2p transition, with the gap $\omega_0$ and the detuning $\Delta$. The selection rules forbid the coupling of 1s to 2s, and every other allowed transition is far from resonance, so the atom behaves as a two-level system.

Hamiltonian of the pair $g$, $r$ that connects to the energy of $g$ when $\Omega_r \to 0$. The rotating-wave approximation requires $|\Omega|, |\Delta| \ll \omega_L$. The dropped terms raise the resonance frequency by $\Omega^2/(4\omega_0)$, the *Bloch–Siegert shift*, again a quoted result. The excited state also decays by spontaneous emission with the rate

$$\Gamma = \frac{\omega_0^3 \, |d_{eg}|^2}{3\pi\varepsilon_0 \hbar c^3}, \tag{3.21}$$

with the speed of light $c$, a result of quantum electrodynamics that we quote. The 2p state of hydrogen, with $d_{eg}$ from (3.19), lives only $1/\Gamma \approx 1.60$ ns, far too short for a computation. This decay returns as the relaxation time $T_1$ in Section 12. Atomic and ionic qubits therefore use pairs of long-lived levels. One choice is two hyperfine levels of the ground state, split by the coupling of the electron to the nuclear spin. Another is a *metastable* excited state, which cannot decay by a dipole transition and lives far longer. Microwaves, pairs of laser beams or narrow optical transitions drive these levels, and the same derivation yields the two-level Hamiltonian (3.15). The qubits of the other platforms of Table 2 obey a Hamiltonian of the same form.

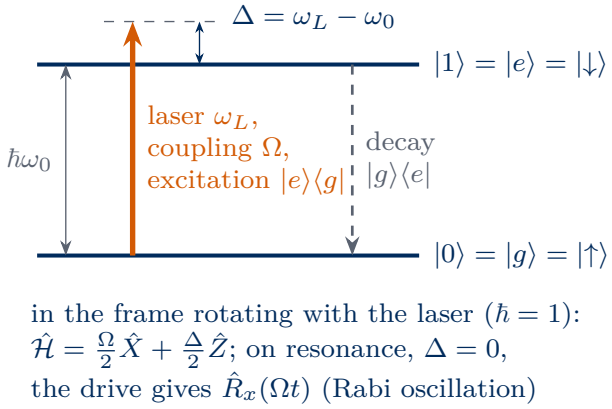



**Figure 6:** A two-level atom with the gap $\omega_0$ driven by a laser of frequency $\omega_L$, detuned by $\Delta = \omega_L - \omega_0$. The Rabi coupling $\Omega$ connects the ground state $|g\rangle = |0\rangle$ and the excited state $|e\rangle = |1\rangle$.

On resonance, $\Delta = 0$ and $\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Omega}{2}\hat{X}$. Its eigenvectors are $|\pm\rangle$ with the energies $\pm\Omega/2$, and the spectral form gives

$$\hat{U}(t) = e^{-i\Omega t/2}|+\rangle\langle+| + e^{i\Omega t/2}|-\rangle\langle-| = \cos\frac{\Omega t}{2}\,\hat{I} - i\sin\frac{\Omega t}{2}\,\hat{X} \equiv \hat{R}_x(\theta), \qquad \theta = \Omega t, \tag{3.22}$$

where the second form uses $|+\rangle\langle+| + |-\rangle\langle-| = \hat{I}$ and $|+\rangle\langle+| - |-\rangle\langle-| = \hat{X}$. The result is the *rotation gate* about $x$ by the angle $\theta$, called the pulse area. Starting in the ground state, the probability of the

excited state is

$$P_e(t) = |\langle e|\hat{R}_x(\Omega t)|g\rangle|^2 = \sin^2\frac{\Omega t}{2}, \tag{3.23}$$

the *Rabi oscillation*. A pulse of area $\pi$ moves the atom from $|g\rangle$ to $|e\rangle$, which is the gate $\hat{X}$ up to the phase $-i$. A pulse of area $\pi/2$ produces the equal superposition $(|g\rangle - i|e\rangle)/\sqrt{2}$. Off resonance the atom oscillates faster, with the generalised Rabi frequency $\sqrt{\Omega^2+\Delta^2}$, and reaches the excited state at most with the probability $\Omega^2/(\Omega^2+\Delta^2)$.

With the laser switched off, $\Omega = 0$, only the detuning remains, $\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z}$, and the spectral form gives

$$\hat{U}(t) = \begin{pmatrix} e^{-i\Delta t/2} & 0 \\ 0 & e^{i\Delta t/2} \end{pmatrix} = e^{-i\Delta t/2}\begin{pmatrix} 1 & 0 \\ 0 & e^{i\Delta t} \end{pmatrix}. \tag{3.24}$$

Up to a global phase, the evolution multiplies the amplitude of $|1\rangle$ by $e^{i\Delta t}$. In the form (2.20) the azimuth grows as $\varphi \to \varphi + \Delta t$ while $\vartheta$ stays fixed. The Bloch vector therefore turns about the $z$ axis with the angular frequency $\Delta$, counterclockwise seen from the north pole, a motion in the rotating frame called *precession*. A spin-1/2 in a static magnetic field along $z$ behaves in the same way. Its direction turns about the field, the *Larmor precession*. Rotations about $y$ and $z$ are defined like $\hat{R}_x$, $\hat{R}_y(\theta) = e^{-i\theta\hat{Y}/2}$ and $\hat{R}_z(\theta) = e^{-i\theta\hat{Z}/2}$.

### Rotation angles set by pulse area

The duration, amplitude and phase of a laser pulse fix the rotation of the qubit. On resonance a laser with the phase $\phi_L$ produces the Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{RF}}(t) = \frac{\Omega(t)}{2}\left(\cos\phi_L\,\hat{X} + \sin\phi_L\,\hat{Y}\right), \tag{3.25}$$

where the Rabi frequency $\Omega(t)$ follows the slowly varying amplitude of the field. This Hamiltonian changes in time only through the number $\Omega(t)$ and therefore commutes with itself at different times. The product of the evolutions over short time steps collapses to one exponential of the integrated coupling,

$$\hat{U} = \hat{R}_{\phi_L}(\theta) = e^{-i\theta(\cos\phi_L\hat{X} + \sin\phi_L\hat{Y})/2}, \qquad \theta = \int_0^T \Omega(t')\,dt', \tag{3.26}$$

for a pulse of duration $T$. The rotation angle is the *pulse area*, $\theta = \Omega T$ for a rectangular pulse of constant $\Omega$. A smooth envelope, preferred in experiments because its narrower frequency spectrum excites fewer neighbouring transitions, produces the same gate whenever its area is the same. The result requires resonance and a constant laser phase. With a detuning the Hamiltonian also contains $\hat{Z}$ and no longer commutes with itself at different times. The gate then depends on the shape of the pulse as well as on its area.

Rotations about $z$ show how the laser phase sets the axis of the rotation. A rotation about $z$ turns $\hat{X}$ into $\hat{R}_z(\phi)\hat{X}\hat{R}_z(-\phi) = \cos\phi\,\hat{X} + \sin\phi\,\hat{Y}$, and the exponential of a transformed matrix is the transformed exponential, so

$$\hat{R}_\phi(\theta) = \hat{R}_z(\phi)\,\hat{R}_x(\theta)\,\hat{R}_z(-\phi). \tag{3.27}$$

Free evolution with a detuning produces rotations about $z$, $\theta = \Delta t$, but in practice they cost no time at all. From (3.27) follows $\hat{R}_\phi(\theta)\hat{R}_z(\alpha) = \hat{R}_z(\alpha)\hat{R}_{\phi-\alpha}(\theta)$. A rotation by $\alpha$ about $z$ followed by pulses with the phases $\phi_1, \phi_2, \ldots$ thus equals the same pulses with the phases $\phi_k - \alpha$, followed by a final rotation about $z$. That last rotation changes only the relative phase of $|0\rangle$ and $|1\rangle$ and leaves the probabilities of a readout in the computational basis unchanged. The control electronics therefore implement a rotation about $z$ without any pulse, by shifting the phase of every later pulse, an operation called a *virtual* $\hat{Z}$ gate. A Rabi frequency, a duration and a phase set every single-qubit gate.

When the Hamiltonian changes in time the solution is no longer a single exponential. For a sequence of pulses, each with a constant Hamiltonian $\hat{\mathcal{H}}_m$ for a duration $t_m$, the evolution is the

product $\hat{U} = e^{-it_M\hat{\mathcal{H}}_M}\cdots e^{-it_2\hat{\mathcal{H}}_2}e^{-it_1\hat{\mathcal{H}}_1}$, with the latest pulse leftmost. The order matters whenever the Hamiltonians do not commute. A circuit is such a product of gates, and a smoothly varying $\hat{\mathcal{H}}(t)$ is the limit of many short pulses, the time-ordered exponential.

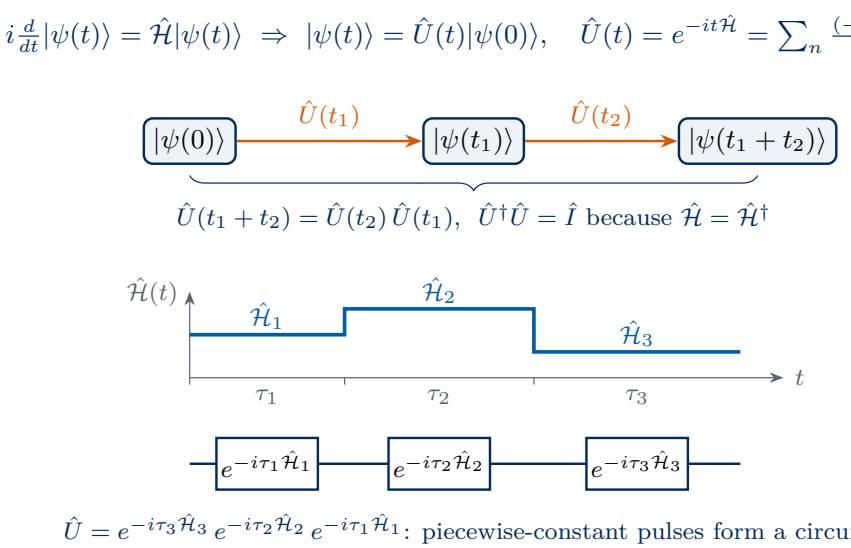



**Figure 7:** The evolution operator $\hat{U}(t) = e^{-it\hat{\mathcal{H}}}$ solves the Schrödinger equation for a constant Hamiltonian, and a sequence of constant pulses gives a product of exponentials, a circuit, with the latest pulse leftmost.

### 3.3 The exponential of a Pauli matrix

For the Pauli matrices and their products the exponential needs no diagonalisation. The single-qubit gates and most two-qubit gates are exponentials $e^{-ia\hat{P}}$ of a matrix $\hat{P}$ that squares to the identity. All even powers of such a matrix equal $\hat{I}$ and all odd powers equal $\hat{P}$. We split the exponential series into even and odd terms,

$$\begin{aligned} e^{-ia\hat{P}} &= \sum_{m\geq 0}\frac{(-ia)^{2m}}{(2m)!}\,\hat{I} + \sum_{m\geq 0}\frac{(-ia)^{2m+1}}{(2m+1)!}\,\hat{P} \\ &= \underbrace{\sum_m \frac{(-1)^m a^{2m}}{(2m)!}}_{\cos a}\,\hat{I} - i\underbrace{\sum_m \frac{(-1)^m a^{2m+1}}{(2m+1)!}}_{\sin a}\,\hat{P} = \cos a\,\hat{I} - i\sin a\,\hat{P}, \end{aligned} \tag{3.28}$$

the matrix version of Euler's formula $e^{-ia} = \cos a - i\sin a$. Every Pauli matrix squares to $\hat{I}$. So does $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} = n_x\hat{X} + n_y\hat{Y} + n_z\hat{Z}$ for a unit vector $\mathbf{n}$, because in its square the cross terms cancel in pairs by anticommutation, $(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}})^2 = (n_x^2+n_y^2+n_z^2)\hat{I} = \hat{I}$. The rotation about the axis $\mathbf{n}$ by the angle $\theta$ is therefore

$$\hat{R}_{\mathbf{n}}(\theta) = e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2} = \cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}. \tag{3.29}$$

It turns the Bloch vector by the angle $\theta$ about $\mathbf{n}$, counterclockwise when $\mathbf{n}$ points towards the viewer. For $\hat{R}_z(\theta)$ the rotation is the precession of Section 3.2, with $\Delta t = \theta$. For any other axis we choose the coordinate axes of the Bloch sphere so that $\mathbf{n}$ becomes the new $z$ axis. In these coordinates $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ plays the role of $\hat{Z}$, and the rotation is again a Larmor precession by $\theta$ about $\mathbf{n}$. Because of the factor $1/2$ in the exponent, the amplitudes acquire phases of only $\theta/2$ while the Bloch vector turns by $\theta$. After a full turn, $\hat{R}_{\mathbf{n}}(2\pi) = -\hat{I}$ returns every Bloch vector to itself and multiplies the state by the unobservable global phase $-1$.

### 3.4 Single-qubit gates as rotations

Up to a global phase that no measurement can see, every single-qubit gate is a rotation (3.29). The standard gates are named after their action. The gate $\hat{X}$ is the *bit flip*, $\hat{X}|0\rangle = |1\rangle$ and $\hat{X}|1\rangle = |0\rangle$. The gate $\hat{Z}$ is the *phase flip*, which leaves $|0\rangle$ alone and multiplies $|1\rangle$ by $-1$. The *phase gates* $\hat{S} = \mathrm{diag}(1, i)$ and $\hat{T} = \mathrm{diag}(1, e^{i\pi/4})$ multiply $|1\rangle$ by $i$ and by $e^{i\pi/4}$. The *Hadamard gate* $\hat{H} = (\hat{X} + \hat{Z})/\sqrt{2}$ turns $|0\rangle$ into $|+\rangle$ and $|1\rangle$ into $|-\rangle$.

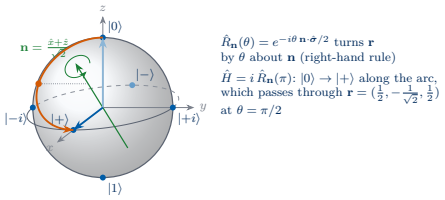



**Figure 8:** The Hadamard gate as a half turn about the axis $(\hat{x} + \hat{z})/\sqrt{2}$, which carries $|0\rangle$ at the north pole to $|+\rangle$ on the equator.

Evaluating (3.29) at the angle of each gate yields its rotation form. At $\theta = \pi$ the cosine vanishes and the sine equals one, hence $\hat{R}_{\mathbf{n}}(\pi) = -i\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ and

$$\hat{X} = i\,\hat{R}_x(\pi), \qquad \hat{Z} = i\,\hat{R}_z(\pi), \qquad \hat{H} = \frac{\hat{X} + \hat{Z}}{\sqrt{2}} = i\,\hat{R}_{\mathbf{n}_H}(\pi) \quad \text{with} \quad \mathbf{n}_H = \frac{\hat{x} + \hat{z}}{\sqrt{2}}. \tag{3.30}$$

For the phase gates, $\hat{R}_z(\phi) = \mathrm{diag}(e^{-i\phi/2}, e^{i\phi/2}) = e^{-i\phi/2}\,\mathrm{diag}(1, e^{i\phi})$, which gives $\hat{S} = e^{i\pi/4}\hat{R}_z(\pi/2)$ and $\hat{T} = e^{i\pi/8}\hat{R}_z(\pi/4)$. Each gate $\hat{G} = e^{i\alpha}\hat{R}_{\mathbf{n}}(\theta)$ is also the evolution under a generator $\hat{\mathcal{H}}_G$ switched on for the unit time, $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$, with

$$\hat{\mathcal{H}}_G = \frac{\theta}{2}\,(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \hat{I}), \tag{3.31}$$

where the term proportional to $\hat{I}$ produces the global phase $e^{i\theta/2}$, which equals $e^{i\alpha}$ for the five named gates of Table 3. In general $\hat{G} = e^{i\alpha}\hat{R}_{\mathbf{n}}(\theta)$ has the generator $\hat{\mathcal{H}}_G = \frac{\theta}{2}\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \alpha\hat{I}$, and the plain rotation in the last row of Table 3 has $\alpha = 0$.

| gate | matrix | rotation | axis $\mathbf{n}$ | angle $\theta$ | generator $\hat{\mathcal{H}}_G$ |
|---|---|---|---|---|---|
| $\hat{X}$ | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ | $i\,\hat{R}_x(\pi)$ | $\hat{x}$ | $\pi$ | $\frac{\pi}{2}(\hat{X} - \hat{I})$ |
| $\hat{Z}$ | $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ | $i\,\hat{R}_z(\pi)$ | $\hat{z}$ | $\pi$ | $\frac{\pi}{2}(\hat{Z} - \hat{I})$ |
| $\hat{H}$ | $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ | $i\,\hat{R}_{\mathbf{n}_H}(\pi)$ | $(\hat{x} + \hat{z})/\sqrt{2}$ | $\pi$ | $\frac{\pi}{2}(\hat{H} - \hat{I})$ |
| $\hat{S}$ | $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ | $e^{i\pi/4}\hat{R}_z(\pi/2)$ | $\hat{z}$ | $\pi/2$ | $\frac{\pi}{4}(\hat{Z} - \hat{I})$ |
| $\hat{T}$ | $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$ | $e^{i\pi/8}\hat{R}_z(\pi/4)$ | $\hat{z}$ | $\pi/4$ | $\frac{\pi}{8}(\hat{Z} - \hat{I})$ |
| $\hat{R}_{\mathbf{n}}(\theta)$ | $\cos\frac{\theta}{2}\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ | $\hat{R}_{\mathbf{n}}(\theta)$ | $\mathbf{n}$ | $\theta$ | $\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ |

**Table 3:** Single-qubit gates as rotations of the Bloch sphere and as evolution operators $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$. For the five named gates the unobservable global phase equals $e^{i\theta/2}$.

We watch a gate act as a rotation by following the evolution that implements it. For $\hat{G} = e^{i\alpha}\hat{R}_{\mathbf{n}}(\theta)$ we switch on $\hat{\mathcal{H}} = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ for the times $0 \le t \le 1$, and $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$ ends at the gate. Figure 9 shows

the path of the Bloch vector of the state with $\vartheta = 2\pi/3$, $\varphi = \pi/3$, chosen away from every rotation axis.

Each path is an arc of a circle about the axis of the gate, spanning the angle of the gate. The component of the Bloch vector along the axis stays fixed. The gates $\hat{X}$, $\hat{Z}$ and $\hat{H}$ turn the state by half a circle, about $x$, about $z$ and about the diagonal axis between $x$ and $z$. The gates $\hat{S}$ and $\hat{T}$ turn it by a quarter and an eighth of a circle about $z$. The three phase gates share their axis. As a result, the path of $\hat{T}$ is the first half of the path of $\hat{S}$, which is the first half of the path of $\hat{Z}$.

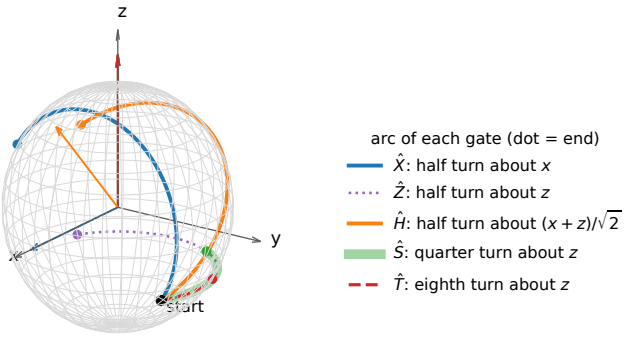



**Figure 9:** Bloch-vector paths during the evolutions $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$, $0 \le t \le 1$, that implement the gates $\hat{X}$, $\hat{Z}$, $\hat{H}$, $\hat{S}$ and $\hat{T}$, started from $\vartheta = 2\pi/3$, $\varphi = \pi/3$. Arrows mark the rotation axes.

Each gate is thus given by its matrix in Table 3, by its action on the basis states and on the Bloch sphere, and by the pulses that implement it. Table 4 collects the last two. The implementations follow from the pulse area (3.26) and the virtual $\hat{Z}$ gate of Section 3.2. A resonant pulse of area $\pi$ and phase $\phi_L = 0$ produces $\hat{R}_x(\pi) = -i\hat{X}$, and the phase gates are rotations about $z$ that cost no time. The Hadamard gate combines both, $\hat{H} = \hat{R}_y(\pi/2)\,\hat{Z}$, a virtual $\hat{Z}$ gate followed by a pulse of area $\pi/2$ with the phase $\phi_L = \pi/2$.

| gate | action on basis states | Bloch sphere | implementation |
|---|---|---|---|
| $\hat{X}$ | $\lvert0\rangle \leftrightarrow \lvert1\rangle$, $\lvert\pm\rangle \to \pm\lvert\pm\rangle$ | half turn about $x$ | pulse of area $\pi$, $\phi_L = 0$ |
| $\hat{Z}$ | $\lvert1\rangle \to -\lvert1\rangle$, $\lvert+\rangle \leftrightarrow \lvert-\rangle$ | half turn about $z$ | virtual $\hat{Z}$ gate |
| $\hat{H}$ | $\lvert0\rangle \to \lvert+\rangle$, $\lvert1\rangle \to \lvert-\rangle$, $\lvert\pm\rangle \to \lvert0\rangle, \lvert1\rangle$ | half turn about $(\hat{x}+\hat{z})/\sqrt{2}$ | virtual $\hat{Z}$, then area $\pi/2$ at $\phi_L = \pi/2$ |
| $\hat{S}$ | $\lvert1\rangle \to i\lvert1\rangle$, $\lvert+\rangle \to \lvert+i\rangle$ | quarter turn about $z$ | virtual $\hat{Z}$ gate |
| $\hat{T}$ | $\lvert1\rangle \to e^{i\pi/4}\lvert1\rangle$ | eighth of a turn about $z$ | virtual $\hat{Z}$ gate |
| $\hat{R}_\phi(\theta)$ | mixes $\lvert0\rangle$ and $\lvert1\rangle$ with amplitudes $\cos\frac{\theta}{2}$, $\sin\frac{\theta}{2}$ | turn by $\theta$ about $(\cos\phi, \sin\phi, 0)$ | pulse of area $\theta$ and phase $\phi_L = \phi$ |
| $\hat{R}_z(\theta)$ | $\lvert1\rangle \to e^{i\theta}\lvert1\rangle$ up to a global phase | turn by $\theta$ about $z$ | detuning for a time $\theta/\Delta$, or virtual $\hat{Z}$ |

**Table 4:** Action of the single-qubit gates on basis states and on the Bloch sphere, and their implementation with resonant pulses and virtual $\hat{Z}$ gates.

### 3.5 A one-qubit circuit and interference

A *circuit* is a sequence of gates applied to a prepared state, followed by a measurement. It is drawn as a horizontal line for the qubit, with time running from left to right, gates as boxes and the measurement

as a meter. Gates act on the state from the left, so the gate applied first stands rightmost in the matrix product. The circuit "first $\hat{H}$, then $\hat{S}$" is the matrix $\hat{S}\hat{H}$, and it differs from $\hat{H}\hat{S}$. Starting from $|0\rangle$, $\hat{S}\hat{H}|0\rangle = \hat{S}|+\rangle = |+i\rangle$ points along $y$, while $\hat{H}\hat{S}|0\rangle = \hat{H}|0\rangle = |+\rangle$ points along $x$, because $\hat{S}$ leaves $|0\rangle$ unchanged.

The circuit "first $\hat{H}$, then $\hat{Z}$, then $\hat{H}$" is the matrix $\hat{H}\hat{Z}\hat{H}$. The Hadamard gate exchanges the $x$ and $z$ axes of the Bloch sphere, so a half turn about $z$ framed by two Hadamard gates is a half turn about $x$,

$$\hat{H}\hat{Z}\hat{H} = \hat{X}. \tag{3.32}$$

The circuit flips $|0\rangle$ to $|1\rangle$ with certainty. The first Hadamard gate prepares $|+\rangle$, the $\hat{Z}$ gate changes only the relative phase and gives $|-\rangle$, and the second Hadamard gate turns this phase difference into a different bit. With $\hat{R}_z(\phi)$ in place of $\hat{Z}$, the circuit becomes the *Ramsey interference* experiment of Figure 10. The amplitude of $|1\rangle$ after $\hat{H}\,\hat{R}_z(\phi)\,\hat{H}|0\rangle$ is

$$\langle 1|\hat{H}\,\hat{R}_z(\phi)\,\hat{H}|0\rangle = \tfrac{1}{2}(e^{-i\phi/2} - e^{i\phi/2}) = -i\sin\frac{\phi}{2}, \qquad \text{so} \qquad p(1) = \sin^2\frac{\phi}{2}. \tag{3.33}$$

Without the second Hadamard gate the state $\hat{R}_z(\phi)|+\rangle$ lies on the equator and $p(1) = 1/2$ for every $\phi$. Interference converts a phase that is invisible in a $\hat{Z}$ measurement into a population.

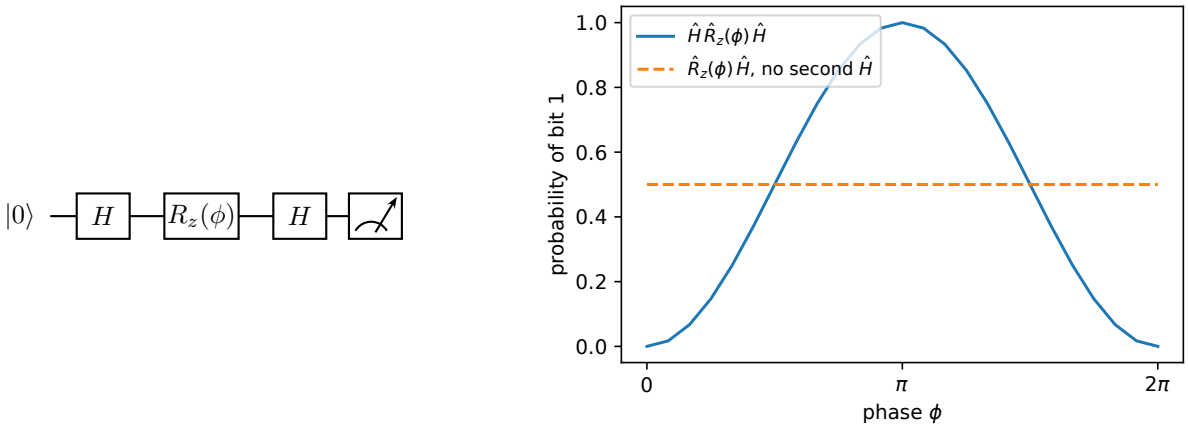



**Figure 10:** Ramsey interference. Left, the circuit. Right, the probability of the bit 1 against the phase $\phi$, with the second Hadamard gate, $\sin^2(\phi/2)$, and without it, a constant $1/2$.

### 3.6 The Euler decomposition (optional)

A product of rotations is again a rotation, and every $2 \times 2$ unitary can be written as

$$\hat{U} = e^{i\alpha}\,\hat{R}_z(\beta)\,\hat{R}_y(\gamma)\,\hat{R}_z(\delta), \tag{3.34}$$

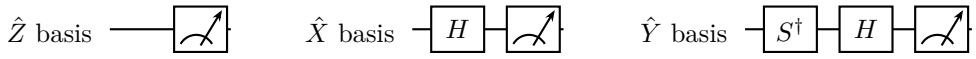
the *Euler decomposition* with three angles $\beta, \gamma, \delta$ and a global phase $\alpha$. The set of single-qubit gates is therefore the set of rotations of the Bloch sphere, and two fixed rotation axes suffice to produce all of them. For the Hadamard gate, $\hat{H} = i\,\hat{R}_y(\pi/2)\,\hat{R}_z(\pi)$. For a diagonal gate such as $\hat{T}$ the angle $\gamma$ vanishes and only the sum $\beta + \delta$ is determined, so the decomposition is not unique.

## 4 Other measurement bases and shots

### 4.1 Measurement along $x$ and $y$

A laboratory reads a qubit in its computational basis. To measure the spin along $x$ we first rotate the state so that the eigenvectors of $\hat{X}$ are carried to $|0\rangle$ and $|1\rangle$, and then measure $\hat{Z}$. The Hadamard gate, a half turn about the axis between $x$ and $z$, performs this rotation, $\hat{H}|+\rangle = |0\rangle$ and $\hat{H}|-\rangle = |1\rangle$.

For the $y$ axis we first apply $\hat{S}^\dagger = \mathrm{diag}(1, -i)$, which turns $|\pm i\rangle$ into $|\pm\rangle$, and then $\hat{H}$. The three measurement circuits are




In every basis the outcome bit 0 stands for the eigenvalue $+1$ of the measured Pauli matrix and the bit 1 for $-1$. From $\langle\hat{\sigma}_\alpha\rangle = p_\alpha(0) - p_\alpha(1)$ and $p_\alpha(0) + p_\alpha(1) = 1$ the probabilities are

$$p_\alpha(0) = \frac{1 + \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad p_\alpha(1) = \frac{1 - \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad \alpha = x, y, z. \tag{4.1}$$

For the example state (2.4) the Bloch vector gives $p_x(0) = 0.806$ and $p_y(0) = 0.806$, in addition to $p_z(0) = 0.750$.

The three measurements are incompatible. A state that gives a certain result in one basis gives random bits in the other two. The state $|0\rangle$ always returns 0 in the $\hat{Z}$ basis and random bits with probability one half each, a *fair coin*, in the $\hat{X}$ and $\hat{Y}$ bases, while $|+\rangle$ does the opposite (Figure 12). This pattern is called *complementarity*. Besides interference, it is the second way in which the relative phase shows up, since $|+\rangle$ and $|-\rangle$ have the same $\hat{Z}$ probabilities and opposite $\hat{X}$ outcomes. On the Bloch sphere a pure state has $|\mathbf{r}| = 1$, which allows at most one component to equal $\pm 1$. A state with a certain outcome along one axis therefore lies on the equator of the other two.

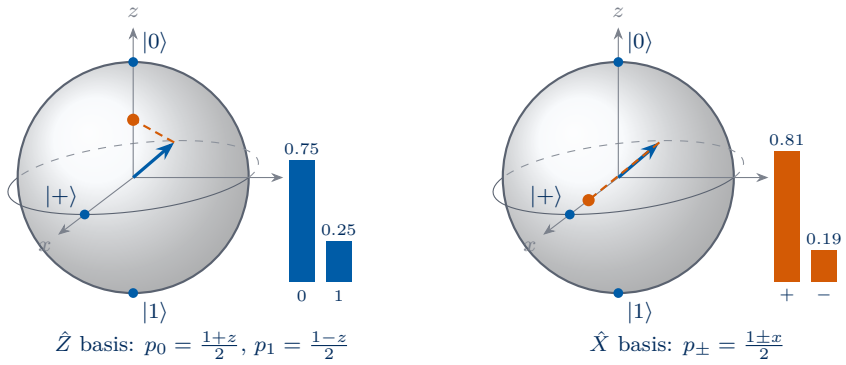



**Figure 11:** Measurement in the $\hat{Z}$ and $\hat{X}$ bases. The probability of the bit 0 is fixed by the projection of the Bloch vector on the measured axis, $p(0) = (1 + r_\alpha)/2$. After the measurement the state sits at the pole that was found.

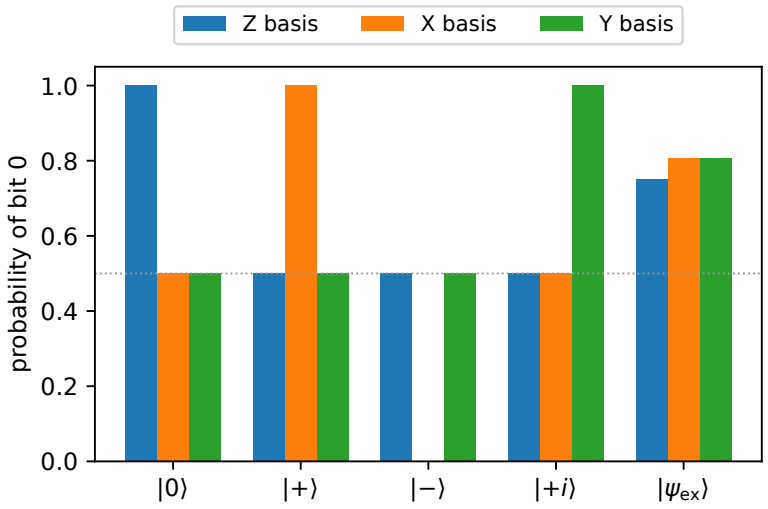



**Figure 12:** Probability of the bit 0 for five states measured in the $\hat{Z}$, $\hat{X}$ and $\hat{Y}$ bases. A bar at 1 or 0 is a certain outcome, a bar at 0.5 (dotted) a fair coin. No state has certain outcomes in two bases.

### 4.2 Shots and the estimate of an expectation value

An experiment repeats preparation and measurement $M$ times. Each repetition is a *shot* and returns a value $z_j = \pm 1$. We estimate $\langle \hat{Z} \rangle$ by the sample mean

$$\bar{z} = \frac{1}{M} \sum_{j=1}^{M} z_j. \tag{4.2}$$

Its average over many records is $\langle \hat{Z} \rangle$, so the estimator is unbiased. Because $z_j^2 = 1$, one outcome has the variance $\langle \hat{Z}^2 \rangle - \langle \hat{Z} \rangle^2 = 1 - \langle \hat{Z} \rangle^2$. Since the shots are independent, the variance of the mean is this single-shot variance divided by $M$, and the standard error is

$$\mathrm{std}(\bar{z}) = \sqrt{\frac{1 - \langle \hat{Z} \rangle^2}{M}}. \tag{4.3}$$

With the error falling as $1/\sqrt{M}$, doubling the precision costs four times as many shots and one more decimal digit one hundred times as many. In practice $\langle \hat{Z} \rangle$ is unknown, and the standard error is estimated by the sample standard deviation of the record divided by $\sqrt{M}$.

A simulated record of 1000 shots of the example state, with the exact value $\langle \hat{Z} \rangle = 1/2$, gives the estimate $0.484 \pm 0.028$, in agreement with the standard error 0.027 predicted by (4.3). The spread of the estimates over 400 repeated records for each record length $M$ follows the $1/\sqrt{M}$ law (Figure 13, left). The Ramsey experiment of Section 3.5, simulated with 200 shots for each phase, follows the curve $\sin^2(\phi/2)$ within about two error bars $\sqrt{p(1-p)/M}$ (Figure 13, right). Computed from the estimated $p$, these bars are largest near $p = 1/2$ and vanish where every shot gave the same bit. Every expectation value measured on a quantum device carries this statistical error, and variational algorithms have to budget it.

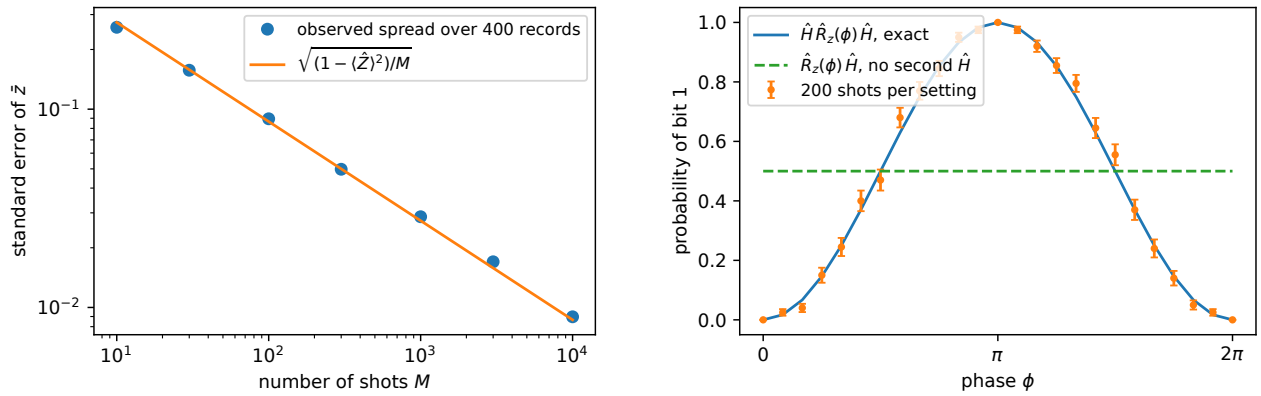



**Figure 13:** Left, the observed spread of the estimate of $\langle \hat{Z} \rangle$ over repeated records against the number of shots $M$, compared with formula (4.3). The slope $-1/2$ in the double-logarithmic plot is the $1/\sqrt{M}$ law. Right, the Ramsey fringe measured with 200 shots per phase, with the exact curve and the curve without the second Hadamard gate.

## 5 The two-spin register in a computer

### 5.1 Two spins and the tensor product

Two spins, qubit 0 and qubit 1, form the smallest register in which the parts can be correlated. Suppose qubit 0 is prepared in $|\psi_1\rangle = a|0\rangle + b|1\rangle = (a, b)^T$ and, independently, qubit 1 in $|\psi_2\rangle = c|0\rangle + d|1\rangle = (c, d)^T$. The labels 1 and 2 count the two states, and the first factor always belongs to

qubit 0. The joint state is the *tensor product*, computed by distributing the product over both sums,

$$|\psi_{12}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}. \quad (5.1)$$

Every amplitude of the pair is a product of one amplitude of each qubit. Probabilities of bit strings factorise in the same way, for example $p(01) = |a|^2|d|^2$, as expected for independently prepared qubits. The symbol $\otimes$ is often omitted, and the shorthands $|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_1\rangle|\psi_2\rangle = |\psi_1\psi_2\rangle$ denote one vector, as in $|0\rangle \otimes |1\rangle = |0\rangle|1\rangle = |01\rangle$. We list the amplitudes of every register in the order $00, 01, 10, 11$ of the column vector in (5.1). In code the vector is `np.kron(psi1, psi2)`, and the unnormalised inputs $(a, b) = (1, 2)$ and $(c, d) = (3, 4)$ return $(3, 4, 6, 8)$.

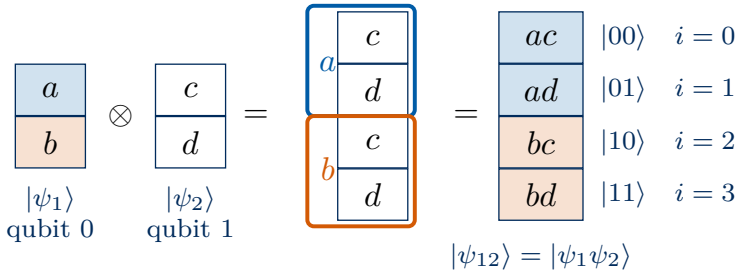



**Figure 14:** The tensor product $(a, b) \otimes (c, d) = (ac, ad, bc, bd)$. Each amplitude of the first qubit multiplies the whole vector of the second, which produces the order $00, 01, 10, 11$.

A state of the form $|a\rangle \otimes |b\rangle$ is a *product state*. The state space of the pair is the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$, spanned by the four products of basis states

$$|00\rangle = |0\rangle \otimes |0\rangle, \quad |01\rangle = |0\rangle \otimes |1\rangle, \quad |10\rangle = |1\rangle \otimes |0\rangle, \quad |11\rangle = |1\rangle \otimes |1\rangle. \quad (5.2)$$

The label of a basis state is a *bit string*. Its first bit is the state of qubit 0 and its second bit the state of qubit 1. A general state of the register is any normalised superposition of the four basis states,

$$|\psi\rangle = \sum_{b_0, b_1 \in \{0,1\}} \psi_{b_0 b_1} |b_0 b_1\rangle, \qquad \sum_{b_0 b_1} |\psi_{b_0 b_1}|^2 = 1, \quad (5.3)$$

with four complex amplitudes. A state that cannot be written as a product of two single-qubit states is called *entangled*, and most states of the register are of this kind.

The spin components of the two qubits along $z$ are $\hat{Z}_0$, the Pauli matrix $\hat{Z}$ acting on qubit 0, and $\hat{Z}_1$, acting on qubit 1. Every basis state is a simultaneous eigenstate of both, and the eigenvalue of $\hat{Z}_q$ on $|b_0 b_1\rangle$ is $(-1)^{b_q}$, equal to $+1$ for $b_q = 0$ and $-1$ for $b_q = 1$. A measurement of both spins along $z$ reads out the bit string, and the computational basis takes its name from this readout. For $N$ qubits the computational basis consists of the $2^N$ common eigenvectors of $\hat{Z}_0, \ldots, \hat{Z}_{N-1}$, labelled by the bit strings $b_0 b_1 \ldots b_{N-1}$.

## 5.2 The vector of $2^N$ amplitudes and its order

A computer stores the state of $N$ qubits as a vector of $2^N$ complex numbers, the amplitudes $\psi_{b_0 \ldots b_{N-1}}$ of the computational basis states. To place them in one array we must agree on an order. We use the order produced by the *Kronecker product*, the matrix form of the tensor product computed in (5.1),

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \begin{pmatrix} c \\ d \end{pmatrix} \\ b \begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix}. \quad (5.4)$$

Each entry of the left vector multiplies a copy of the right vector. Down the column the left factor, qubit 0, changes slowest and the right factor, qubit 1, fastest. The four entries are the amplitudes of 00, 01, 10 and 11, in binary counting order. The basis vector of a bit string is the Kronecker product $|b_0\rangle \otimes |b_1\rangle \otimes \cdots \otimes |b_{N-1}\rangle$ of single-qubit basis vectors, the unit vector with entry 1 at the index

$$i = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q}. \tag{5.5}$$

Qubit 0 is the most significant bit, and Table 5 lists the order for two and three qubits. For three qubits the bit string 101 sits at the index $1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 5$. Product states are Kronecker products of single-qubit states, and the running product state of the register is

$$|0\rangle \otimes |+\rangle = \tfrac{1}{\sqrt{2}}(1, 1, 0, 0)^T, \tag{5.6}$$

with amplitude $1/\sqrt{2}$ on 00 and on 01, the two bit strings in which qubit 0 is 0. A general state is any normalised vector of length $2^N$.

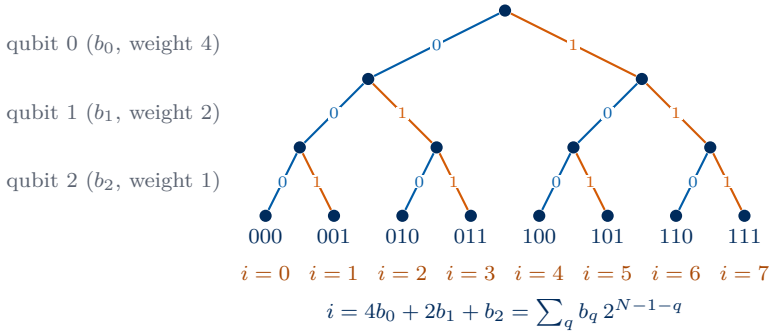



**Figure 15:** The storage order for $N = 3$. Reading the bits of qubits 0, 1 and 2 from the top of the tree to the bottom gives the binary index $i$, with qubit 0 as the most significant bit.

| $N = 2$ | | | $N = 3$ | | |
|---|---|---|---|---|---|
| index $i$ | bit string | Kronecker product | index $i$ | bit string | Kronecker product |
| 0 | 00 | $\|0\rangle \otimes \|0\rangle$ | 0 | 000 | $\|0\rangle \otimes \|0\rangle \otimes \|0\rangle$ |
| 1 | 01 | $\|0\rangle \otimes \|1\rangle$ | 1 | 001 | $\|0\rangle \otimes \|0\rangle \otimes \|1\rangle$ |
| 2 | 10 | $\|1\rangle \otimes \|0\rangle$ | 2 | 010 | $\|0\rangle \otimes \|1\rangle \otimes \|0\rangle$ |
| 3 | 11 | $\|1\rangle \otimes \|1\rangle$ | 3 | 011 | $\|0\rangle \otimes \|1\rangle \otimes \|1\rangle$ |
| | | | 4 | 100 | $\|1\rangle \otimes \|0\rangle \otimes \|0\rangle$ |
| | | | 5 | 101 | $\|1\rangle \otimes \|0\rangle \otimes \|1\rangle$ |
| | | | 6 | 110 | $\|1\rangle \otimes \|1\rangle \otimes \|0\rangle$ |
| | | | 7 | 111 | $\|1\rangle \otimes \|1\rangle \otimes \|1\rangle$ |

**Table 5:** Storage order of the amplitudes. The index of a bit string is its value as a binary number with qubit 0 as the most significant bit, equation (5.5).

The memory grows exponentially with the number of qubits. With 16 bytes per complex number in double precision, the state of $N$ qubits needs $16 \cdot 2^N$ bytes. Thirty qubits fill 16 GiB, the memory of a large workstation. Forty qubits need 16 TiB, and every further qubit doubles the requirement. Classical simulation of quantum systems is hard because of this cost, and tensor networks are one way around it.

Other software libraries may use the opposite convention. Qiskit, for example, treats qubit 0 as the least significant bit and prints bit strings with qubit 0 on the right. Before results from such a library are compared with ours, their bit strings must be reversed and their amplitudes reordered.

### 5.3 The state tensor

The flat index $i$ hides the qubits, and a reshape of the vector of length $2^N$ into an array with $N$ axes of length 2 recovers them,

$$\psi[b_0, b_1, \ldots, b_{N-1}] = \psi_i, \qquad i = \sum_q b_q 2^{N-1-q}. \tag{5.7}$$

The reshaped array is the *state tensor*, with one index per qubit. NumPy and JAX store arrays in row-major or C order, where the last axis changes fastest, as in equation (5.5). The call `psi.reshape((2,)*N)` produces the state tensor without moving any number in memory, and axis $q$ of the result is qubit $q$. For two qubits the state tensor is a $2 \times 2$ matrix with qubit 0 labelling the rows,

$$\psi[b_0, b_1] = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{5.8}$$

A reshape only regroups indices, whereas a transpose permutes the axes. Applied to the two-qubit state tensor, a transpose exchanges the roles of the qubits, and flattening the transposed array lists the amplitudes in the order $00, 10, 01, 11$.

### 5.4 An operator on one qubit of the register

Operators on the register are built with the Kronecker product, in the same way as states. For two $2 \times 2$ matrices the Kronecker product is a $4 \times 4$ block matrix. Every entry of the left factor multiplies a full copy of the right factor,

$$\hat{A} \otimes \hat{B} = \begin{pmatrix} A_{00}\hat{B} & A_{01}\hat{B} \\ A_{10}\hat{B} & A_{11}\hat{B} \end{pmatrix}, \qquad (\hat{A} \otimes \hat{B})_{(b_0 b_1),(b_0' b_1')} = A_{b_0 b_0'}\, B_{b_1 b_1'}, \tag{5.9}$$

where the row label $(b_0 b_1)$ is the bit string of the two qubits, at the index $2b_0 + b_1$ of Table 5, and likewise for the column label. By the element formula, a Kronecker product acts on product states factor by factor, $(\hat{A} \otimes \hat{B})(|u\rangle \otimes |v\rangle) = \hat{A}|u\rangle \otimes \hat{B}|v\rangle$, and two Kronecker products multiply in the same way, $(\hat{A} \otimes \hat{B})(\hat{C} \otimes \hat{D}) = \hat{A}\hat{C} \otimes \hat{B}\hat{D}$. An operator $\hat{U}$ on qubit 0 alone is $\hat{U} \otimes \hat{I}$, and on qubit 1 alone it is $\hat{I} \otimes \hat{U}$. Four examples written out are

$$\hat{Z} \otimes \hat{I} = \mathrm{diag}(1, 1, -1, -1), \qquad \hat{I} \otimes \hat{Z} = \mathrm{diag}(1, -1, 1, -1), \tag{5.10}$$

$$\hat{X} \otimes \hat{I} = \begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \hat{I} \otimes \hat{X} = \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{5.11}$$

The matrix $\hat{X} \otimes \hat{I}$ flips qubit 0 by exchanging the amplitudes of $00 \leftrightarrow 10$ and $01 \leftrightarrow 11$, and $\hat{I} \otimes \hat{X}$ flips qubit 1 by exchanging $00 \leftrightarrow 01$ and $10 \leftrightarrow 11$. The diagonal of $\hat{Z}_0 = \hat{Z} \otimes \hat{I}$ holds $+1$ at the indices 0 and 1, the bit strings 00 and 01, and $-1$ at the indices 2 and 3. It reads bit 0 of the index. The alternating diagonal of $\hat{Z}_1 = \hat{I} \otimes \hat{Z}$ reads bit 1. Every operator on the register has its rows and columns labelled by bit strings in the order of Table 5.

In index notation, where a matrix–vector product reads $w_i = \sum_j A_{ij} v_j$, an operator on one qubit changes only the index of that qubit,

$$\big((\hat{U} \otimes \hat{I})\psi\big)[b_0', b_1] = \sum_{b_0} U_{b_0' b_0}\, \psi[b_0, b_1], \qquad \big((\hat{I} \otimes \hat{U})\psi\big)[b_0, b_1'] = \sum_{b_1} U_{b_1' b_1}\, \psi[b_0, b_1]. \tag{5.12}$$

For three qubits the operator on qubit 1 changes only the middle index and equals $\hat{I} \otimes \hat{U} \otimes \hat{I}$. For $N$ qubits the matrix $\hat{U}$ acts on index $q$ of the state tensor alone.

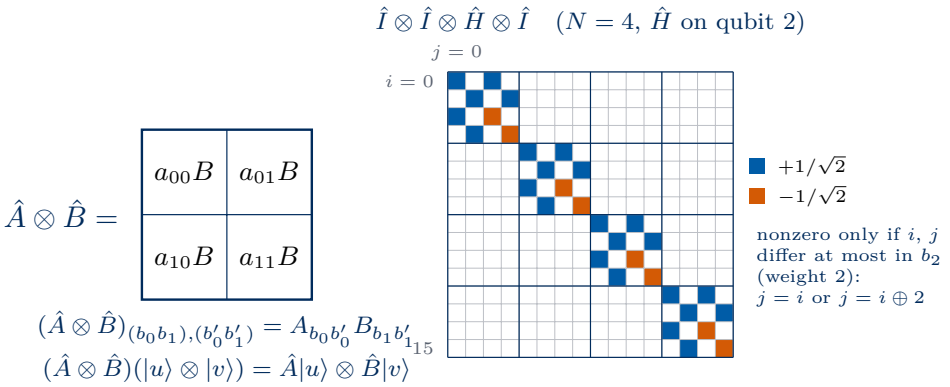



**Figure 16:** Left, the Kronecker product $\hat{A} \otimes \hat{B}$ as a block matrix in which every entry of $\hat{A}$ multiplies a copy of $\hat{B}$. Right, the pattern of nonzero entries of the $16 \times 16$ matrix $\hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, two in every row, a gate on one qubit of four.

### Index sums on a computer

A computer evaluates the index sums of the Kronecker matrices without ever building a $2^N \times 2^N$ matrix. The function `einsum` of NumPy and JAX turns an index formula into code through a string of index letters. Letters that appear in two inputs and not in the output are summed over. Letters in the output are kept, in the order written after `->`. Table 6 lists the basic operations. The two sums of (5.12) are `"Aa,ab->Ab"` and `"Bb,ab->aB"`, where an upper-case letter labels the new index produced by the operator, the primed index of the formula. A *tensor-network diagram* pictures the same sum with a box for each tensor and a line, a *leg*, for each index, and joined legs are summed over (Figure 18). The contraction and the Kronecker matrix give the same vector. Its cost of about $2 \cdot 2^N$ multiplications per one-qubit gate, against $4^N$ for the full matrix, makes it the method used for many qubits.

The Kronecker matrix $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ of a one-qubit gate is mostly zeros. A computer stores it as a *sparse matrix*, a record of the nonzero entries and their positions. A common format is the compressed sparse row (CSR) format of `scipy.sparse`, and the function `scipy.sparse.kron` builds $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ directly in this format. A one-qubit gate on $N$ qubits has at most $2 \cdot 2^N$ nonzero entries, two in each row, out of $4^N$. The cost of a sparse matrix–vector product is proportional to the number of nonzero entries.

| operation | index notation | `einsum` string |
|---|---|---|
| matrix times vector | $w_i = \sum_j A_{ij} v_j$ | `"ij,j->i"` |
| inner product $\langle a\|b\rangle$ | $\sum_i a_i^* b_i$ | `"i,i->"` on $(a^*, b)$ |
| outer product $\|a\rangle\langle b\|$ | $a_i b_j^*$ | `"i,j->ij"` on $(a, b^*)$ |
| trace | $\sum_i A_{ii}$ | `"ii->"` |
| Kronecker product $a \otimes b$ | $a_i b_j$, then reshape | `"i,j->ij"`, then `reshape(-1)` |

**Table 6:** Index notation and `einsum` strings for basic operations.

# 6 Bit strings of two qubits

## 6.1 Probabilities, marginals and correlators

Measuring both qubits in the computational basis returns one of the four bit strings. By the Born rule the bit string $b_0 b_1$ appears with the probability

$$p(b_0 b_1) = |\psi[b_0, b_1]|^2, \tag{6.1}$$

the squared modulus of the corresponding entry of the state tensor. A record of $M$ shots is a list of $M$ bit strings drawn independently from these four probabilities, and from it we compute the following

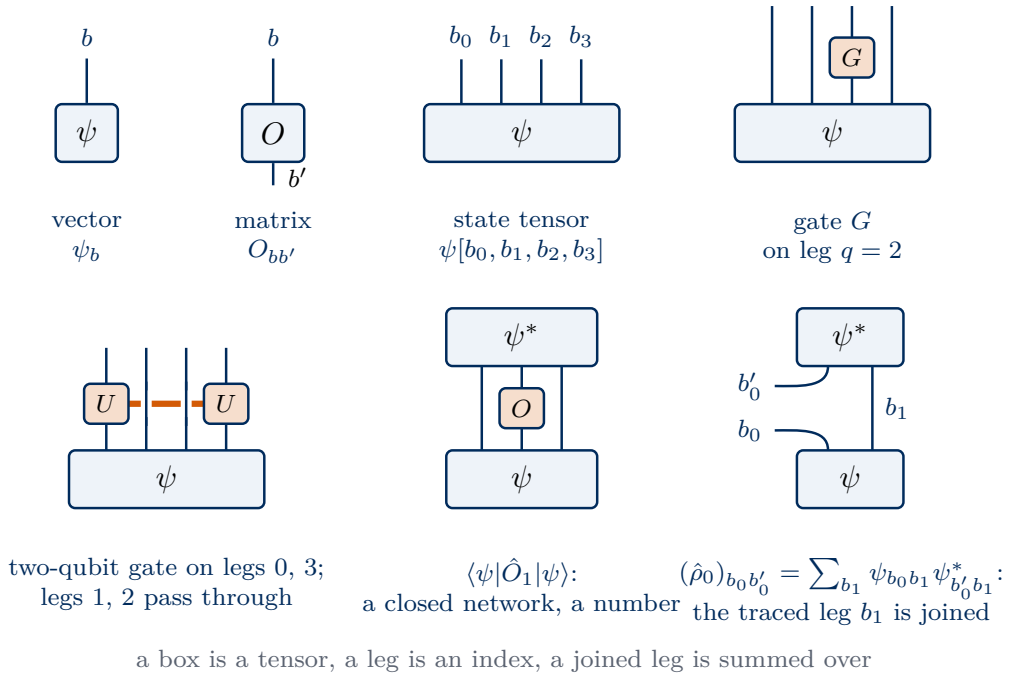



**Figure 17:** Tensor-network diagrams, the picture of index sums. A vector has one leg, a matrix two, the state tensor of $N$ qubits $N$. Joining two legs sums over the shared index, as in a gate on one leg, an expectation value with all legs joined and a partial trace.

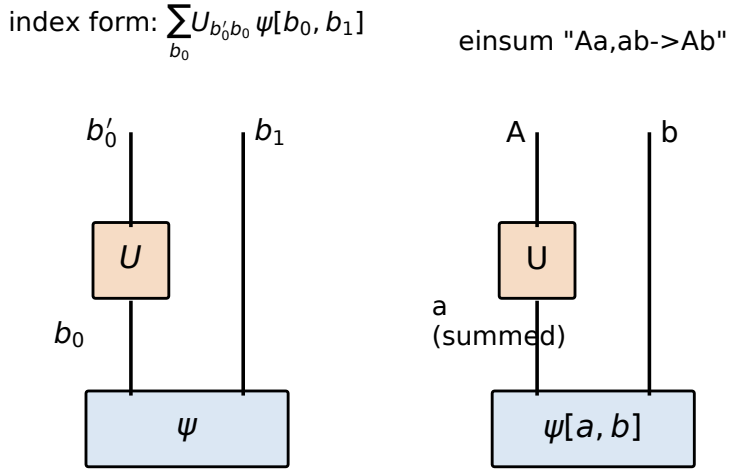



**Figure 18:** Tensor-network diagrams of a one-qubit operator $\hat{U}$ acting on qubit 0 (`"Aa,ab->Ab"`) and on qubit 1 (`"Bb,ab->aB"`) of the two-qubit state tensor $\psi$. The joined leg is summed, and the open legs are the indices of the new state.

quantities.

- The *marginal* of qubit 0, $p_0(b_0) = \sum_{b_1} p(b_0 b_1)$, is what an observer who looks only at qubit 0 sees, and the marginal $p_1(b_1) = \sum_{b_0} p(b_0 b_1)$ of qubit 1 is defined in the same way.
- The single-qubit expectation values are $\langle \hat{Z}_q \rangle = p_q(0) - p_q(1)$.
- The *correlator*

$$\langle \hat{Z}_0 \hat{Z}_1 \rangle = \sum_{b_0 b_1} (-1)^{b_0 + b_1} \, p(b_0 b_1) \tag{6.2}$$

  is $+1$ when the two bits always agree, $-1$ when they always differ and 0 when equal and different bits are equally frequent. The factor $(-1)^{b_0+b_1}$ is the eigenvalue of $\hat{Z} \otimes \hat{Z}$ on $|b_0 b_1\rangle$, the *parity* of the bit string.

We estimate each quantity as the mean of the $\pm 1$ values of the shots, the parities $(-1)^{b_0+b_1}$ in the case of the correlator. The standard error is the sample standard deviation divided by $\sqrt{M}$, as in Section 4.2.

The two bits are statistically independent when the joint probabilities factorise, $p(b_0 b_1) = p_0(b_0)\, p_1(b_1)$, and every correlator is then a product of single-qubit expectation values, $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \langle \hat{Z}_0 \rangle \langle \hat{Z}_1 \rangle$. A vanishing correlator is weaker than independence. The distribution $p(00) = 1/2$, $p(01) = p(10) = 1/4$, $p(11) = 0$ has $\langle \hat{Z}_0 \hat{Z}_1 \rangle = 0$, while $p(00) = 1/2$ differs from $p_0(0)\, p_1(0) = 9/16$, so its bits are dependent. Product states always give independent bits. In the second example, the product state $\hat{R}_y(2\pi/3)|0\rangle \otimes |0\rangle$, qubit 0 is rotated by $2\pi/3$ about $y$ and qubit 1 stays in $|0\rangle$. Qubit 1 always reads 0, so $\langle \hat{Z}_1 \rangle = 1$ and the correlator equals $\langle \hat{Z}_0 \rangle = \cos(2\pi/3) = -1/2$. A simulated record of 2000 shots reproduces this value within its standard error.

### 6.2 Bit strings in different bases

Each qubit of the register can be measured in its own basis by applying the rotation of Section 4.1 to that qubit before the $\hat{Z}$ readout. The rotation is $\hat{H}$ for the $\hat{X}$ basis and $\hat{H}\hat{S}^\dagger$ for the $\hat{Y}$ basis. A choice of basis for every qubit is a *measurement setting*, written as a string of letters such as $\hat{Z}\hat{Z}$, $\hat{Z}\hat{X}$ or $\hat{X}\hat{X}$, with the first letter for qubit 0. Outcome bit 0 means the eigenvalue $+1$ of the measured Pauli matrix.

The same state produces different records in different settings. Take $|0\rangle \otimes |+\rangle$, qubit 0 up along $z$ and qubit 1 along $+x$. In the setting $\hat{Z}\hat{Z}$ qubit 0 always reads 0 and qubit 1 is a fair coin, so only 00 and 01 occur, with equal frequency. Every shot returns 00 in the setting $\hat{Z}\hat{X}$, where both qubits are measured along the axes of their own states. In the setting $\hat{X}\hat{X}$ qubit 0 is a fair coin and qubit 1 always reads 0, so 00 and 10 occur. Simulated records of 1000 shots in each setting show these patterns. A record in one setting reveals only part of the state, and quantum state tomography combines records from several settings to reconstruct it.

## 7 Two-qubit gates as rotations

### 7.1 Interactions switched on for a time

Single-qubit gates map product states to product states, and the bits stay uncorrelated. Correlated qubits need an interaction. In the laboratory a two-qubit gate is a coupling between the spins switched on for a time $t$. An Ising-type coupling is an energy proportional to the product of the same spin component of the two qubits. Its Hamiltonian $\hat{\mathcal{H}}_{\alpha\alpha} = \frac{J}{2}\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$, $\alpha \in \{x, y, z\}$, produces the evolution operator

$$\hat{R}_{\alpha\alpha}(\theta) = e^{-i\hat{\mathcal{H}}_{\alpha\alpha} t} = e^{-i\theta\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha / 2}, \qquad \theta = Jt. \tag{7.1}$$

The Kronecker product of two Pauli matrices squares to the identity, $(\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha)^2 = \hat{\sigma}_\alpha^2 \otimes \hat{\sigma}_\alpha^2 = \hat{I}$, and identity (3.28) applies unchanged,

$$\hat{R}_{\alpha\alpha}(\theta) = \cos\frac{\theta}{2}\, \hat{I} - i \sin\frac{\theta}{2}\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha. \tag{7.2}$$

We also use three fixed two-qubit gates. With the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, the *controlled NOT* gate with control qubit 0 and target qubit 1 is

$$\widehat{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{X} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{7.3}$$

Qubit 1 is left alone when qubit 0 is 0 and flipped when qubit 0 is 1, which exchanges the amplitudes of 10 and 11. Circuit diagrams draw each qubit as a horizontal line, qubit 0 on top. CNOT is a

vertical line from a filled dot on the control qubit to the symbol $\oplus$ on the target qubit (Figure 20). The symbol $\oplus$ stands for addition of bits modulo two, the XOR operation, because CNOT maps $|b_0 b_1\rangle$ to $|b_0,\, b_1 \oplus b_0\rangle$. CNOT is a reversible XOR. It keeps the control bit and can therefore be undone, and applied twice it gives the identity. The *controlled* $\hat{Z}$ gate $\widehat{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z} = \mathrm{diag}(1, 1, 1, -1)$ multiplies $|11\rangle$ by $-1$ and is symmetric in the two qubits. The $\widehat{\mathrm{SWAP}}$ gate exchanges the states of the qubits, $|01\rangle \leftrightarrow |10\rangle$,

$$\widehat{\mathrm{SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{7.4}$$

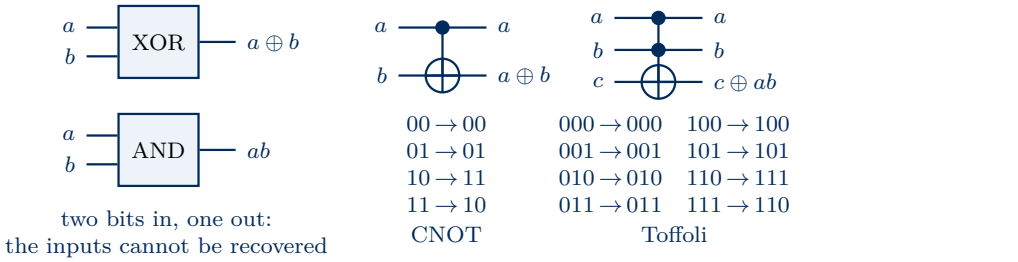



**Figure 19:** Classical XOR and AND lose information, since two input bits give one output bit. CNOT and Toffoli keep their inputs and write the XOR or the AND into the target bit, so they are reversible and can act as unitary gates. Truth tables for basis states.

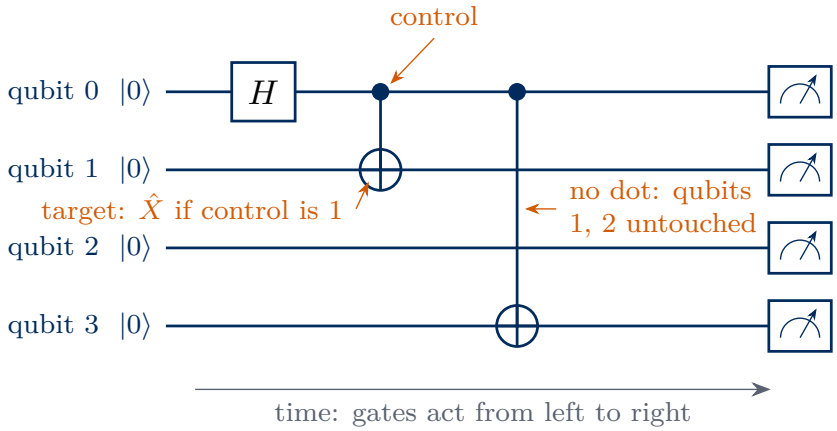



**Figure 20:** Circuit conventions. Each qubit is a wire, qubit 0 on top, and time runs from left to right. A box is a one-qubit gate, and CNOT joins a filled control dot to the target $\oplus$, also between qubits that are not neighbours on the diagram.

### Physical origin of the couplings

The coupling comes from different physics on each platform. Table 7 lists the mechanism, the effective coupling written with Pauli operators and the gate that the platform applies most directly, its *native gate*. The iSWAP gate exchanges $|01\rangle$ and $|10\rangle$ with the phase $i$, and $\sqrt{\widehat{\mathrm{SWAP}}}$ applied twice gives SWAP. Rydberg states have a large principal quantum number $n$, energies $-\mathrm{Ry}/n^2$ of (3.16) close to the ionisation threshold and large electron orbits. Neutral atoms excited to Rydberg states repel each other with a van der Waals energy $V$ that falls as the inverse sixth power of their distance [6]. Trapped

ions share their vibrational modes through the Coulomb repulsion, and a laser with two frequency components converts the shared motion into an effective $\hat{X}\hat{X}$ coupling [7,8]. Superconducting circuits coupled by a capacitor or by a resonator, a circuit that stores one electrical oscillation mode, exchange an excitation. A microwave drive of one qubit at the frequency of the other produces the cross-resonance coupling $\hat{Z}\hat{X}$ [9]. Electron spins in neighbouring quantum dots feel the exchange interaction, whose strength $J(t)$ is set by the overlap of the electron orbitals and controlled by gate voltages [10].

| platform | physical mechanism | effective coupling | native gate |
|---|---|---|---|
| neutral atoms | van der Waals interaction of two Rydberg atoms | $V\|rr\rangle\langle rr\|$, blockade of double excitation | $\widehat{\mathrm{CZ}}$ by a laser-pulse sequence that uses the blockade |
| trapped ions | Coulomb repulsion and shared vibrations | $\hat{X}\hat{X}$ | $\hat{R}_{xx}(\theta)$ |
| superconducting circuits | capacitive or resonator coupling | $\hat{X}\hat{X}+\hat{Y}\hat{Y}$, or $\hat{Z}\hat{X}$ with a cross-resonance drive | iSWAP or its inverse, or CNOT up to one-qubit gates |
| quantum-dot spins | exchange interaction | $J(t)\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$ | $\sqrt{\widehat{\mathrm{SWAP}}}$ |

**Table 7:** Coupling mechanisms of four platforms, their effective two-qubit couplings in Pauli form and the gates they produce most directly.

### Interactions in Pauli language

Writing the projectors on the two levels with Pauli operators, $|0\rangle\langle 0| = (\hat{I}+\hat{Z})/2$ and $|1\rangle\langle 1| = (\hat{I}-\hat{Z})/2$, turns each physical interaction into a combination of Pauli strings. When atoms are used to simulate an Ising model, each atom stores its qubit in the ground state $|g\rangle = |0\rangle$ and a Rydberg state $|r\rangle = |1\rangle$. The interaction acts only on $|rr\rangle = |11\rangle$, and with one projector per atom it becomes

$$V|11\rangle\langle 11| = V\,\frac{\hat{I}-\hat{Z}_0}{2}\,\frac{\hat{I}-\hat{Z}_1}{2} = \frac{V}{4}(\hat{I}-\hat{Z}_0-\hat{Z}_1+\hat{Z}_0\hat{Z}_1). \tag{7.5}$$

The Rydberg interaction is an Ising coupling $\frac{V}{4}\hat{Z}_0\hat{Z}_1$ plus a shift $-\frac{V}{4}\hat{Z}$ of each atom, which acts as a detuning, and a constant. In the same way the exchange of one excitation is $|01\rangle\langle 10| + |10\rangle\langle 01| = \frac{1}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$, and the Heisenberg coupling of two spins is $\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1 = \frac{1}{4}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1)$.

## 7.2 The rotation picture of each gate

Like the one-qubit gates, each two-qubit gate is a rotation or an evolution operator.

### $\hat{R}_{xx}$ as a rotation in a two-dimensional subspace

The operator $\hat{X}\otimes\hat{X}$ maps $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$. By (7.2), in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\hat{R}_{xx}(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & 0 & 0 & -i\sin\frac{\theta}{2} \\ 0 & \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} & 0 \\ 0 & -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} & 0 \\ -i\sin\frac{\theta}{2} & 0 & 0 & \cos\frac{\theta}{2} \end{pmatrix}, \tag{7.6}$$

and $\hat{R}_{xx}(\theta)$ couples $|00\rangle$ only with $|11\rangle$. On the pair $\{|00\rangle, |11\rangle\}$ its matrix is that of an ordinary $\hat{R}_x(\theta)$, a rotation of a Bloch sphere with $|00\rangle$ at the north pole and $|11\rangle$ at the south pole. The pair

$\{|01\rangle, |10\rangle\}$ rotates in the same way. At $\theta = \pi/2$ the state $|00\rangle$ reaches the equator of that sphere,

$$\hat{R}_{xx}(\pi/2)|00\rangle = \frac{|00\rangle - i|11\rangle}{\sqrt{2}}. \tag{7.7}$$

In trapped ions the Mølmer–Sørensen drive produces an $\hat{X}_0\hat{X}_1$ coupling of strength $J$, and a drive of duration $t$ applies a rotation by the angle $Jt$. At $Jt = \pi/2$ it turns $|00\rangle$ into an equal superposition of $|00\rangle$ and $|11\rangle$, a Bell state (Section 8).

### The Mølmer–Sørensen gate of two trapped ions

Two ions in one trap repel each other, and their joint oscillation about the trap centre is a *motional mode*, a harmonic oscillator of frequency $\nu$ shared by both ions. Its quanta are *phonons*, removed by the operator $\hat{a}$ and created by $\hat{a}^\dagger$. A laser tuned to the qubit frequency plus or minus $\nu$ flips a qubit and adds or removes one phonon in the same step. The two transitions are the blue and the red *sideband*. Their strength is reduced by the *Lamb–Dicke parameter* $\eta$, the spread of the ground-state motion of an ion divided by the reduced wavelength of the light, typically $\eta \approx 0.1$. The gate of Refs. [7,8] drives both ions with two tones detuned by $\pm(\nu - \delta)$ from the qubit frequency, close to both sidebands and off them by a small $\delta$. To first order in $\eta$ and after the rotating-wave approximation the two tones exert a force on the motion whose sign depends on the qubits,

$$\hat{\mathcal{H}}(t) = g\,\hat{S}_x(\hat{a}\,e^{-i\delta t} + \hat{a}^\dagger e^{i\delta t}), \qquad \hat{S}_x = \tfrac{1}{2}(\hat{X}_0 + \hat{X}_1), \qquad g = \eta\Omega, \tag{7.8}$$

with the Rabi frequency $\Omega$ of each tone. The collective spin $\hat{S}_x$ has the eigenvalues $m = 1, 0, -1$. For a fixed $m$ the force displaces the motion by the complex amplitude $m\,\alpha(t)$ with

$$\alpha(t) = \frac{g}{\delta}(1 - e^{i\delta t}), \tag{7.9}$$

a circle that starts at the origin and returns to it at the times $t = 2\pi K/\delta$ for every integer $K$. At these times the motion is back in its initial state, whatever that state was, and the qubits are no longer entangled with it. The loop leaves a phase proportional to the enclosed area and to $m^2$. The commutator of the Hamiltonian (7.8) at two different times is proportional to $\hat{S}_x^2$ and commutes with everything else. The Magnus expansion of the evolution operator therefore ends after its second term. With $\hat{S}_x^2 = \frac{1}{2}(\hat{I} + \hat{X}_0\hat{X}_1)$ the result after $K$ loops is

$$\hat{U}\Big(\frac{2\pi K}{\delta}\Big) = e^{\,i\Phi\hat{S}_x^2} = e^{i\Phi/2}\,\hat{R}_{xx}(-\Phi), \qquad \Phi = \frac{2\pi K g^2}{\delta^2}. \tag{7.10}$$

The ions feel the effective coupling $-\frac{J}{2}\hat{X}_0\hat{X}_1$ with $J = g^2/\delta$, and the angle $\Phi = Jt$ grows with the gate time like that of every other coupling of this section. The gate rotates by $-\Phi$ for $\delta > 0$ and in the opposite direction for the opposite sign of $\delta$. One loop with $\delta = 2g$ yields $\Phi = \pi/2$, the maximally entangling gate that takes $|00\rangle$ to $(|00\rangle + i|11\rangle)/\sqrt{2}$. Because every motional state travels around the same closed loop, the gate works for ions that are not cooled to the motional ground state. The simulated gate takes $|00\rangle$ to the target entangled state $(|00\rangle + i|11\rangle)/\sqrt{2}$ with probability 0.9999 also for a mean phonon number of two. This probability is the squared overlap $|\langle\psi_{\text{target}}|\psi\rangle|^2$ with the simulated two-qubit state $|\psi\rangle$. A gate time off by ten per cent leaves the loop open. The error is then 0.045 for ions in the motional ground state and 0.182 for a mean phonon number of two (Figure 21).

### $\hat{R}_{zz}$ as a conditional rotation

The operator $\hat{Z}\otimes\hat{Z} = |0\rangle\langle 0|\otimes\hat{Z} - |1\rangle\langle 1|\otimes\hat{Z}$ is diagonal with the entries $(1, -1, -1, 1)$. The rotation $\hat{R}_{zz}(\theta) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2})$ only attaches phases to the four basis states, and in block form

$$\hat{R}_{zz}(\theta) = |0\rangle\langle 0|\otimes\hat{R}_z(\theta) + |1\rangle\langle 1|\otimes\hat{R}_z(-\theta). \tag{7.11}$$

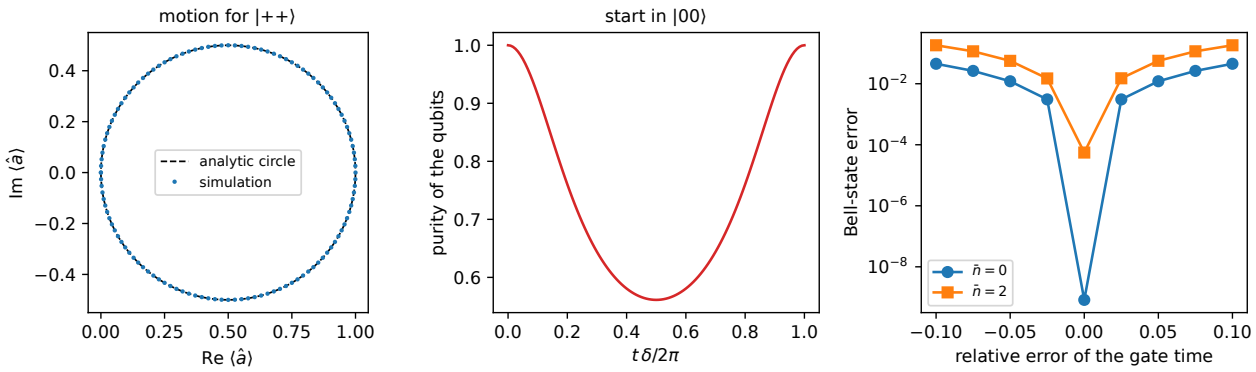



**Figure 21:** The Mølmer–Sørensen gate simulated with two qubits and one motional mode. Left, the mean motional amplitude for the qubits in $|{+}{+}\rangle$ follows the circle (7.9) and returns to the origin. Middle, the purity of the qubit state starting from $|00\rangle$ drops while the qubits are entangled with the motion and returns to one at the gate time. Right, one minus the probability of finding the qubits in the target state when the gate time misses the closing time, for cold and for thermal ions.

Qubit 1 rotates about $z$ by $+\theta$ when qubit 0 is $|0\rangle$ and by $-\theta$ when qubit 0 is $|1\rangle$, a rotation of one qubit conditioned on the other. Physically, $\hat{R}_{zz}(\theta)$ is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ switched on for the time $t = \theta/J$, for example the Rydberg interaction in the form (7.5) or a tunable coupler between superconducting circuits.

### CNOT as a conditional half turn

Since $\hat{X} = i\hat{R}_x(\pi)$, the conditional rotation $|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)$ equals CNOT up to the phase $-i$ in the branch where qubit 0 is 1. The phase gate $\hat{S}$ on qubit 0 supplies the missing factor $i$ in that branch, so

$$\widehat{\mathrm{CNOT}} = (\hat{S} \otimes \hat{I})\,\big[|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)\big]. \tag{7.12}$$

CNOT is also the evolution operator of one Hamiltonian. The matrix $\hat{P} = \frac{\hat{I}-\hat{Z}}{2} \otimes \frac{\hat{I}-\hat{X}}{2}$, a product of the projector on $|1\rangle$ of qubit 0 and the projector on $|-\rangle$ of qubit 1, is itself a projector, $\hat{P}^2 = \hat{P}$. Its exponential series reduces to $e^{-ia\hat{P}} = \hat{I} + (e^{-ia} - 1)\hat{P}$, and for $a = \pm\pi$

$$e^{\mp i\pi\hat{P}} = \hat{I} - 2\hat{P} = \exp\Big[\mp\frac{i\pi}{4}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})\Big] = \widehat{\mathrm{CNOT}}. \tag{7.13}$$

The operator $\hat{I} - 2\hat{P}$ flips the sign of $|1\rangle|-\rangle$ and leaves the three orthogonal states $|0\rangle|\pm\rangle$ and $|1\rangle|+\rangle$ unchanged. Since $\hat{X}|\pm\rangle = \pm|\pm\rangle$, this sign flip is the action of CNOT written in the $\hat{X}$ basis of the target.

### CZ as a conditional half turn

In the same way CZ is the conditional half turn about $z$ and the evolution generated by the projector on $|11\rangle$,

$$\widehat{\mathrm{CZ}} = (\hat{S} \otimes \hat{I})\,\big[|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_z(\pi)\big] = \exp\Big[-\frac{i\pi}{4}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z})\Big]. \tag{7.14}$$

The projector form shows how the Ising coupling makes CZ. Expanding $(\hat{I} - \hat{Z}_0)(\hat{I} - \hat{Z}_1)$ as in (7.5) turns the exponential into $\hat{R}_{zz}(\pi/2)$ times one $z$ rotation on each qubit and a global phase,

$$\widehat{\mathrm{CZ}} = e^{-i\pi/4}\,\big[\hat{R}_z(-\tfrac{\pi}{2}) \otimes \hat{R}_z(-\tfrac{\pi}{2})\big]\,\hat{R}_{zz}(\tfrac{\pi}{2}), \qquad \widehat{\mathrm{CNOT}} = (\hat{I} \otimes \hat{H})\,\widehat{\mathrm{CZ}}\,(\hat{I} \otimes \hat{H}). \tag{7.15}$$

A device applies CZ by switching the Ising coupling on for the calibrated time $t = \pi/(2J)$ and adding the two $z$ rotations as virtual $\hat{Z}$ gates. The second relation follows from $\hat{H}\hat{Z}\hat{H} = \hat{X}$, which turns the conditional sign flip into a conditional bit flip. CNOT is therefore applied as CZ between two Hadamard gates on the target. Superconducting circuits also produce CNOT from the cross-resonance coupling $\hat{Z}\hat{X}$ switched on for a calibrated time, followed by single-qubit corrections.

**SWAP and the exchange interaction**

SWAP is generated by the Heisenberg exchange coupling $\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}=4\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$, the interaction of neighbouring spins in a magnet. It is proportional to the scalar product of the two spin vectors, with the spin operators $\hat{S}_a=\hat{\sigma}_a/2$. The three symmetric states $|00\rangle$, $|11\rangle$ and $(|01\rangle+|10\rangle)/\sqrt{2}$ do not change when the qubits are exchanged and are eigenstates of this coupling with the eigenvalue $+1$. The antisymmetric *singlet* $(|01\rangle-|10\rangle)/\sqrt{2}$ changes sign under the exchange and has the eigenvalue $-3$. SWAP has the eigenvalue $+1$ on the symmetric states and $-1$ on the singlet. On these eigenvectors the operator $\frac{1}{2}(\hat{I}+\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z})$ has the same eigenvalues as SWAP, $\frac{1}{2}(1+1)=1$ and $\frac{1}{2}(1-3)=-1$, and the two operators are therefore equal,

$$\widehat{\mathrm{SWAP}}=\tfrac{1}{2}(\hat{I}+\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z})=e^{i\pi/4}\exp\Big[-\frac{i\pi}{4}(\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z})\Big]. \quad (7.16)$$

In the exponential form the symmetric states acquire the phase $e^{i\pi/4}e^{-i\pi/4}=1$ and the singlet the phase $e^{i\pi/4}e^{3i\pi/4}=-1$. Since $\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}=4\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$, the Heisenberg coupling $J\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$ of two quantum-dot spins applies SWAP at $Jt=\pi$ and a square root of SWAP at $Jt=\pi/2$. An exchange pulse with the area $\int J\,dt=\pi/2$ therefore produces $\sqrt{\widehat{\mathrm{SWAP}}}$. The exchange of one excitation, $\frac{J}{2}(\hat{X}_0\hat{X}_1+\hat{Y}_0\hat{Y}_1)$, swaps $|01\rangle$ and $|10\rangle$ with the phase $-i$ at $Jt=\pi/2$. The result is the inverse of the iSWAP gate, and the opposite sign of $J$ produces iSWAP itself. In the basis $|00\rangle,|01\rangle,|10\rangle,|11\rangle$ the two gates are

$$\mathrm{iSWAP}=\begin{pmatrix}1&0&0&0\\0&0&i&0\\0&i&0&0\\0&0&0&1\end{pmatrix},\qquad \sqrt{\widehat{\mathrm{SWAP}}}=\begin{pmatrix}1&0&0&0\\0&\frac{1+i}{2}&\frac{1-i}{2}&0\\0&\frac{1-i}{2}&\frac{1+i}{2}&0\\0&0&0&1\end{pmatrix}, \quad (7.17)$$

both act only in the pair $\{|01\rangle,|10\rangle\}$, and $\sqrt{\widehat{\mathrm{SWAP}}}$ applied twice gives SWAP.

## 7.3 Parametrised gates and the physical processes behind them

A *parametrised gate* is a family of unitary operators $\hat{U}(\theta)$ labelled by a continuous angle $\theta$, such as $\hat{R}_x(\theta)$, $\hat{R}_y(\theta)$, $\hat{R}_z(\theta)$, $\hat{R}_\phi(\theta)$, $\hat{R}_{xx}(\theta)$ and $\hat{R}_{zz}(\theta)$. A *fixed gate*, such as $\hat{H}$, $\hat{S}$, $\hat{T}$, CNOT or CZ, is one member of such a family at a calibrated value of its angle, together with fixed one-qubit phases. Every angle is set by a physical process in the device. For a single-qubit rotation the angle is the pulse area (3.26), and for a two-qubit rotation the interaction sets it in the same way. A coupling $\frac{J(t)}{2}\hat{\sigma}_\alpha\otimes\hat{\sigma}_\alpha$ with a time-dependent strength commutes with itself at all times, and switching it on for the duration $T$ produces

$$\hat{U}=\hat{R}_{\alpha\alpha}(\theta),\qquad \theta=\int_0^T J(t')\,dt', \quad (7.18)$$

and $\theta=JT$ for a constant coupling. Experiments control $J(t)$ with a tunable coupler between superconducting qubits or with the strength and duration of the laser drive that produces the ion coupling. Between quantum-dot spins the gate voltage sets the exchange. Any angle is reached by choosing the duration. For a fixed gate the duration that gives the required angle is calibrated once and reused, and its local phases are added as virtual $\hat{Z}$ gates. Table 8 connects each gate of these notes with the process that sets its parameter.

Every gate of a circuit is a Hamiltonian of the device switched on for a time. A circuit of $K$ steps is then a pulse sequence of Section 3.2 acting on the whole register,

$$\hat{U}=e^{-it_K\hat{\mathcal{H}}_K}\cdots e^{-it_2\hat{\mathcal{H}}_2}\,e^{-it_1\hat{\mathcal{H}}_1}, \quad (7.19)$$

where each $\hat{\mathcal{H}}_k$ is a drive of one qubit or a coupling of a pair. Gates on different qubits act simultaneously within one layer, and the circuit lasts for the sum of its layer durations. On many platforms the couplings between qubits are weaker than the drives, so two-qubit gates take longer than

| gate | parameter | physical process | control setting |
|---|---|---|---|
| $\hat{R}_\phi(\theta)$ | $\theta = \int \Omega\, dt$, axis angle $\phi_L$ | resonant laser or microwave pulse | pulse duration or amplitude, and phase |
| $\hat{R}_z(\theta)$ | $\theta = \Delta t$, or a phase shift | detuning and free evolution, or a virtual $\hat{Z}$ gate | waiting time, or the phase of later pulses |
| $\hat{S}$, $\hat{T}$ | $\hat{R}_z(\pi/2)$, $\hat{R}_z(\pi/4)$ up to a phase | virtual $\hat{Z}$ gates | phase of later pulses |
| $\hat{H}$, $\hat{X}$ | calibrated angles $\pi/2$ and $\pi$ | resonant pulses and virtual $\hat{Z}$ gates | calibrated durations |
| $\hat{R}_{zz}(\theta)$ | $\theta = \int J\, dt$ | Ising coupling, Rydberg interaction or tunable coupler | interaction time |
| $\hat{R}_{xx}(\theta)$ | $\theta = \int J\, dt$ | Mølmer–Sørensen drive of trapped ions | drive duration |
| $\sqrt{\widehat{\text{SWAP}}}$, iSWAP | $\int J\, dt = \pi/2$ | exchange of quantum-dot spins or of superconducting circuits | exchange pulse $J(t)$ |
| CZ, CNOT | $\hat{R}_{zz}(\pi/2)$ and fixed phases (7.15) | Ising coupling, Rydberg blockade, or cross-resonance drive | calibrated duration |

**Table 8:** Parametrised and fixed gates and the physical processes that set their parameters.

single-qubit gates. Each entangling gate of Table 8, together with arbitrary single-qubit rotations, is *universal*, meaning that every unitary operation on the register decomposes into it and single-qubit gates [1]. Different platforms therefore run the same circuits once each gate is compiled into their native operations.

## 7.4 Gates and expectation values on a computer

On paper a two-qubit gate is the $4 \times 4$ matrix $\hat{G}$ multiplying the vector of amplitudes. An expectation value is the product $\langle\psi|\hat{O}|\psi\rangle$ of a row vector, a matrix and a column vector. A computer evaluates the same sums as contractions. For this the $4 \times 4$ gate is reshaped into a tensor $G[b_0', b_1', b_0, b_1]$ with two output and two input legs, which acts on the register as

$$(\hat{G}\psi)[b_0', b_1'] = \sum_{b_0, b_1} G[b_0', b_1', b_0, b_1]\, \psi[b_0, b_1], \qquad \texttt{"ABab,ab->AB"}, \tag{7.20}$$

drawn as a box joined to both legs of the state (Figure 22, left). The expectation value of an observable $\hat{O}$ is a closed network,

$$\langle\psi|\hat{O}|\psi\rangle = \sum_{a, b, a', b'} \psi^*[a', b']\, O[a', b', a, b]\, \psi[a, b], \qquad \texttt{"AB,ABab,ab->"}, \tag{7.21}$$

with the conjugate state $\psi^*$ on top, the operator in the middle and $\psi$ below (Figure 22, right). With every leg joined, the network evaluates to one number. For an operator on one qubit the box on the other qubit is a plain line, the identity.

## 7.5 Pauli strings and gates on $N$ qubits

The Kronecker product of Section 5.4 extends to a register of $N$ qubits with one factor per qubit. Factor $q$ acts on qubit $q$, qubit 0 being the leftmost, and an operator that leaves a qubit alone carries the identity in that position.

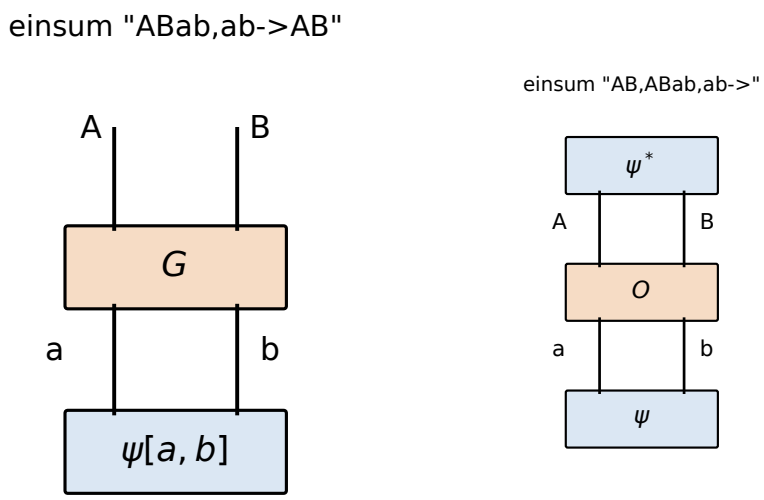



**Figure 22:** Left, a two-qubit gate as a box with two input and two output legs contracted with both legs of the state, `"ABab,ab->AB"`. Right, the expectation value $\langle\psi|\hat{O}|\psi\rangle$ as a closed network, `"AB,ABab,ab->"`.

### Pauli strings

A *Pauli string* is a Kronecker product of one Pauli matrix or identity for each qubit,

$$\hat{P} = \hat{P}_0 \otimes \hat{P}_1 \otimes \cdots \otimes \hat{P}_{N-1}, \qquad \hat{P}_q \in \{\hat{I}, \hat{X}, \hat{Y}, \hat{Z}\}. \tag{7.22}$$

We write only the factors that differ from the identity and label them by their qubit. For three qubits $\hat{X}_0\hat{Z}_2 = \hat{X} \otimes \hat{I} \otimes \hat{Z}$, and $\hat{Z}_0\hat{Z}_1$ of Section 6 is $\hat{Z} \otimes \hat{Z}$. There are $4^N$ Pauli strings, and their properties follow from the rule $(\hat{A} \otimes \hat{B})(\hat{C} \otimes \hat{D}) = \hat{A}\hat{C} \otimes \hat{B}\hat{D}$ applied factor by factor.

- Every Pauli string is Hermitian and squares to the identity, $\hat{P}^2 = \hat{P}_0^2 \otimes \cdots \otimes \hat{P}_{N-1}^2 = \hat{I}$, and its eigenvalues are therefore $\pm 1$. One shot returns one of these two values.
- Two Pauli strings either commute or anticommute. Every qubit on which they carry two different non-identity Pauli matrices contributes a sign $-1$, because the two factors anticommute. An even number of such qubits makes the strings commute, and an odd number makes them anticommute. For example $\hat{X}_0\hat{X}_1$ and $\hat{Z}_0\hat{Z}_1$ commute, while $\hat{X}_0$ and $\hat{Z}_0\hat{Z}_1$ anticommute.
- Different strings are orthogonal, $\mathrm{Tr}(\hat{P}\hat{Q}) = 2^N\delta_{PQ}$, because the trace of a Kronecker product is the product of the traces and every Pauli matrix is traceless. The $4^N$ strings therefore form a basis of all $2^N \times 2^N$ matrices, and every Hermitian matrix is a real combination

  $$\hat{\mathcal{H}} = \sum_{\hat{P}} c_P\, \hat{P}, \qquad c_P = \frac{\mathrm{Tr}(\hat{P}\hat{\mathcal{H}})}{2^N}. \tag{7.23}$$

  The Hamiltonian $J(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h(\hat{Z}_0 + \hat{Z}_1)$ of two coupled spins in a field has $c_{XX} = c_{YY} = c_{ZZ} = J$, $c_{ZI} = c_{IZ} = h$ and all eleven other coefficients zero.
- A string made of $\hat{Z}$ factors and identities is diagonal in the computational basis. Its entry on $|b_0 b_1 \ldots b_{N-1}\rangle$ is the parity $(-1)^{b_{q_1}+\cdots+b_{q_k}}$ of the bits of the qubits $q_1, \ldots, q_k$ that carry a $\hat{Z}$. For three qubits $\hat{Z}_0\hat{Z}_2$ has the entry $(-1)^{b_0+b_2}$, a generalisation of the parity of equation (6.2).

### A one-qubit gate on qubit $q$

A gate $\hat{G}$ on qubit $q$ of an $N$-qubit register is the Kronecker product with $\hat{G}$ in position $q$ and identities elsewhere, $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$. For $N = 4$ the Hadamard gate on qubit 2 is the $16 \times 16$ matrix $\hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, whose pattern of nonzero entries is drawn in Figure 16. By the element formula (5.9) it

changes only the bit $b_2$ and mixes each amplitude with the one whose bit string differs in bit 2 alone. The two amplitudes sit at the indices $i$ and $i \oplus 2^{N-1-q}$, where $\oplus$ flips the bit of value $2^{N-1-q} = 2$ in the binary index. The pair 0000 and 0010, at the indices 0 and 2, becomes

$$\psi'_{0000} = \frac{\psi_{0000} + \psi_{0010}}{\sqrt{2}}, \qquad \psi'_{0010} = \frac{\psi_{0000} - \psi_{0010}}{\sqrt{2}}, \tag{7.24}$$

and the same holds for the seven other pairs, for example 1101 and 1111 at the indices 13 and 15. In index notation the update reads $\psi'[b_0, b_1, b'_2, b_3] = \sum_{b_2} \hat{H}_{b'_2 b_2} \psi[b_0, b_1, b_2, b_3]$. On a computer it is the contraction `"Cc,abcd->abCd"`, a box on the third leg of the diagram of Figure 18. Alternatively, the vector is reshaped into an array of shape $(2^q, 2, 2^{N-q-1})$, here $(4, 2, 2)$, and the middle axis is contracted with `"Bb,abc->aBc"`. Both routes need $2 \cdot 2^N$ multiplications, as in Section 5.4, and the sparse matrix $\hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$ keeps $2 \cdot 2^N = 32$ nonzero entries out of $4^N = 256$. Each factor of a Pauli string has one nonzero entry in each row, and so does the string, with $2^N$ in total.

Every gate on a chain of $N$ qubits is the exponential of its generator, a Pauli string padded with identities on the untouched qubits. The rotation $\hat{R}^{\alpha}_j(\theta) = e^{-i\theta\hat{\sigma}^{\alpha}_j/2}$, the two-qubit rotation $e^{-i\theta\hat{X}_i\hat{X}_j/2}$ and the controlled gates of Section 7.2 are built this way for any qubits $i$ and $j$ of the chain.

### A two-qubit gate on a distant pair

CNOT with control qubit $i$ and target qubit $j$ is a sum of two Kronecker products of single-qubit operators, with identities on all other qubits,

$$\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i + |1\rangle\langle 1|_i \, \hat{X}_j, \tag{7.25}$$

where a subscript names the qubit of each factor. For $N = 4$,

$$\widehat{\mathrm{CNOT}}_{0\to 3} = |0\rangle\langle 0| \otimes \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{I} \otimes \hat{I} \otimes \hat{X}. \tag{7.26}$$

The gate permutes the basis states by flipping $b_3$ whenever $b_0 = 1$. The four exchanged pairs are $1000 \leftrightarrow 1001$, $1010 \leftrightarrow 1011$, $1100 \leftrightarrow 1101$ and $1110 \leftrightarrow 1111$, at the indices $8 \leftrightarrow 9$, $10 \leftrightarrow 11$, $12 \leftrightarrow 13$ and $14 \leftrightarrow 15$, and the eight bit strings with $b_0 = 0$ stay in place. With the control below the target, $\widehat{\mathrm{CNOT}}_{3\to 0} = \hat{I} \otimes \hat{I} \otimes \hat{I} \otimes |0\rangle\langle 0| + \hat{X} \otimes \hat{I} \otimes \hat{I} \otimes |1\rangle\langle 1|$ flips $b_0$ whenever $b_3 = 1$ and exchanges $0001 \leftrightarrow 1001$, $0011 \leftrightarrow 1011$, $0101 \leftrightarrow 1101$ and $0111 \leftrightarrow 1111$.

A general $4 \times 4$ gate on the qubits $i$ and $j$ acts in the same way, through its matrix elements on the bits $b_i, b_j$, with the other bits as spectators. On a computer the gate tensor $G[b'_0, b'_1, b_0, b_1]$ of Section 7.4 is contracted with the legs $i$ and $j$ of the state tensor while the other legs pass through (Figure 23). No $16 \times 16$ matrix is built. The contraction strings are `"ADad,abcd->AbcD"` for $\widehat{\mathrm{CNOT}}_{0\to 3}$, `"DAda,abcd->AbcD"` for $\widehat{\mathrm{CNOT}}_{3\to 0}$ and `"BDbd,abcd->aBcD"` for a gate on the qubits 1 and 3. As a sparse matrix, CNOT on any pair of qubits is a permutation matrix with $2^N$ nonzero entries, one in each row and column.

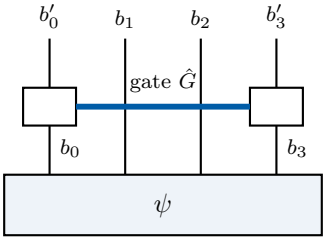



**Figure 23:** A two-qubit gate on the distant qubits 0 and 3 of a four-qubit state tensor. The gate box joins the legs $b_0$ and $b_3$, which are summed and replaced by the output legs $b'_0$ and $b'_3$. The legs $b_1$ and $b_2$ pass the gate unchanged. The two boxes joined by a bar are one tensor with four legs, `"ADad,abcd->AbcD"`.

Many devices couple only neighbouring qubits directly. A distant gate is then compiled with SWAP gates that move the qubits next to each other and back, for example $\widehat{\mathrm{CNOT}}_{0\to 3} = \widehat{\mathrm{SWAP}}_{23}\widehat{\mathrm{SWAP}}_{12}\,\widehat{\mathrm{CNOT}}_{0\to 1}\,\widehat{\mathrm{SWAP}}_{12}\widehat{\mathrm{SWAP}}_{23}$ (optional remark).

A chain of CNOT gates prepares the *Greenberger–Horne–Zeilinger* (GHZ) state, the equal superposition of all qubits in 0 and all qubits in 1,

$$|\mathrm{GHZ}_N\rangle = \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt{2}} = \widehat{\mathrm{CNOT}}_{N-2\to N-1}\cdots\widehat{\mathrm{CNOT}}_{1\to2}\,\widehat{\mathrm{CNOT}}_{0\to1}\,\hat{H}_0|0\ldots0\rangle. \tag{7.27}$$

The Hadamard gate creates $(|0\rangle + |1\rangle)|0\ldots0\rangle/\sqrt{2}$, and each CNOT copies the bit of the previous qubit into the next one in both branches. The fan-out $\widehat{\mathrm{CNOT}}_{0\to N-1}\cdots\widehat{\mathrm{CNOT}}_{0\to2}\,\widehat{\mathrm{CNOT}}_{0\to1}\,\hat{H}_0$, with qubit 0 as the control of every gate, prepares the same state.

## 7.6 A register of interacting qubits

The register of Section 2.1 consists of $N$ two-level systems driven one at a time and coupled in pairs. Examples are atoms held in an array of optical tweezers, ions in a linear trap and superconducting circuits on a chip. Driving qubit $q$ with the Rabi frequency $\Omega_q$ and the detuning $\Delta_q$ adds the one-qubit Hamiltonian $\frac{\Omega_q}{2}\hat{X}_q + \frac{\Delta_q}{2}\hat{Z}_q$ of Section 3.2. Qubits that are close enough, or wired together, interact in pairs, and these pairs are the edges of the *connectivity graph* of the device (Figure 24). With an Ising coupling of strength $J_{qq'}$ on each edge the device Hamiltonian is

$$\hat{\mathcal{H}}_{\mathrm{dev}} = \sum_{q=0}^{N-1}\left(\frac{\Omega_q}{2}\hat{X}_q + \frac{\Delta_q}{2}\hat{Z}_q\right) + \sum_{(q,q')\in\mathrm{edges}}\frac{J_{qq'}}{2}\,\hat{Z}_q\hat{Z}_{q'}, \tag{7.28}$$

a sum of Pauli strings whose coefficients the control electronics switch on and off. The factor $\frac{1}{2}$ matches the two-qubit rotations of Section 7. A quantum circuit is a schedule of such pulses. For a chain of 4 qubits with nearest-neighbour couplings the Hamiltonian contains 11 Pauli strings. Its sparse matrix has 80 nonzero entries out of $4^4 = 256$, one diagonal entry and one entry for each single bit flip in every row.

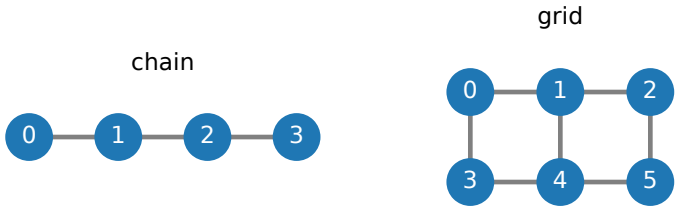



**Figure 24:** Two connectivity graphs of a register. Each disc is a qubit and each line a pair of qubits that interact directly. In a chain every qubit has at most two neighbours, and in a grid the inner qubits have three or four.

### A blockade gate

The blockade gate of Ref. [5] uses the Rydberg interaction without encoding the qubit in the Rydberg state. Each atom stores its qubit in two long-lived ground levels $|0\rangle$ and $|1\rangle$, and a laser couples only $|1\rangle$ to the Rydberg state $|r\rangle$ with the Rabi frequency $\Omega$. The Hamiltonian of the two three-level atoms is

$$\hat{\mathcal{H}} = \frac{\Omega_0(t)}{2}\left(|1\rangle\langle r| + |r\rangle\langle 1|\right)_0 + \frac{\Omega_1(t)}{2}\left(|1\rangle\langle r| + |r\rangle\langle 1|\right)_1 + V|rr\rangle\langle rr|, \tag{7.29}$$

where the subscript names the atom on which the bracket acts and $\Omega_q(t)$ equals $\Omega$ while the laser on atom $q$ is on. The gate consists of a $\pi$ pulse on the control atom 0, a $2\pi$ pulse on the target atom 1 and a second $\pi$ pulse on the control. A $2\pi$ pulse takes $|1\rangle$ to $|r\rangle$ and back and multiplies it by $-1$. The state $|00\rangle$ is untouched, while $|01\rangle$ and $|10\rangle$ each pass one full cycle and change sign. In $|11\rangle$ the first pulse excites the control, and the interaction $V$ shifts the target transition out of resonance. The

target therefore stays in $|1\rangle$, the *blockade*, and the third pulse returns the control with a sign change. The resulting gate is $\mathrm{diag}(1,-1,-1,-1) = (\hat{Z}\otimes\hat{Z})\,\widehat{\mathrm{CZ}}$, a controlled-Z gate up to a $\hat{Z}$ gate on each qubit.

A finite interaction leaves the blockade imperfect. The off-resonant target drive shifts the energy of $|r1\rangle$ by about $\Omega^2/(4V)$, the light shift (3.20). During the $2\pi$ pulse of duration $2\pi/\Omega$ the state $|11\rangle$ acquires the extra phase $\pi\Omega/(2V)$, and the gate error grows as its square, in proportion to $(\Omega/V)^2$. The doubly excited state $|rr\rangle$ is populated only transiently, with a probability of order $(\Omega/V)^2$, and at $V = 20\,\Omega$ it never holds more than $2.5\times10^{-3}$ of the population. We measure the error by the *average gate fidelity*, the overlap $|\langle\psi|\hat{U}_{\mathrm{ideal}}^{\dagger}\hat{U}|\psi\rangle|^2$ of the simulated and the ideal output averaged over all pure input states $|\psi\rangle$. We allow one $\hat{Z}$ rotation on each qubit before the comparison, applied as a virtual $\hat{Z}$ gate at no cost in time. With this correction the error $1-F$ is $1.3\times10^{-3}$ at $V = 10\,\Omega$ and $7.7\times10^{-5}$ at $V = 40\,\Omega$, falling with the slope $-2.0$ on logarithmic axes (Figure 25). An error of $10^{-3}$ requires $V \approx 11\,\Omega$, a condition that fixes how close the atoms must be and which Rydberg level is used.

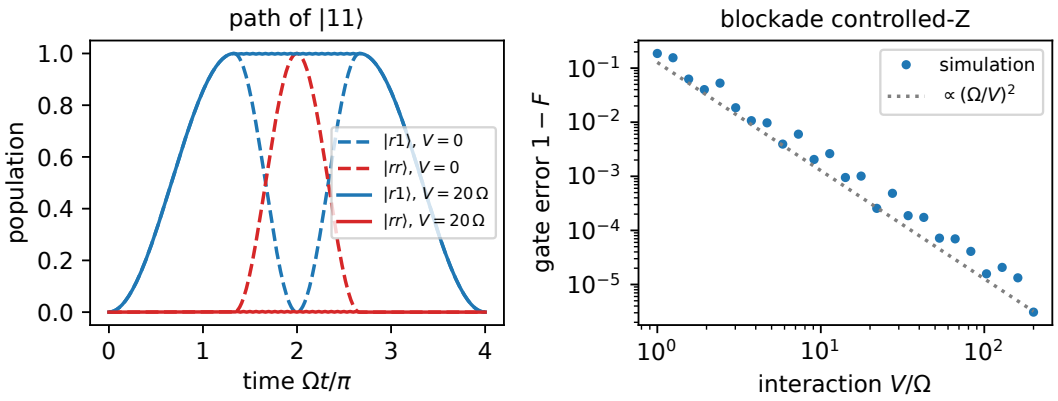



**Figure 25:** The Rydberg blockade gate. Left, the populations of $|r1\rangle$ (control excited) and $|rr\rangle$ during the three pulses applied to $|11\rangle$. Without interaction (dashed) the target pulse carries the population through $|rr\rangle$, and with $V = 20\,\Omega$ (solid) $|rr\rangle$ stays nearly empty. Right, the gate error after the $\hat{Z}$ corrections against $V/\Omega$, which follows the line $\propto (\Omega/V)^2$.

On a chain, where only neighbours interact, a gate between distant qubits is routed with the SWAP chain of Section 7.5. Qubits at distance $d$ need $2(d-1)$ SWAP gates, and each SWAP costs three CNOT gates. A controlled-Z gate between the ends of a four-qubit chain therefore costs 4 SWAP gates, or 12 CNOT gates, in addition to the gate itself. Richer connectivity graphs reduce this overhead. So do qubits that can be moved, such as atoms in tweezers or ions in segmented traps, whose electrodes are divided into zones that shuttle ions between them.

# 8 Entanglement

## 8.1 The Bell state and its predictions

The CNOT and $\hat{R}_{xx}$ gates of Section 7 correlate the two qubits, and the simplest state they produce from $|00\rangle$ is the *Bell state*

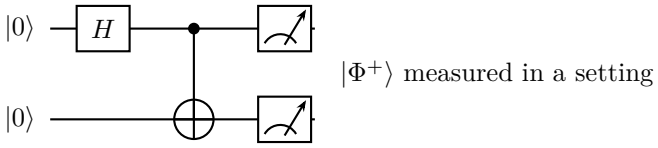
$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{8.1}$$

Its predictions split into local ones, which concern one qubit, and joint ones, which concern both. Measured alone in any basis, each qubit gives a fair coin, since its Bloch vector, computed with $\hat{X}\otimes\hat{I}$, $\hat{Y}\otimes\hat{I}$ and $\hat{Z}\otimes\hat{I}$, vanishes. The parity of each joint measurement is certain. In the settings $\hat{Z}\hat{Z}$ and $\hat{X}\hat{X}$ the two bits agree in every shot, and in the setting $\hat{Y}\hat{Y}$ they disagree in every shot. The Bell state is an eigenstate of $\hat{Z}\otimes\hat{Z}$ and $\hat{X}\otimes\hat{X}$ with the eigenvalue $+1$ and of $\hat{Y}\otimes\hat{Y}$ with the

eigenvalue $-1$. The last sign comes from $\hat{Y}|0\rangle = i|1\rangle$ and $\hat{Y}|1\rangle = -i|0\rangle$, whence $\hat{Y}\otimes\hat{Y}|00\rangle = -|11\rangle$ and $\hat{Y}\otimes\hat{Y}|11\rangle = -|00\rangle$.

No product state has these predictions, because for a product state the bits are independent in every setting and $\langle\hat{Z}_0\hat{Z}_1\rangle = \langle\hat{Z}_0\rangle\langle\hat{Z}_1\rangle$. The value $\langle\hat{Z}_0\hat{Z}_1\rangle = 1$ would require $|\langle\hat{Z}_0\rangle| = |\langle\hat{Z}_1\rangle| = 1$, with both qubits at a pole of the Bloch sphere. Their $\hat{X}$ outcomes would then be independent fair coins with $\langle\hat{X}_0\hat{X}_1\rangle = \langle\hat{X}_0\rangle\langle\hat{X}_1\rangle = 0$, against the value one of the Bell state. The Bell state is the simplest example of an entangled state.

We prepare it from $|00\rangle$ by a Hadamard gate on qubit 0 followed by CNOT with qubit 0 as control,




The Hadamard gate gives $(|00\rangle + |10\rangle)/\sqrt{2}$, and CNOT flips qubit 1 in the second branch,

$$\widehat{\mathrm{CNOT}}\,(\hat{H}\otimes\hat{I})|00\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \equiv |\Phi^+\rangle. \tag{8.2}$$

The interaction gate of Section 7.2 offers a second route. Acting on $|00\rangle$, $\hat{R}_{xx}(\pi/2)$ produces $(|00\rangle - i|11\rangle)/\sqrt{2}$, and the phase gate $\hat{S}$ on qubit 0 multiplies the $|11\rangle$ term by $i$ and turns the state into $|\Phi^+\rangle$.

## 8.2 The product test and the Schmidt decomposition

We decide whether a state of the register is a product state $|a\rangle\otimes|b\rangle$ or entangled, the two cases defined in Section 5, by reading the state tensor as the $2\times 2$ matrix

$$\Psi = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}, \tag{8.3}$$

with qubit 0 labelling the rows. A product state has the amplitudes $\psi[b_0, b_1] = a_{b_0} b_{b_1}$, so $\Psi = ab^T$ is an outer product and has rank one. For the running product state $|0\rangle\otimes|+\rangle$,

$$\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = 1\cdot\begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \tag{8.4}$$

whose second row vanishes. The Bell state has $\Psi = \hat{I}/\sqrt{2}$ of rank two, which no outer product reproduces, and is therefore entangled. For two qubits the test reduces to $\det\Psi = \psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$ for a product state.

The *singular value decomposition* (SVD), which generalises diagonalisation to any matrix, makes the test quantitative. Every complex matrix can be written as

$$\Psi = U\,\mathrm{diag}(s_1, s_2)\,V^T = \sum_{k=1}^{2} s_k\, u_k v_k^T, \qquad s_1 \geq s_2 \geq 0, \tag{8.5}$$

with two unitary matrices $U$ and $V$, whose columns $u_1, u_2$ and $v_1, v_2$ form two orthonormal sets of vectors, and the non-negative *singular values* $s_k$. Geometrically the matrix first changes the basis unitarily and stretches the two new basis directions by the factors $s_1$ and $s_2$. A second unitary change of basis follows. The squares $s_k^2$ are the eigenvalues of the Hermitian matrix $\Psi\Psi^\dagger$, so the singular values follow from an ordinary eigenvalue problem. Read back as a state, the decomposition is the *Schmidt decomposition*

$$|\psi\rangle = \sum_k s_k\,|u_k\rangle\otimes|v_k\rangle, \qquad s_1^2 + s_2^2 = 1, \tag{8.6}$$

a sum of at most two product terms, normalised because $\sum_k s_k^2 = \sum_{ab} |\psi_{ab}|^2 = 1$. The *Schmidt coefficients* $s_k$ quantify entanglement. The vectors $|u_k\rangle$ are orthonormal, and so are the $|v_k\rangle$. A measurement of qubit 0 in the basis $\{|u_k\rangle\}$ therefore returns the outcome $k$ with the probability $s_k^2$ and leaves qubit 1 in $|v_k\rangle$. The number of nonzero Schmidt coefficients, the *Schmidt rank*, equals the rank of $\Psi$, one for a product state and two for an entangled state. Among the entangled states of two qubits the equal spread $s_1 = s_2 = 1/\sqrt{2}$ carries the largest possible entanglement, since the local outcome is then a fair coin.

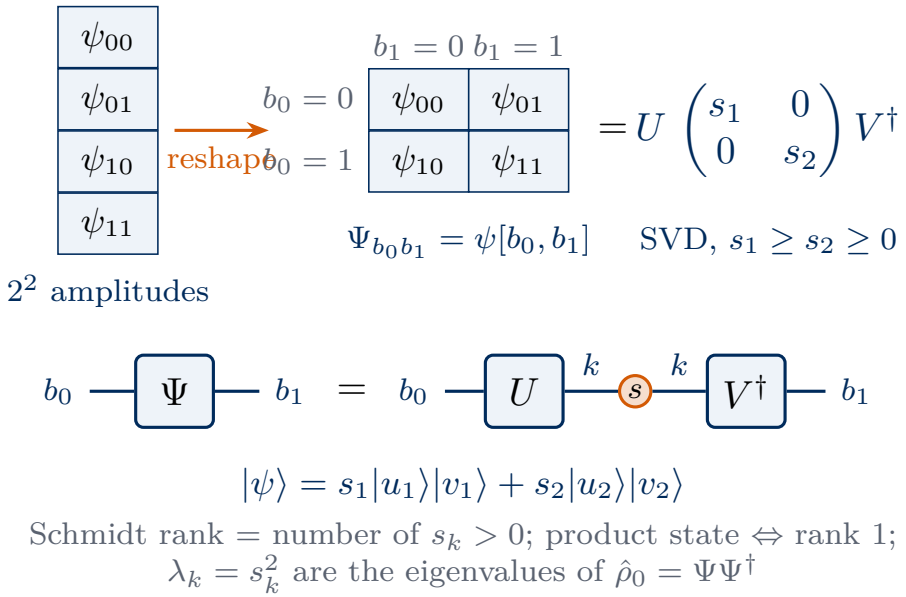



**Figure 26:** The product test and the Schmidt decomposition. The four amplitudes are reshaped into the $2 \times 2$ matrix $\Psi$, and its SVD gives the Schmidt coefficients $s_1 \geq s_2$. The number of nonzero coefficients, the Schmidt rank, tells a product state from an entangled one.

For $|0\rangle \otimes |+\rangle$ the decomposition has $s_1 = 1$, $s_2 = 0$, $u_1 = |0\rangle$ and $v_1 = |+\rangle$. For the Bell state $\Psi = \hat{I}/\sqrt{2}$ is already diagonal with orthonormal rows and columns, so $s_1 = s_2 = 1/\sqrt{2} = 0.7071$ with $u_k = v_k$ the two basis vectors.

Software libraries write the decomposition in their own convention. The function `np.linalg.svd` returns three arrays $U$, $s$ and $V_h$ with $\Psi = U\,\mathrm{diag}(s)\,V_h$. The matrix $V$ of (8.5) is therefore $V_h^T$, and the Schmidt vector $|v_k\rangle$ of qubit 1 has the components of the $k$-th row of $V_h$. In the mathematical convention $V_h = V_{\mathrm{std}}^\dagger$, so the same vector is the complex conjugate of the $k$-th column of $V_{\mathrm{std}}$. Real examples hide this conjugation. For the complex product state with $|\psi_1\rangle = (1, i)^T/\sqrt{2}$ on qubit 0 and $|\psi_2\rangle = (\sqrt{3}/2,\, e^{i\pi/3}/2)^T$ on qubit 1, the first row of $V_h$ equals $|\psi_2\rangle$ up to a global phase. The first column of $V_{\mathrm{std}}$ equals its complex conjugate.

A gate $\hat{A}$ on qubit 0 and a gate $\hat{B}$ on qubit 1 change the matrix to $\hat{A}\Psi\hat{B}^T$, and multiplication by unitary matrices leaves the singular values unchanged. Local gates can change individual correlators while the Schmidt coefficients stay fixed. Entanglement is therefore unchanged by local gates, and only an interaction between the qubits creates it. The two preparations of the Bell state, by CNOT and by $\hat{R}_{xx}(\pi/2)$ followed by $\hat{S}$, accordingly produce equally entangled states.

The family

$$|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle \tag{8.7}$$

interpolates between the product state $|00\rangle$ at $\theta = 0$ and the Bell state at $\theta = \pi/4$. Its matrix is $\mathrm{diag}(\cos\theta, \sin\theta)$, and the Schmidt coefficients are $|\cos\theta|$ and $|\sin\theta|$ (Figure 27).

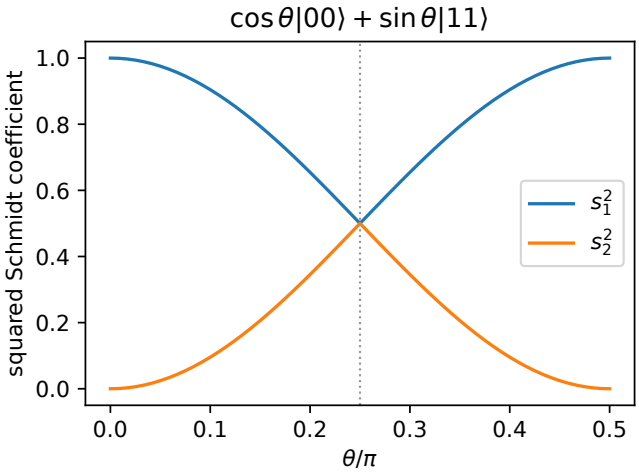



**Figure 27:** Squared Schmidt coefficients of $\cos\theta|00\rangle + \sin\theta|11\rangle$. They start at 1 and 0 for the product state and meet at 1/2 at $\theta = \pi/4$ (dotted), the Bell state. Beyond it the two terms exchange their roles, and the state returns to the product $|11\rangle$ at $\theta = \pi/2$.

### Concurrence (optional)

For two qubits the Schmidt coefficients follow from the determinant without an SVD. With the amplitudes $\psi = (\alpha, \beta, \gamma, \delta)^T$ the matrix is $\Psi = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$. The determinant of a product is the product of the determinants, $\det \Psi = \det U \, s_1 s_2 \, \det V$ by (8.5), and a unitary matrix has a determinant of modulus one. The product of the singular values is therefore the modulus of the determinant, and their squares sum to one,

$$s_1 s_2 = |\alpha\delta - \beta\gamma|, \qquad s_1^2 + s_2^2 = 1 \qquad \Longrightarrow \qquad s_{1,2}^2 = \frac{1 \pm \sqrt{1 - C^2}}{2}, \qquad C = 2|\alpha\delta - \beta\gamma|. \tag{8.8}$$

The number $C$, between 0 and 1, is the *concurrence* of the pure state. It vanishes for a product state, where $\alpha\delta = \beta\gamma$ as for the amplitudes $ac, ad, bc, bd$ of (5.1), and equals one for the Bell state, where $\alpha = \delta = 1/\sqrt{2}$ and $\beta = \gamma = 0$.

## 8.3 Bell correlations against a classical coin

A *correlated coin*, a source that prepares $|00\rangle$ or $|11\rangle$ with probability one half each, imitates the Bell state classically and reproduces its $\hat{Z}\hat{Z}$ record. In each run, however, the coin holds a definite product state along $z$. Its bits in the $\hat{X}$ and $\hat{Y}$ bases are then independent and random, and its correlators $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ vanish. Simulated records of 4000 shots of both preparations in each setting confirm this (Table 9 and Figure 28). The coin matches the Bell state in $\hat{Z}\hat{Z}$ and fails in $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$, and no mixture of $|00\rangle$ and $|11\rangle$ reproduces all three records. To treat the coin and the Bell state as different objects with the same $\hat{Z}\hat{Z}$ statistics we need a description of incompletely known preparations, the density matrix.

| preparation | $\langle \hat{Z}\hat{Z} \rangle$ | $\langle \hat{X}\hat{X} \rangle$ | $\langle \hat{Y}\hat{Y} \rangle$ |
|---|---|---|---|
| Bell state $\|\Phi^+\rangle$ | $+1.000 \pm 0.000$ | $+1.000 \pm 0.000$ | $-1.000 \pm 0.000$ |
| correlated coin | $+1.000 \pm 0.000$ | $-0.006 \pm 0.016$ | $-0.002 \pm 0.016$ |

**Table 9:** Correlators estimated from 4000 simulated shots per setting, with standard errors. The Bell state has definite parities in all three settings, so its standard errors vanish.

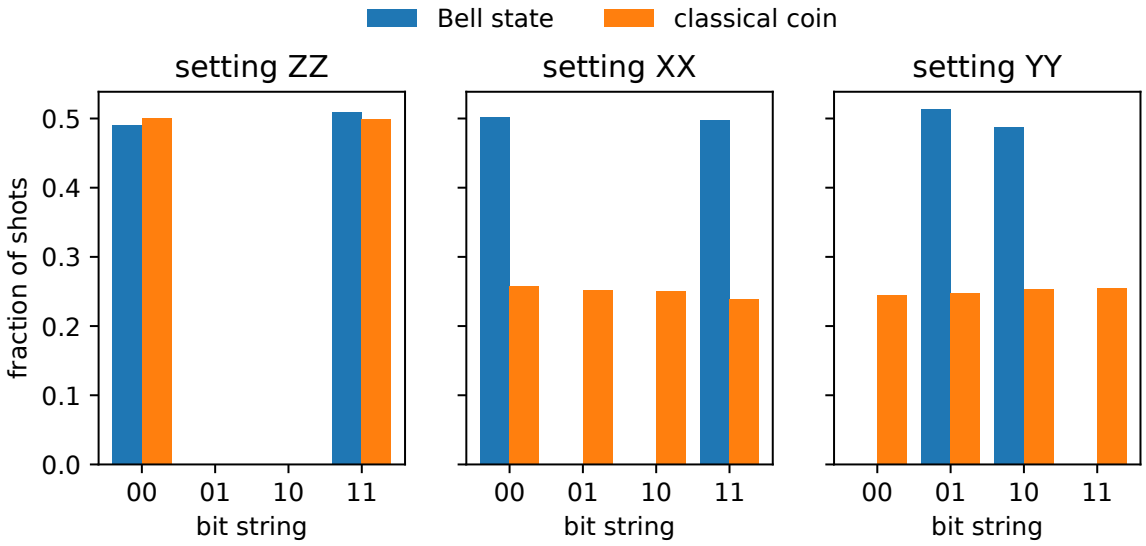



**Figure 28:** Frequencies of the four bit strings for the Bell state and the correlated coin in the settings $\hat{Z}\hat{Z}$, $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$. The Bell state produces only 00 and 11 in $\hat{Z}\hat{Z}$ and $\hat{X}\hat{X}$, and only 01 and 10 in $\hat{Y}\hat{Y}$. The coin produces all four bit strings in $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$.

## 8.4 Single-qubit Bloch vectors during an entangling rotation

The Bloch vector of qubit 0, computed with $\hat{X}\otimes\hat{I}$, $\hat{Y}\otimes\hat{I}$ and $\hat{Z}\otimes\hat{I}$ as for the Bell state, shows how entanglement builds up during an entangling rotation. We write the rotation angle of $\hat{R}_{xx}$ as $\theta_{xx}$, to keep it apart from the angle $\theta$ of the family (8.7). During the rotation

$$\hat{R}_{xx}(\theta_{xx})|00\rangle = \cos\tfrac{\theta_{xx}}{2}|00\rangle - i\sin\tfrac{\theta_{xx}}{2}|11\rangle \tag{8.9}$$

the Bloch vector of qubit 0 is $(0, 0, \cos\theta_{xx})$. Its length shrinks from one to 0.000 at the Bell point $\theta_{xx} = \pi/2$ and grows again, pointing down, towards $|11\rangle$ at $\theta_{xx} = \pi$ (Figure 29). A vector inside the sphere belongs to no pure state of one qubit. While the qubits are entangled, qubit 0 alone is therefore in no pure state, and its description needs a density matrix.

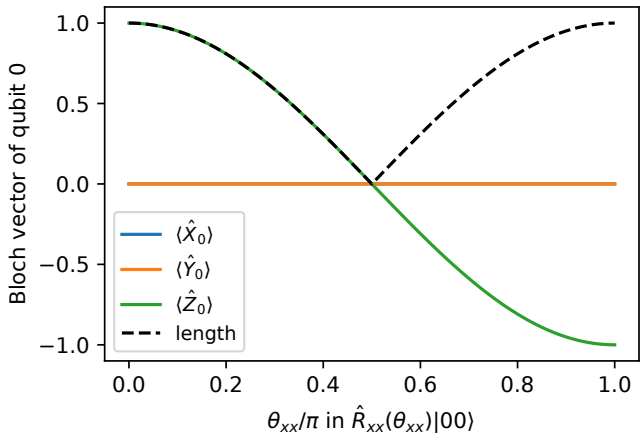



**Figure 29:** Components and length (dashed) of the Bloch vector of qubit 0 during $\hat{R}_{xx}(\theta_{xx})|00\rangle$, against the rotation angle $\theta_{xx}$. The length $|\cos\theta_{xx}|$ equals one only for the product states $|00\rangle$ and $|11\rangle$ and vanishes at the Bell point.

# 9 The register's Hamiltonian

## 9.1 Pauli strings and exact diagonalisation

Physical qubits interact through a Hamiltonian, and quantum machine-learning methods prepare its ground state, learn it from data and map its phases. For two spins a common model couples them by the Heisenberg exchange of Section 7.2 and places both in a magnetic field along $z$,

$$\hat{\mathcal{H}} = J\,(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h\,(\hat{Z}_0 + \hat{Z}_1), \qquad \hat{X}_0\hat{X}_1 \equiv \hat{X}\otimes\hat{X}, \quad \hat{Z}_0 \equiv \hat{Z}\otimes\hat{I}, \quad \hat{Z}_1 \equiv \hat{I}\otimes\hat{Z}. \tag{9.1}$$

Each term is a Pauli string of Section 7.5 with a real coefficient. By (7.23) every Hamiltonian of two qubits is a real combination of the sixteen Pauli strings, and (9.1) uses five of them. The Hamiltonian is a $4\times 4$ Hermitian matrix built from Kronecker products. Its eigenvalues are the allowed energies and its eigenvectors the stationary states, and *exact diagonalisation* computes all of them numerically.

For longer spin chains we assemble the Hamiltonian from the Kronecker products of its Pauli strings as a sparse matrix. Each string contributes one nonzero entry in each row, so a Hamiltonian with $L$ Pauli-string terms has at most $L\cdot 2^N$ nonzero entries, against $4^N$ entries of the full matrix.

A sparse eigensolver based on the Lanczos method, `scipy.sparse.linalg.eigsh`, finds the ground state from products of the matrix with vectors alone. The time evolution $e^{-it\hat{\mathcal{H}}}|\psi\rangle$ is computed with `expm_multiply`, which applies the exponential to the vector without forming the matrix $e^{-it\hat{\mathcal{H}}}$.

The rotation picture of Section 7.2 predicts the spectrum. The exchange term has the eigenvalue $+1$ on the three symmetric states and $-3$ on the singlet, so it contributes $J$ and $-3J$. The field term counts the spins along $z$, since $\hat{Z}_0+\hat{Z}_1$ equals $+2$ on $|00\rangle$, $-2$ on $|11\rangle$ and 0 on the two states with a single 1. The symmetric and singlet states are eigenvectors of both terms, and the four energies are

$$E_{|00\rangle} = J + 2h, \qquad E_{|11\rangle} = J - 2h, \qquad E_{(|01\rangle+|10\rangle)/\sqrt{2}} = J, \qquad E_{\text{singlet}} = -3J. \tag{9.2}$$

For $J>0$ the coupling is *antiferromagnetic*, with the exchange energy lowest when the two spins point in opposite directions, whereas a coupling with $J<0$ favours parallel spins and is ferromagnetic. At $J>0$ and zero field the ground state is the singlet, an entangled state with $s_1 = s_2 = 1/\sqrt{2}$, at the energy $-3J$ found by exact diagonalisation. The singlet level is flat (Figure 30), because the two spins of the singlet point in opposite directions in both components (total spin zero) and the field term vanishes on it.

The level of $|11\rangle$ falls with $h$ and crosses the singlet where $J-2h=-3J$, at $h=2J$. A strong positive field lowers the energy of spins in $|1\rangle$ and makes the product state $|11\rangle$ the ground state for $h>2J$, while for $h<-2J$ the ground state is $|00\rangle$. Between the two crossings the ground state is entangled, and outside them it is a product state. One parameter of the Hamiltonian thus switches the ground state between an entangled and a product state, the smallest example of the ground-state problems addressed by variational quantum algorithms.

The sign of this field is opposite to that of the free-qubit Hamiltonian $-\frac{\omega_0}{2}\hat{Z}$ of Section 2.1, which corresponds to $h=-\omega_0/2<0$.

## 9.2 An excitation hopping between the spins

A gate is the evolution under a Hamiltonian for a fixed time, and following the state continuously in time shows how entanglement builds up. We keep only the transverse part of the exchange,

$$\hat{\mathcal{H}}_{XX} = \frac{J}{2}\,(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1) = J\,(|01\rangle\langle 10| + |10\rangle\langle 01|). \tag{9.3}$$

The second form follows from $\hat{X}\otimes\hat{X}|01\rangle = |10\rangle$ and $\hat{Y}\otimes\hat{Y}|01\rangle = (i|1\rangle)\otimes(-i|0\rangle) = |10\rangle$, together with the same relations for $|10\rangle$. Both terms exchange $|00\rangle$ and $|11\rangle$ with opposite signs and cancel. In the ladder operators of Section 2.6 the same term reads $\hat{\sigma}_0^+\hat{\sigma}_1^- + \hat{\sigma}_0^-\hat{\sigma}_1^+ = 2(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$. Hence $\hat{\mathcal{H}}_{XX} = \frac{J}{4}(\hat{\sigma}_0^+\hat{\sigma}_1^- + \hat{\sigma}_0^-\hat{\sigma}_1^+) = J(\hat{S}_0^+\hat{S}_1^- + \hat{S}_0^-\hat{S}_1^+)$ lowers one spin and raises the other, moving one

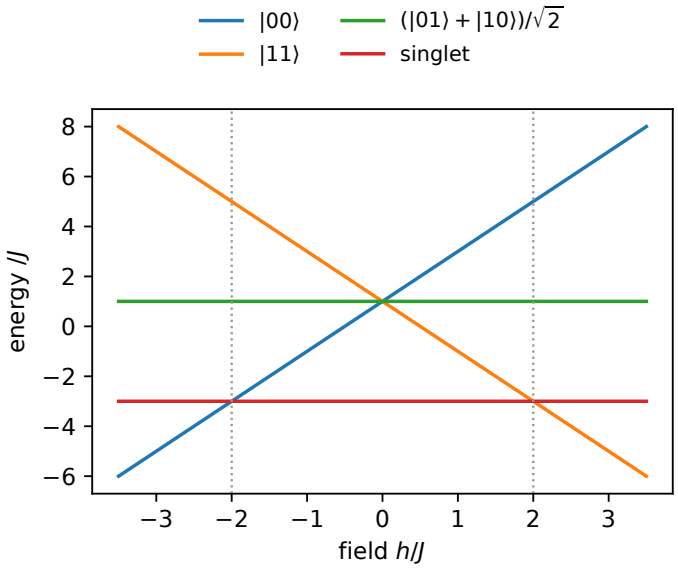



**Figure 30:** Energy levels of the two-spin Hamiltonian (9.1) against the field $h$ for $J = 1$. The flat singlet level is the ground state for $|h| < 2J$, and the product states $|11\rangle$ and $|00\rangle$ take over beyond the crossings at $h = \pm 2J$ (dotted).

excitation from spin to spin. In the subspace $\{|01\rangle, |10\rangle\}$ it acts as $J\hat{X}$, and the evolution from $|01\rangle$, qubit 0 up and qubit 1 down, is the rotation $\hat{R}_x(2Jt)$ of that subspace,

$$|\psi(t)\rangle = e^{-i\hat{\mathcal{H}}_{XX}t}|01\rangle = \cos(Jt)|01\rangle - i\sin(Jt)|10\rangle. \tag{9.4}$$

The magnetisations oscillate in opposite phase,

$$\langle\hat{Z}_0(t)\rangle = \cos^2(Jt) - \sin^2(Jt) = \cos(2Jt) = -\langle\hat{Z}_1(t)\rangle, \tag{9.5}$$

and the Schmidt coefficients are $|\cos Jt|$ and $|\sin Jt|$. The excitation is shared equally, and the register maximally entangled, at $Jt = \pi/4$ and $3\pi/4$, where the smaller squared Schmidt coefficient reaches 0.500 (Figure 31). We compute the evolution numerically from the spectral form (3.5). Records of $\langle\hat{Z}_0(t)\rangle$ and $\langle\hat{Z}_1(t)\rangle$ at many times are the data from which Hamiltonian learning infers the unknown Hamiltonian of a device.

The three couplings of the two spins differ in their evolution operators and in the times at which they entangle maximally (Table 10). The Ising coupling $\frac{J}{2}\hat{X}_0\hat{X}_1$ produces $\hat{R}_{xx}(\theta)$ with $\theta = Jt$, so its Bell point $\theta = \pi/2$ is reached at $Jt = \pi/2$. The hopping Hamiltonian $\hat{\mathcal{H}}_{XX}$ entangles $|01\rangle$ maximally at $Jt = \pi/4$.

| Hamiltonian | evolution after the time $t$ | maximal entanglement |
|---|---|---|
| Ising coupling $\frac{J}{2}\hat{X}_0\hat{X}_1$ | $\hat{R}_{xx}(\theta)$ with $\theta = Jt$ | $\|00\rangle \to (\|00\rangle - i\|11\rangle)/\sqrt{2}$ at $Jt = \pi/2$ |
| hopping $\frac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$ | $\hat{R}_x(2Jt)$ in $\{\|01\rangle, \|10\rangle\}$ | from $\|01\rangle$ at $Jt = \pi/4$ |
| exchange $J(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1)$, equal to $4J\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$ | phase $e^{-iJt}$ on the symmetric states and $e^{3iJt}$ on the singlet | the singlet ground state for $J > 0$ |

**Table 10:** The two-spin couplings, their evolution operators and the times or states of maximal entanglement.

## 9.3 Energies from bit strings

Every term of $\hat{\mathcal{H}}$ is a Pauli string whose expectation value we estimate from measurement records as in Section 6. One shot of $\hat{Z}_0$ yields the value $(-1)^{b_0}$, and one shot of $\hat{Z}_0\hat{Z}_1$ the parity $(-1)^{b_0+b_1}$ of

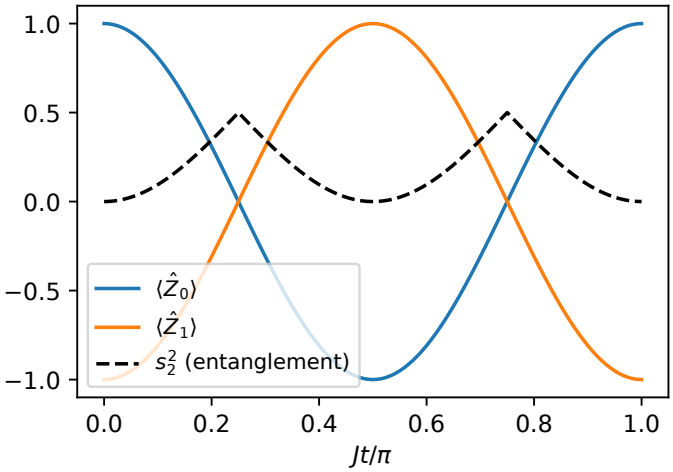



**Figure 31:** An excitation hopping between the two spins under $\hat{\mathcal{H}}_{XX}$, started in $|01\rangle$. The magnetisations $\langle\hat{Z}_0\rangle$ and $\langle\hat{Z}_1\rangle$ oscillate in opposite phase, and the smaller squared Schmidt coefficient (dashed) reaches $1/2$ whenever they cross zero.

equation (6.2). A Pauli string with $\hat{X}$ or $\hat{Y}$ factors is measured in the matching setting of Section 6.2 and evaluated with the same parity.

One setting measures several strings at once, and the setting $\hat{Z}\hat{Z}$ gives $\hat{Z}_0$, $\hat{Z}_1$ and $\hat{Z}_0\hat{Z}_1$ from the same shots. In general a setting measures every Pauli string that carries, on each qubit, either the identity or the Pauli matrix of the setting, and such strings commute with each other. The Hamiltonian (9.1) therefore needs three settings, $\hat{X}\hat{X}$ for $\hat{X}_0\hat{X}_1$, $\hat{Y}\hat{Y}$ for $\hat{Y}_0\hat{Y}_1$, and $\hat{Z}\hat{Z}$ for $\hat{Z}_0\hat{Z}_1$, $\hat{Z}_0$ and $\hat{Z}_1$. Each shot in the setting $\hat{Z}\hat{Z}$ yields one value of $J\hat{Z}_0\hat{Z}_1 + h(\hat{Z}_0 + \hat{Z}_1)$. Because the three parts of the energy come from independent records, their standard errors add in quadrature,

$$\bar{E} = \overline{\langle J\hat{X}_0\hat{X}_1\rangle} + \overline{\langle J\hat{Y}_0\hat{Y}_1\rangle} + \overline{\langle J\hat{Z}_0\hat{Z}_1 + h(\hat{Z}_0+\hat{Z}_1)\rangle}, \qquad \mathrm{std}(\bar{E}) = \Big(\sum_{\text{settings}} \mathrm{std}^2\Big)^{1/2}. \tag{9.6}$$

A bar denotes the sample mean over the shots of the setting that measures the term. We estimate the energy at $J = 1$, $h = 0.5$ from 1000 simulated shots in each setting. In the Bell state the three exchange strings take the definite values $+1$, $-1$ and $+1$, and the exchange part equals $J$ in every shot. The field part has mean zero but takes the values $\pm 2h$ from shot to shot, because in the setting $\hat{Z}\hat{Z}$ the two bits agree and each is a fair coin. Its standard error is therefore $2h/\sqrt{M} \approx 0.032$, and the estimate $+1.030$ compares with the exact $+1.000$. The product state $|0\rangle|+\rangle$ has $\langle\hat{X}_0\hat{X}_1\rangle = \langle\hat{Y}_0\hat{Y}_1\rangle = 0$ and $\langle\hat{Z}_0\hat{Z}_1\rangle = \langle\hat{Z}_0\rangle\langle\hat{Z}_1\rangle = 0$, hence the exact energy $h = +0.500$. Its bits are random in more settings, and the estimate $+0.590 \pm 0.065$ carries the larger standard error. Both estimates agree with the exact energies within about 1.4 standard errors. Variational quantum algorithms use this estimate of the energy of a prepared state from a few measurement settings as their measurement step.

# 10 Density matrices

## 10.1 The density matrix of a pure state

The correlated coin of Section 8.3 and one qubit of an entangled register have no state vector, and both call for a description of incompletely known states. We build it from the expectation value (2.11) rewritten with the trace. The trace is cyclic for any matrices, $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$, and therefore

$$\langle\psi|\hat{O}|\psi\rangle = \mathrm{Tr}(\hat{O}|\psi\rangle\langle\psi|) = \mathrm{Tr}(\hat{O}\hat{\rho}), \qquad \hat{\rho} = |\psi\rangle\langle\psi|. \tag{10.1}$$

The outer product $\hat{\rho}$ is the *density matrix* of the pure state $|\psi\rangle$. For one qubit with $|\psi\rangle = (a, b)^T$,

$$\hat{\rho} = \begin{pmatrix} a \\ b \end{pmatrix}\begin{pmatrix} a^* & b^* \end{pmatrix} = \begin{pmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{pmatrix}. \tag{10.2}$$

The diagonal holds the Born probabilities of the computational basis, and the off-diagonal elements, the *coherences*, carry the relative phase. The global phase cancels in $\hat{\rho}$, since $e^{i\alpha}|\psi\rangle\langle\psi|e^{-i\alpha} = \hat{\rho}$, and the density matrix keeps everything observable about the state. For the two-qubit product state $|\psi_1\rangle \otimes |\psi_2\rangle$ of (5.1), with $|\psi_1\rangle$ on qubit 0 and $|\psi_2\rangle$ on qubit 1, the rule $(\hat{A}\otimes\hat{B})(\hat{C}\otimes\hat{D}) = \hat{A}\hat{C}\otimes\hat{B}\hat{D}$ yields

$$\hat{\rho} = \big(|\psi_1\rangle \otimes |\psi_2\rangle\big)\big(\langle\psi_1| \otimes \langle\psi_2|\big) = \hat{\rho}_0 \otimes \hat{\rho}_1 = \begin{pmatrix} |a|^2\hat{\rho}_1 & ab^*\hat{\rho}_1 \\ a^*b\,\hat{\rho}_1 & |b|^2\hat{\rho}_1 \end{pmatrix}, \qquad \hat{\rho}_1 = \begin{pmatrix} |c|^2 & cd^* \\ c^*d & |d|^2 \end{pmatrix}, \tag{10.3}$$

where $\hat{\rho}_0 = |\psi_1\rangle\langle\psi_1|$ is the density matrix of qubit 0 and $\hat{\rho}_1 = |\psi_2\rangle\langle\psi_2|$ that of qubit 1. The register matrix $\hat{\rho}$ is $4\times 4$, with rows and columns in the order $00, 01, 10, 11$. A pure-state density matrix is a projector of rank one, since $\langle\psi|\psi\rangle = 1$ and

$$\hat{\rho}^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = \hat{\rho}, \qquad \mathrm{Tr}\hat{\rho}^2 = \mathrm{Tr}\hat{\rho} = 1. \tag{10.4}$$

## 10.2 Mixtures and superpositions

A state vector describes a system whose preparation we know completely, and often we know less. A source may emit $|\psi_1\rangle$ with probability $p_1$ and $|\psi_2\rangle$ with probability $p_2$ without revealing which one it emitted. The expectation value of an observable is then the average of the expectation values of the possible states, and by (10.1)

$$\langle\hat{O}\rangle = \sum_i p_i\langle\psi_i|\hat{O}|\psi_i\rangle = \sum_i p_i \mathrm{Tr}(\hat{O}|\psi_i\rangle\langle\psi_i|) = \mathrm{Tr}(\hat{O}\hat{\rho}), \qquad \hat{\rho} = \sum_i p_i|\psi_i\rangle\langle\psi_i|. \tag{10.5}$$

One matrix, the *density matrix* or density operator $\hat{\rho}$ of the mixture, contains all predictions of the source. The probability of the outcome $k$ of a measurement in an orthonormal basis $\{|k\rangle\}$ is $p_k = \langle k|\hat{\rho}|k\rangle$, the diagonal of $\hat{\rho}$ in that basis.

To see how a mixture differs from a superposition, we compare the superposition $|+\rangle$ with a source that emits $|0\rangle$ or $|1\rangle$ with probability one half each,

$$\hat{\rho}_+ = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{mix}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{10.6}$$

Sharing their diagonal, the two matrices give 0 and 1 with probability one half in the $\hat{Z}$ basis. In the $\hat{X}$ basis the superposition always gives the outcome $+$, while the mixture still gives $+$ and $-$ with probability one half. The coherences of $\hat{\rho}_+$ carry this difference, and simulated records of 2000 shots in each basis show it. The $\hat{X}$ basis thus distinguishes these two particular states. One fixed basis cannot determine an arbitrary density matrix, because its outcome probabilities fix only the diagonal of $\hat{\rho}$ in that basis.

## 10.3 Properties, purity and the Bloch ball

The mixture $\hat{\rho} = \sum_i p_i|\psi_i\rangle\langle\psi_i|$ gives every density matrix three properties.

- It is Hermitian, $\hat{\rho}^\dagger = \hat{\rho}$, because each projector $|\psi_i\rangle\langle\psi_i|$ is Hermitian.
- Its trace is one, $\mathrm{Tr}\hat{\rho} = \sum_i p_i\langle\psi_i|\psi_i\rangle = 1$, so the outcome probabilities sum to one.
- It is positive, $\langle v|\hat{\rho}|v\rangle = \sum_i p_i|\langle v|\psi_i\rangle|^2 \geq 0$ for every vector $v$, so every outcome probability is non-negative.

Its eigenvalues $\lambda_k$ are therefore non-negative and sum to one, so they are probabilities. The *purity*

$$\mathrm{Tr}\hat{\rho}^2 = \sum_k \lambda_k^2 \tag{10.7}$$

equals one if and only if one eigenvalue is one and the others vanish, that is, for a pure state. It is smallest, $1/d$ in dimension $d$, for the *maximally mixed state* $\hat{I}/d$. Table 11 contrasts the two cases.

| | pure state | mixed state |
|---|---|---|
| form | $\hat\rho = \lvert\psi\rangle\langle\psi\rvert$ | $\hat\rho = \sum_i p_i \lvert\psi_i\rangle\langle\psi_i\rvert$, at least two $p_i > 0$ with $\lvert\psi_i\rangle$ not all equal |
| rank | 1 | $\geq 2$ |
| $\hat\rho^2$ | $= \hat\rho$ | $\neq \hat\rho$ |
| purity $\mathrm{Tr}\hat\rho^2$ | 1 | $< 1$, down to $1/d$ for $\hat I/d$ |
| one-qubit Bloch vector | on the sphere, $\lvert\mathbf{r}\rvert = 1$ | inside the ball, $\lvert\mathbf{r}\rvert < 1$ |
| example | $\lvert+\rangle$, $\lvert\Phi^+\rangle$ | $\hat\rho_{\mathrm{mix}} = \hat I/2$, the correlated coin |

**Table 11:** Pure and mixed density matrices.

For one qubit the four matrices $\hat I, \hat X, \hat Y, \hat Z$ form a basis of the Hermitian $2 \times 2$ matrices, and the three properties allow only the form

$$\hat\rho = \tfrac{1}{2}(\hat I + r_x \hat X + r_y \hat Y + r_z \hat Z), \qquad r_\alpha = \mathrm{Tr}(\hat\rho\,\hat\sigma_\alpha) = \langle\hat\sigma_\alpha\rangle, \qquad |\mathbf{r}| \leq 1, \qquad \mathrm{Tr}\hat\rho^2 = \tfrac{1}{2}(1 + |\mathbf{r}|^2). \tag{10.8}$$

The trace condition fixes the coefficient of $\hat I$. The coefficients $r_\alpha$ are the expectation values because $\mathrm{Tr}(\hat\sigma_\alpha\hat\sigma_\beta) = 2\delta_{\alpha\beta}$, and the eigenvalues $\frac{1}{2}(1 \pm |\mathbf{r}|)$ are non-negative only for $|\mathbf{r}| \leq 1$. Pure states lie on the Bloch sphere $|\mathbf{r}| = 1$ of Section 2.7, and mixed states fill its interior, the *Bloch ball*. The maximally mixed state $\hat I/2$ sits at the centre, with purity 0.5. A mixture averages Bloch vectors with its weights, so the mixture of two pure states lies on the straight segment between them (Figure 33).

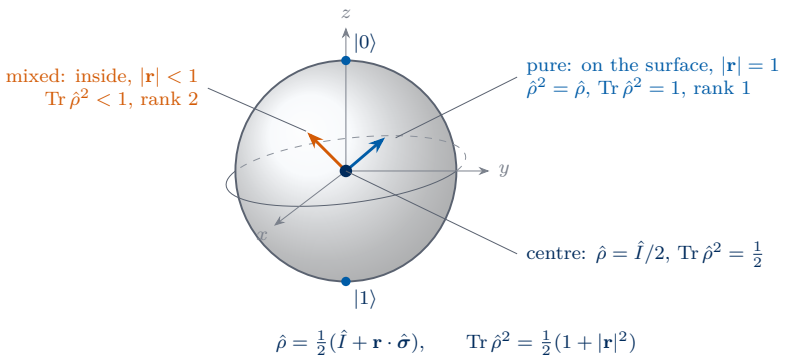



**Figure 32:** Pure states lie on the Bloch sphere and have purity one, mixed states lie inside the ball with purity $(1 + |\mathbf{r}|^2)/2 < 1$, and the maximally mixed state $\hat I/2$ sits at the centre.

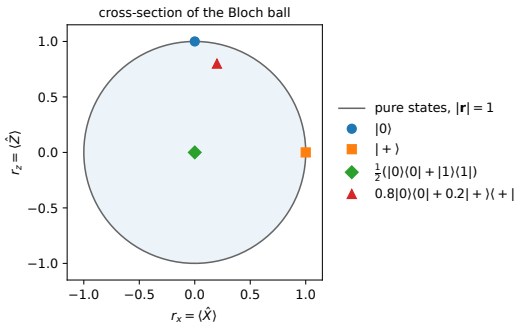



**Figure 33:** Cross-section of the Bloch ball in the $x$–$z$ plane. Pure states lie on the circle, mixtures inside it, and the mixture of $|0\rangle$ and $|1\rangle$ with equal weights at the centre.

## 10.4 The correlated coin as a density matrix

The Bell state and the correlated coin of Section 8.3 now become two density matrices,

$$\hat{\rho}_{\text{Bell}} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\text{coin}} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{10.9}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. They share the diagonal and differ only in the coherences between $|00\rangle$ and $|11\rangle$, the entries responsible for $\langle \hat{X}\hat{X}\rangle$ and $\langle \hat{Y}\hat{Y}\rangle$. Evaluating $\langle \hat{O}\rangle = \mathrm{Tr}(\hat{O}\hat{\rho})$ with these two matrices reproduces the correlators of Table 9. The Bell register is pure, with purity one, whereas the coin register is mixed, with rank two and purity $1/2$.

## 10.5 The partial trace

Qubit 0 of an entangled register has a Bloch vector inside the ball (Section 8), so it is in no pure state. The density matrix that describes one qubit of the register follows from the register state written with one index for each qubit, $|\psi\rangle = \sum_{a,b} \psi_{ab}|a\rangle|b\rangle$. Here $a$ is the bit of qubit 0 and $b$ that of qubit 1. A measurement of qubit 0 alone corresponds to an observable $\hat{O}_0 \otimes \hat{I}$, and in its expectation value the bra carries its own indices, which we write with primes, $a'$ and $b'$,

$$\langle\psi|\hat{O}_0 \otimes \hat{I}|\psi\rangle = \sum_{a,a',b,b'} \psi^*_{a'b'}\,(\hat{O}_0)_{a'a} \underbrace{\delta_{b'b}}_{\text{identity on qubit 1}} \psi_{ab} = \sum_{a,a'} (\hat{O}_0)_{a'a} \underbrace{\sum_b \psi_{ab}\,\psi^*_{a'b}}_{(\hat{\rho}_0)_{aa'}} = \mathrm{Tr}(\hat{O}_0\hat{\rho}_0). \tag{10.10}$$

The identity on qubit 1 produced the Kronecker delta $\delta_{b'b}$, which sets the index of qubit 1 in $\psi$ and in $\psi^*$ equal before it is summed. The result depends on the state only through the $2 \times 2$ matrix

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\,\psi^*_{a'b}, \tag{10.11}$$

the *reduced density matrix* of qubit 0, which contains everything that measurements on qubit 0 alone can reveal. It is produced by the *partial trace* over qubit 1, written $\hat{\rho}_0 = \mathrm{Tr}_1|\psi\rangle\langle\psi|$, a linear map from $4 \times 4$ matrices to $2 \times 2$ matrices. On the diagonal the partial trace sums the joint probabilities over the bit of qubit 1, as marginalisation does for a joint distribution in probability theory. The reduced state of qubit 1 follows in the same way, $(\hat{\rho}_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'}$, as the partial trace over qubit 0. Element by element for two qubits,

$$\hat{\rho}_0 = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\psi^*_{10} + \psi_{01}\psi^*_{11} \\ \psi_{10}\psi^*_{00} + \psi_{11}\psi^*_{01} & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}. \tag{10.12}$$

The diagonal holds the marginals of Section 6, the probabilities that qubit 0 is found in 0 or in 1, summed over both outcomes of qubit 1. Each off-diagonal element pairs amplitudes that agree on qubit 1.

On paper the same matrix follows from the $4 \times 4$ density matrix $\hat{\rho} = |\psi\rangle\langle\psi|$ of the register, with rows and columns labelled by the bit strings $ab$ at the index $2a + b$. Grouping them by the bit of qubit 0 divides $\hat{\rho}$ into four $2 \times 2$ blocks. The partial trace over qubit 1 replaces each block by its trace,

$$\hat{\rho} = \begin{pmatrix} \rho^{(00)} & \rho^{(01)} \\ \rho^{(10)} & \rho^{(11)} \end{pmatrix}, \qquad \hat{\rho}_0 = \mathrm{Tr}_1\hat{\rho} = \begin{pmatrix} \mathrm{Tr}\rho^{(00)} & \mathrm{Tr}\rho^{(01)} \\ \mathrm{Tr}\rho^{(10)} & \mathrm{Tr}\rho^{(11)} \end{pmatrix}, \tag{10.13}$$

where the block $\rho^{(aa')}$ holds the elements $\psi_{ab}\psi^*_{a'b'}$. Its trace sets $b' = b$ and sums over $b$, reproducing formula (10.11). The partial trace over qubit 0 instead adds the two diagonal blocks, $\hat{\rho}_1 = \rho^{(00)} + \rho^{(11)}$. For a product state $\hat{\rho} = \hat{\rho}_0 \otimes \hat{\rho}_1$ the blocks are $(\hat{\rho}_0)_{aa'}\hat{\rho}_1$, and the block traces return $\hat{\rho}_0$, since $\mathrm{Tr}\hat{\rho}_1 = 1$. A third form uses the $2 \times 2$ matrix $\Psi_{ab} = \psi_{ab}$ of Section 8.2, with rows labelled by qubit 0, in which the sum formula becomes a matrix product,

$$\hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^T. \tag{10.14}$$

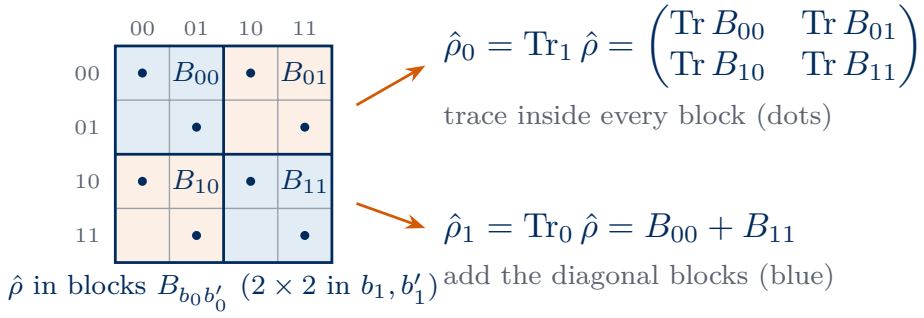



**Figure 34:** The partial traces of a $4 \times 4$ density matrix as block operations. Tracing out qubit 1 replaces each $2 \times 2$ block by its trace, and tracing out qubit 0 adds the two diagonal blocks.

On a computer the partial trace contracts the state tensor $\psi[a, b]$ with its conjugate over the index of qubit 1 and leaves both indices of qubit 0 open. With the operands `(psi, psi.conj())` the partial trace over qubit 1 is `"ab,Ab->aA"` and over qubit 0 `"ab,aB->bB"`. An upper-case letter stands for a primed index. On the density tensor $\hat\rho[a, b, a', b']$, the $4 \times 4$ matrix reshaped to shape $(2, 2, 2, 2)$, the block trace (10.13) reads `"abAb->aA"`. In the first diagram of Figure 35 the legs of qubit 0 stay open and become the two indices of $\hat\rho_0$, while the joined legs of qubit 1 form the sum over $b$. In the second the density tensor carries all four legs, and the trace over qubit 1 joins its ket leg $b$ to its bra leg $b'$. For three qubits, keeping qubit 0 is `"abc,Abc->aA"`. Keeping qubits 0 and 2 is `"abc,AbC->acAC"`, whose four open indices are regrouped into a $4 \times 4$ matrix by a reshape. The contraction never builds the $2^N \times 2^N$ matrix and is the form used for many qubits.

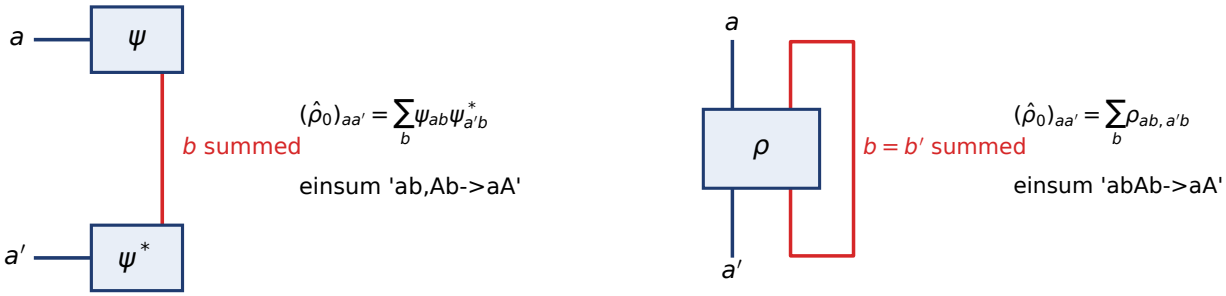



**Figure 35:** The partial trace over qubit 1 as tensor networks. Left, the state tensor $\psi$ and its conjugate with the legs of qubit 1 joined, `"ab,Ab->aA"`. Right, the density tensor with the ket and bra legs of qubit 1 joined, `"abAb->aA"`.

We apply formula (10.11) to three register states.

- The running product state $|0\rangle \otimes |+\rangle$ has the amplitudes $\psi_{00} = \psi_{01} = 1/\sqrt{2}$ and $\psi_{10} = \psi_{11} = 0$. Only $a = a' = 0$ gives a nonzero term, and the two reduced states are

$$\hat\rho_0 = \sum_b \psi_{ab}\psi^*_{a'b} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = |0\rangle\langle 0|, \qquad (\hat\rho_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'} = \psi_{0b}\psi^*_{0b'} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = |+\rangle\langle +|, \tag{10.15}$$

  the pure states of the two qubits themselves.

- The Bell state has $\psi_{00} = \psi_{11} = 1/\sqrt{2}$, so $\Psi = \hat I/\sqrt{2}$ and $\hat\rho_0 = \Psi\Psi^\dagger = \hat I/2$. The register is in a completely known pure state, while each qubit alone is the maximally mixed state of rank two. The local state $\hat I/2$ fixes every prediction for one qubit, a fair coin in every basis, and the correlations of Section 8.3 appear only in joint measurements. A pure state of the whole with maximally mixed parts has no classical counterpart and is the signature of entanglement.

- The family $|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$ has $\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ and $\hat\rho_0 = \mathrm{diag}(\cos^2\theta, \sin^2\theta)$. It interpolates between the pure $|0\rangle\langle 0|$ at $\theta = 0$ and $\hat I/2$ at $\theta = \pi/4$.

By (10.14) the rank of the reduced state equals the rank of $\Psi$, the Schmidt rank. The reduced state of an entangled pure state is therefore always mixed, with rank two for two qubits.

## 10.6 Purity of the reduced state

The purity $\mathrm{Tr}\hat{\rho}_0^2$ measures how mixed qubit 0 looks on its own. For the three cases it is 1 for the product state, $\mathrm{Tr}(\hat{I}/4) = 1/2$ for the Bell state, and $\cos^4\theta + \sin^4\theta$ for the family. The last falls from 1 at $\theta = 0$ to 1/2 at $\theta = \pi/4$ and equals 0.75 at $\theta = \pi/8$. In terms of the optional concurrence of Section 8.2, a general pure state of two qubits has $\mathrm{Tr}\hat{\rho}_0^2 = 1 - C^2/2$. The register is pure in all three cases, so a reduced purity below one comes entirely from the correlations between the qubits.

## 10.7 Entanglement entropy

The eigenvalues $\lambda_k$ of $\hat{\rho}_0$ are probabilities, and their Shannon entropy in bits is the *von Neumann entropy*

$$S(\hat{\rho}_0) = -\mathrm{Tr}\,\hat{\rho}_0 \log_2 \hat{\rho}_0 = -\sum_k \lambda_k \log_2 \lambda_k, \tag{10.16}$$

with the convention $0 \log 0 = 0$. For a pure register it is called the *entanglement entropy*, zero for a product state and one bit for the Bell state. For the family $|\psi_\theta\rangle$ it is the binary entropy $S_{\mathrm{bin}}(\cos^2\theta)$, with $S_{\mathrm{bin}}(p) = -p\log_2 p - (1-p)\log_2(1-p)$, which equals 0.60 bit at $\theta = \pi/8$ (Figure 36).

We can compute the entropy without forming $\hat{\rho}_0$. The SVD (8.5) of the reshaped state, $\Psi = U\,\mathrm{diag}(s_k)\,V^T$, gives with $V^TV^* = (V^\dagger V)^* = \hat{I}$

$$\hat{\rho}_0 = \Psi\Psi^\dagger = U\,\mathrm{diag}(s_k)\,V^TV^*\,\mathrm{diag}(s_k)\,U^\dagger = U\,\mathrm{diag}(s_k^2)\,U^\dagger, \tag{10.17}$$

so the eigenvalues of $\hat{\rho}_0$ are the squared Schmidt coefficients, $\lambda_k = s_k^2$, and

$$S = -\sum_k s_k^2 \log_2 s_k^2. \tag{10.18}$$

The matrix $\Psi^\dagger\Psi$ has the same nonzero eigenvalues as $\Psi\Psi^\dagger$, so the two qubits of a pure register always have equal entropies, $S(\hat{\rho}_0) = S(\hat{\rho}_1)$. A randomly drawn state, for example, has 0.36 bit on each side. Table 12 collects the criteria for the three cases. The singular-value route extends to a cut of $N$ qubits into two blocks, where $\Psi$ becomes a $2^{N_A} \times 2^{N-N_A}$ matrix, and underlies matrix product states.

| pure register state | Schmidt coefficients | Schmidt rank | $\hat{\rho}_0$ | $\mathrm{Tr}\hat{\rho}_0^2$ | $S(\hat{\rho}_0)$ [bit] |
|---|---|---|---|---|---|
| product $\|0\rangle\|+\rangle$ | $1,\ 0$ | 1 | $\|0\rangle\langle 0\|$ | 1 | 0 |
| $\|\psi_\theta\rangle$, $0 < \theta < \frac{\pi}{2}$ | $\|\cos\theta\|,\ \|\sin\theta\|$ | 2 | $\mathrm{diag}(\cos^2\theta, \sin^2\theta)$ | $\cos^4\theta + \sin^4\theta$ | $S_{\mathrm{bin}}(\cos^2\theta)$ |
| Bell $\|\Phi^+\rangle$ | $\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}$ | 2 | $\hat{I}/2$ | 1/2 | 1 |

**Table 12:** Entanglement criteria for pure two-qubit states. All follow from the singular values of the reshaped amplitudes, with $\lambda_k = s_k^2$.

## 10.8 Mixed registers

The entropy of a reduced state measures entanglement only when the whole register is pure. In the correlated coin, a mixed register, qubit 0 is also a fair coin, $\hat{\rho}_0 = \hat{I}/2$, with one bit of entropy as for the Bell state. That bit is ordinary ignorance about which of $|00\rangle$ and $|11\rangle$ was emitted, and the coin contains no entanglement. The two registers look identical on one qubit. They differ in the correlations $\langle\hat{X}\hat{X}\rangle$ and $\langle\hat{Y}\hat{Y}\rangle$ and in the purity of the register, one for the Bell state and one half for the coin. For mixed registers the reduced entropy combines classical uncertainty with entanglement, and other measures are needed to quantify entanglement. Table 13 lists rank, purity and von Neumann entropy of the density matrices met so far. The reduced state of the Bell state shares all three numbers with the coin register.

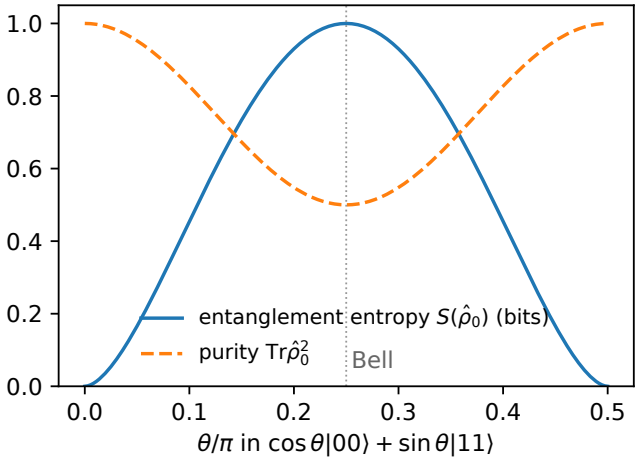



**Figure 36:** Entanglement entropy and purity of qubit 0 for $\cos\theta|00\rangle + \sin\theta|11\rangle$. The entropy rises from zero for $|00\rangle$ to one bit at the Bell state, $\theta = \pi/4$, and returns to zero at $|11\rangle$. The purity mirrors it between one and one half.

| density matrix | rank | purity $\mathrm{Tr}\hat{\rho}^2$ | entropy $S(\hat{\rho})$ [bit] |
|---|---|---|---|
| product register $\|0\rangle\|+\rangle$ | 1 | 1 | 0 |
| Bell register $\|\Phi^+\rangle\langle\Phi^+\|$ | 1 | 1 | 0 |
| qubit 0 of the Bell register, $\hat{I}/2$ | 2 | 1/2 | 1 |
| correlated coin register | 2 | 1/2 | 1 |
| maximally mixed register $\hat{I}/4$ | 4 | 0.25 | 2 |

**Table 13:** Rank, purity and entropy of pure and mixed density matrices. A pure state has rank one, purity one and zero entropy.

# 11 Measurement in general

## 11.1 Projective measurement and the Lüders rule

So far we have read the qubits in a basis and recorded bit strings. The general description of a measurement assigns a matrix $\hat{P}_k$ to every outcome $k$. For a *projective measurement* these matrices satisfy three conditions.

- Hermiticity, $\hat{P}_k^\dagger = \hat{P}_k$.
- Orthogonality, $\hat{P}_k\hat{P}_l = \delta_{kl}\hat{P}_k$. For $k = l$ this says that each $\hat{P}_k$ is a projector, and for $k \neq l$ that different outcomes exclude each other.
- Completeness, $\sum_k \hat{P}_k = \hat{I}$, so that one of the outcomes always occurs.

By the Born rule each outcome occurs with the probability $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$, or $\langle\psi|\hat{P}_k|\psi\rangle$ for a pure state. Completeness makes the probabilities sum to $\mathrm{Tr}\hat{\rho} = 1$. After the outcome $k$ the *Lüders rule* updates the state,

$$\hat{\rho} \;\longrightarrow\; \hat{\rho}_k = \frac{\hat{P}_k\,\hat{\rho}\,\hat{P}_k}{p_k}, \tag{11.1}$$

where the denominator restores the unit trace. A repeated measurement with the same projectors returns the same outcome, since $\hat{P}_k\hat{P}_k = \hat{P}_k$. For one qubit in the state (2.4), measured in the $\hat{Z}$ basis, the projector $\hat{P}_0 = |0\rangle\langle 0|$ yields $p_0 = \langle\psi_{\mathrm{ex}}|\hat{P}_0|\psi_{\mathrm{ex}}\rangle = 3/4$ and the updated state $\hat{P}_0|\psi_{\mathrm{ex}}\rangle\langle\psi_{\mathrm{ex}}|\hat{P}_0/p_0 = |0\rangle\langle 0|$. On the register, a $\hat{Z}$-basis measurement of qubit 0 uses $\hat{P}_0 = |0\rangle\langle 0|\otimes\hat{I}$ and $\hat{P}_1 = |1\rangle\langle 1|\otimes\hat{I}$. The $\hat{X}$ basis has the projectors $|\pm\rangle\langle\pm|\otimes\hat{I}$, with outcome bit 0 for $+$, and the $\hat{Y}$ basis has $|\pm i\rangle\langle\pm i|\otimes\hat{I}$.

## 11.2 Measuring an observable and measuring a Pauli string

An observable $\hat{O}$ with distinct eigenvalues $o$ has the spectral decomposition $\hat{O} = \sum_o o\,\hat{P}_o$, where $\hat{P}_o$ projects onto all eigenvectors with the eigenvalue $o$. Measuring the observable is the projective measurement $\{\hat{P}_o\}$, and its mean is $\sum_o o\,\mathrm{Tr}(\hat{P}_o\hat{\rho}) = \mathrm{Tr}(\hat{O}\hat{\rho})$, the rule (10.1). Measuring in a basis is the special case in which every projector has rank one. When an eigenvalue is degenerate, measuring the observable reveals less than measuring in a basis and disturbs the state less.

A Pauli string $\hat{P}$ has only the eigenvalues $\pm 1$. Its measurement has two outcomes, with the projectors

$$\hat{P}_\pm = \frac{\hat{I} \pm \hat{P}}{2}, \qquad \hat{P} = \hat{P}_+ - \hat{P}_-. \tag{11.2}$$

For $\hat{Z}_0\hat{Z}_1$ the projector $\hat{P}_+$ keeps the bit strings 00 and 11 of even parity and $\hat{P}_-$ keeps 01 and 10. We obtain the same outcome statistics by measuring both qubits in the $\hat{Z}$ basis and multiplying the two values $(-1)^{b_0}(-1)^{b_1}$, but the post-measurement states differ. The measurement of $\hat{Z}_0\hat{Z}_1$ returns $+1$ with certainty on $0.6|00\rangle + 0.8|11\rangle$ and leaves this state unchanged, coherence included, whereas reading both bits leaves $|00\rangle$ or $|11\rangle$. For $\hat{X}_0\hat{X}_1$ we apply Hadamard gates to both qubits before the same projectors, because $(\hat{H}\otimes\hat{H})(\hat{Z}\otimes\hat{Z})(\hat{H}\otimes\hat{H}) = \hat{X}\otimes\hat{X}$, the basis rotation of Section 6.2.

## 11.3 A parity measurement with an ancilla

A device can measure a Pauli string without reading out the qubits themselves. It copies the parity onto an extra qubit prepared in $|0\rangle$, an *ancilla*, and reads only the ancilla (Figure 37). On a basis state the two CNOT gates add both bits to the ancilla modulo two, $|b_0b_1\rangle|0\rangle \to |b_0b_1\rangle|b_0 \oplus b_1\rangle$. A general register state $\sum \psi_{b_0b_1}|b_0b_1\rangle$ becomes $\hat{P}_+|\psi\rangle|0\rangle + \hat{P}_-|\psi\rangle|1\rangle$. The ancilla reads 0 with the probability $\langle\psi|\hat{P}_+|\psi\rangle$, after which the register is in $\hat{P}_+|\psi\rangle$, normalised. The circuit performs the projective measurement (11.2) of $\hat{Z}_0\hat{Z}_1$ without reading either qubit. Quantum error correction is built on such *mid-circuit measurements*, followed by further gates that may depend on the outcome.

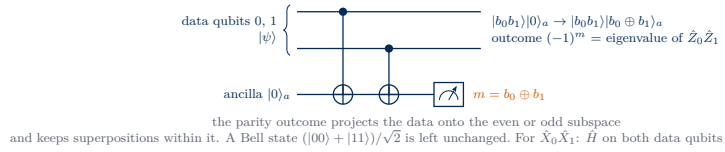



**Figure 37:** Measuring the Pauli string $\hat{Z}_0\hat{Z}_1$ with an ancilla. Two CNOT gates add the bits of qubits 0 and 1 to the ancilla, whose readout gives the parity $b_0 \oplus b_1$ while the two qubits are never read.

## 11.4 Back-action in the register

In an entangled register a measurement on one qubit also changes the other. Measuring qubit 0 of the Bell state in the $\hat{Z}$ basis gives the outcome 0 with probability 1/2 and leaves the register in $|00\rangle$, with qubit 1 in $|0\rangle$. The outcome $+$ in the $\hat{X}$ basis leaves qubit 1 in $|+\rangle$, as the Bell state written in the $\hat{X}$ basis shows,

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|++\rangle + |--\rangle}{\sqrt{2}}. \tag{11.3}$$

Before the measurement the partner qubit is a fair coin. Afterwards it is in a definite pure state, the eigenstate of the chosen basis that is correlated with the observed outcome. We simulate records by measuring qubit 0 first and qubit 1 afterwards in the same basis, sampling each outcome from the updated state. The two outcomes of the Bell state agree in every shot in $\hat{Z}$ and $\hat{X}$ and disagree in every shot in $\hat{Y}$, as $\langle\hat{Y}\hat{Y}\rangle = -1$ predicts. For the correlated coin they agree in every shot only in $\hat{Z}$, whereas in $\hat{X}$ they agree in a fraction 0.48 of the shots, like two independent coins. Only the entangled register shows back-action in every basis.

## 11.5 No signalling

The back-action transmits no signal. An observer at qubit 1 who does not know the outcome on qubit 0 describes qubit 1 by the average of the post-measurement states, weighted with the outcome probabilities. The average is independent of the basis chosen at qubit 0,

$$\sum_k p_k \,\mathrm{Tr}_0\hat\rho_k = \sum_k \mathrm{Tr}_0\big[(\hat\Pi_k\otimes\hat I)\hat\rho(\hat\Pi_k\otimes\hat I)\big] = \mathrm{Tr}_0\Big[\Big(\sum_k\hat\Pi_k\Big)\otimes\hat I\,\hat\rho\Big] = \mathrm{Tr}_0\hat\rho = \hat\rho_1, \tag{11.4}$$

where $\hat\Pi_k$ is the one-qubit projector and $\hat P_k = \hat\Pi_k\otimes\hat I$. In the middle step the right factor $\hat\Pi_k\otimes\hat I$ moves cyclically to the left inside the partial trace over qubit 0. The cyclic move is allowed for any operator $\hat A$ acting on the traced qubit alone. In index form, for any register matrix $\hat R$,

$$\big(\mathrm{Tr}_0[\hat R\,(\hat A\otimes\hat I)]\big)_{bb'} = \sum_{a,a''} R_{ab,a''b'}A_{a''a} = \sum_{a,a''} A_{a''a}R_{ab,a''b'} = \big(\mathrm{Tr}_0[(\hat A\otimes\hat I)\,\hat R]\big)_{bb'}, \tag{11.5}$$

where the second equality only reorders two numbers, and the last one renames the summed indices $a\leftrightarrow a''$. With $\hat A=\hat\Pi_k$ the two projectors meet in $\hat\Pi_k^2=\hat\Pi_k$, and the sum over $k$ collapses by $\sum_k\hat\Pi_k=\hat I$. Qubit 1 of the Bell state remains $\hat I/2$ whichever basis is measured at qubit 0. Measurements at one place cannot detect the choice of measurement at another place. The correlations appear only when the two records are compared.

## 11.6 Unread measurements and dephasing

When a measurement happens but its outcome is not recorded, the state is the average of the post-measurement states weighted with their probabilities,

$$\hat\rho \;\longrightarrow\; \sum_k p_k\hat\rho_k = \sum_k \hat P_k\,\hat\rho\,\hat P_k. \tag{11.6}$$

The map removes the coherences between different outcome sectors, the ranges of different projectors, and keeps those inside each sector. An unread measurement in a basis keeps the diagonal of $\hat\rho$ in that basis and removes all off-diagonal elements. The parity measurement of $\hat Z_0\hat Z_1$ keeps the coherence between $|00\rangle$ and $|11\rangle$. An unread $\hat Z$ measurement turns $|+\rangle$ into the mixture $\hat\rho_{\rm mix}$ and lowers the purity from 1 to 1/2. On qubit 0 of the Bell state an unread $\hat Z$ measurement produces the correlated coin, in which $\langle\hat X\hat X\rangle$ drops from 1 to 0 and $\langle\hat Z\hat Z\rangle$ stays at 1. Loss of coherence in a fixed basis is called *dephasing*.

## 11.7 General measurements

Projective measurements describe ideal detectors. A real $\hat Z$-basis detector may report the wrong bit, say with the probability $\epsilon$ for either input, and then reads 0 with the probability $(1-\epsilon)\langle\psi|0\rangle\langle0|\psi\rangle + \epsilon\langle\psi|1\rangle\langle1|\psi\rangle = \langle\psi|\hat\pi_0|\psi\rangle$ with the matrices

$$\hat\pi_0 = (1-\epsilon)|0\rangle\langle0| + \epsilon|1\rangle\langle1|, \qquad \hat\pi_1 = \hat I - \hat\pi_0 = \epsilon|0\rangle\langle0| + (1-\epsilon)|1\rangle\langle1|. \tag{11.7}$$

With $\epsilon=0.1$ the detector yields $p(0)=0.9$ for the state $|0\rangle$ and $p(0)=0.5$ for $|+\rangle$. Although $\hat\pi_0$ and $\hat\pi_1$ are no projectors, since $\hat\pi_0^2\neq\hat\pi_0$ for $0<\epsilon<1$, they reproduce all outcome probabilities through $p_k=\mathrm{Tr}(\hat\pi_k\hat\rho)$. Matrices of this kind form a *positive operator-valued measure* (POVM), a set of matrices $\hat\pi_k$, the POVM elements, with two conditions.

- Positivity, $\langle v|\hat\pi_k|v\rangle\geq0$ for every vector $v$, written $\hat\pi_k\geq0$, so that every probability is non-negative.
- Completeness, $\sum_k\hat\pi_k=\hat I$, so that the probabilities sum to one.

The elements need be neither orthogonal nor projectors, and a POVM may have more outcomes than the dimension of the system. The noisy detector describes the readout error of a device.

A POVM fixes only the statistics. The state after the outcome $k$ is described by a *measurement instrument*, a set of matrices $\hat{M}_{k,\alpha}$ with

$$\hat{\pi}_k = \sum_\alpha \hat{M}_{k,\alpha}^\dagger \hat{M}_{k,\alpha}, \qquad \hat{\rho} \;\longrightarrow\; \hat{\rho}^{(k)} = \frac{1}{p_k} \sum_\alpha \hat{M}_{k,\alpha}\, \hat{\rho}\, \hat{M}_{k,\alpha}^\dagger. \tag{11.8}$$

We use the *square-root instrument*, with one matrix $\hat{M}_k = \sqrt{\hat{\pi}_k}$ per outcome, computed in the eigenbasis of $\hat{\pi}_k$. For a projective measurement $\sqrt{\hat{P}_k} = \hat{P}_k$ and the update reduces to the Lüders rule (11.1). Other instruments realise the same POVM. With any unitary $\hat{V}_k$ the matrices $\hat{M}'_k = \hat{V}_k \sqrt{\hat{\pi}_k}$ satisfy $\hat{M}_k'^\dagger \hat{M}'_k = \hat{\pi}_k$ and yield the same probabilities with a different post-measurement state. The state update therefore depends on how the device is built.

Every POVM can be realised as a projective measurement on a larger system (optional, Naimark's theorem). The map $|\psi\rangle \to \sum_k \hat{M}_k|\psi\rangle|k\rangle_A$ with an ancilla $A$ preserves the norm, because $\sum_k \hat{M}_k^\dagger \hat{M}_k = \hat{I}$. A $\hat{Z}$-basis measurement of the ancilla then gives the outcome $k$ with the probability $\langle\psi|\hat{M}_k^\dagger \hat{M}_k|\psi\rangle = \langle\psi|\hat{\pi}_k|\psi\rangle$.

### 11.8 Informationally complete measurements

A measurement is *informationally complete* when its outcome probabilities determine the density matrix. Its elements must then span the real space of Hermitian $d \times d$ matrices, of dimension $d^2$. An informationally complete POVM therefore has at least $d^2$ outcomes and a minimal one has $d^2$. The state follows from the probabilities as a linear function. Two such families are the starting point of quantum state tomography.

The Pauli strings of Section 7.5 form a basis of the Hermitian matrices, and their expectation values determine $\hat{\rho} = \frac{1}{2^N} \sum_{\hat{P}} \langle \hat{P} \rangle \hat{P}$. A setting of Section 6.2 assigns $\hat{X}$, $\hat{Y}$ or $\hat{Z}$ to every qubit. It measures every string that carries, on each qubit, the identity or the Pauli matrix of that setting (Section 9.3), and the $3^N$ settings together yield all $4^N$ expectation values. The Pauli strings themselves are not positive and are therefore no POVM elements. A POVM appears when we choose the setting at random with the probability $3^{-N}$. The setting and the bit string then label the outcome together, and the projectors of all settings, each weighted with $3^{-N}$, form the POVM.

The second family is the *symmetric informationally complete* POVM (SIC-POVM) of one qubit, built on the four Bloch vectors of a regular tetrahedron,

$$\mathbf{s}_1 = (0,0,1), \quad \mathbf{s}_2 = \big(\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}\big), \quad \mathbf{s}_3 = \big(-\tfrac{\sqrt{2}}{3}, \sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}\big), \quad \mathbf{s}_4 = \big(-\tfrac{\sqrt{2}}{3}, -\sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}\big), \tag{11.9}$$

and the elements $\hat{\pi}_a = \frac{1}{4}(\hat{I} + \mathbf{s}_a \cdot \hat{\boldsymbol{\sigma}})$. Each $\hat{\pi}_a$ is half of a pure-state projector and hence positive, and $\sum_a \mathbf{s}_a = 0$ ensures $\sum_a \hat{\pi}_a = \hat{I}$. A state with Bloch vector $\mathbf{r}$ yields the probabilities $p_a = \frac{1}{4}(1 + \mathbf{s}_a \cdot \mathbf{r})$, which are non-negative and sum to one. Since $\mathbf{s}_a \cdot \mathbf{s}_b = -1/3$ for $a \neq b$, we recover the Bloch vector as $\mathbf{r} = 3 \sum_a p_a \mathbf{s}_a$. The four elements are linearly independent, and their number is $d^2 = 4$ for one qubit. For $N$ qubits the Kronecker products $\hat{\pi}_{a_0} \otimes \cdots \otimes \hat{\pi}_{a_{N-1}}$ provide $4^N$ outcomes in one setting.

## 12 Qubit errors and noise channels

### 12.1 An environment that records the qubit

A qubit in a laboratory is never completely isolated. It interacts with its surroundings, the *environment* $E$, which we do not observe, and the qubit alone then evolves from a pure state into a mixed one. The smallest example is an environment that keeps a record of the qubit's value. A perfect record acts like the unread measurement of Section 11, and a partial record produces partial dephasing. As

the environment $E$ we take a second qubit, prepared in $|0\rangle_E$. It couples to the system qubit through the Hamiltonian $\hat{\mathcal{H}} = \frac{\Omega}{2}|1\rangle\langle 1| \otimes \hat{Y}$, which rotates the environment about $y$ only when the system is in $|1\rangle$. Switched on for a time $t$, the coupling produces the controlled rotation

$$\hat{U} = e^{-i\hat{\mathcal{H}}t} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_y(g), \qquad g = \Omega t. \tag{12.1}$$

A system state $\alpha|0\rangle + \beta|1\rangle$ becomes

$$\hat{U}(\alpha|0\rangle + \beta|1\rangle)|0\rangle_E = \alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle, \qquad |e_0\rangle = |0\rangle, \quad |e_1\rangle = \cos\tfrac{g}{2}|0\rangle + \sin\tfrac{g}{2}|1\rangle. \tag{12.2}$$

The partial trace (10.11) over the environment, written $\mathrm{Tr}_E$, sums over the bit of the environment qubit and gives the state of the system,

$$\hat{\rho}_S = \begin{pmatrix} |\alpha|^2 & \alpha\beta^*\langle e_1|e_0\rangle \\ \alpha^*\beta\langle e_0|e_1\rangle & |\beta|^2 \end{pmatrix}, \qquad \langle e_0|e_1\rangle = \cos\tfrac{g}{2}. \tag{12.3}$$

The populations are unchanged, while the coherence is multiplied by the overlap of the two environment states. We take $0 \le g \le \pi$, where the overlap is non-negative. At $g = \pi$ the environment states are orthogonal, the environment holds a perfect record of the system's bit, and the result is the unread measurement (11.6). For $\pi < g \le 2\pi$ the negative overlap adds a phase flip $\hat{Z}$ to the dephasing. At $g = 2\pi$ the environment is back in $|0\rangle_E$ and the system has acquired the gate $\hat{Z}$. A finite environment such as one qubit can therefore give the information back, and irreversible decay requires an environment with many degrees of freedom.

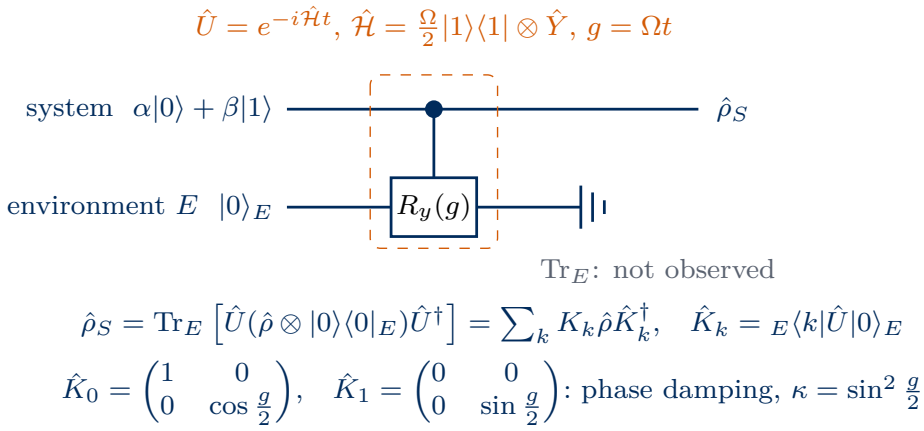



**Figure 38:** A qubit recorded by an environment qubit. A controlled rotation correlates the environment with the system, and tracing out the environment leaves a mixed state of the system. The partial matrix elements of the joint unitary are the Kraus operators.

## 12.2 Kraus operators from an unobserved environment

The general rule follows the same steps. The environment starts in a state $|0\rangle_E$ and evolves jointly with the system by a unitary $\hat{U}$. Afterwards the partial trace $\mathrm{Tr}_E$ over the environment describes the system, and we take it in an orthonormal basis $\{|k\rangle_E\}$ of the environment. The *partial matrix element* ${}_E\langle k|\hat{U}|0\rangle_E$ takes the matrix element of $\hat{U}$ between the environment states $|0\rangle_E$ and $|k\rangle_E$ only. It is a matrix that still acts on the system, with the entries $\left({}_E\langle k|\hat{U}|0\rangle_E\right)_{ss'} = \langle s|\langle k|_E\,\hat{U}\,|s'\rangle|0\rangle_E$. With it

$$\hat{\rho} \;\longrightarrow\; \mathcal{E}(\hat{\rho}) = \mathrm{Tr}_E[\hat{U}(\hat{\rho} \otimes |0\rangle\langle 0|_E)\hat{U}^\dagger] = \sum_k {}_E\langle k|\hat{U}|0\rangle_E\,\hat{\rho}\,{}_E\langle 0|\hat{U}^\dagger|k\rangle_E = \sum_k \hat{K}_k\,\hat{\rho}\,\hat{K}_k^\dagger, \tag{12.4}$$

with the *Kraus operators* $\hat{K}_k = {}_E\langle k|\hat{U}|0\rangle_E$, one for each basis state of the environment. Unitarity of $\hat{U}$ gives the completeness relation

$$\sum_k \hat{K}_k^\dagger \hat{K}_k = \sum_k {}_E\langle 0|\hat{U}^\dagger|k\rangle_E\,{}_E\langle k|\hat{U}|0\rangle_E = {}_E\langle 0|\hat{U}^\dagger\hat{U}|0\rangle_E = \hat{I}, \tag{12.5}$$

which keeps the trace of $\hat\rho$ equal to one. A map of the form (12.4) with (12.5) is a *quantum channel*, a completely positive and trace-preserving (CPTP) map [12]. Trace preservation, $\mathrm{Tr}\mathcal{E}(\hat\rho) = \mathrm{Tr}\hat\rho$, is the completeness relation. Complete positivity requires that $\mathcal{E}$ map density matrices to density matrices also when it acts on one part of a larger system. The condition reads $(\mathcal{E}\otimes\mathrm{id})(\hat\rho_{SA}) \geq 0$ for every state of the system and an ancilla $A$. Every map of the Kraus form has this property, and conversely every completely positive map has a Kraus form. Positivity alone is weaker. The transpose $\hat\rho \to \hat\rho^T$ maps every one-qubit density matrix to a density matrix. Applied to one qubit of the Bell state, however, it produces a matrix with the eigenvalue $-1/2$ and is therefore no physical operation.

Every physical noise process has the Kraus form. One Kraus operator, $\hat K_0 = \hat U_S$, is ordinary unitary evolution. Several Kraus operators describe a mixture of things that may have happened to the qubit, the branch $k$ occurring with the probability $\mathrm{Tr}(\hat K_k\hat\rho\hat K_k^\dagger)$. The Kraus operators of a channel are not unique. Any unitary $r\times r$ matrix $u$ produces a new set $\hat{\tilde K}_j = \sum_l u_{jl}\hat K_l$ with the same $\mathcal{E}(\hat\rho)$, because the sum $\sum_j \hat{\tilde K}_j\hat\rho\hat{\tilde K}_j^\dagger$ contains $\sum_j u_{jl}u^*_{jl'} = \delta_{ll'}$. Different physical environments can thus produce identical noise on the qubit. For the environment of Section 12.1 with $0\leq g\leq\pi$, the controlled rotation yields, with $\kappa = \sin^2\frac{g}{2}$,

$$\hat K_0 = {}_E\langle 0|\hat U|0\rangle_E = \begin{pmatrix}1 & 0\\ 0 & \cos\frac{g}{2}\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & \sqrt{1-\kappa}\end{pmatrix}, \qquad \hat K_1 = {}_E\langle 1|\hat U|0\rangle_E = \begin{pmatrix}0 & 0\\ 0 & \sin\frac{g}{2}\end{pmatrix} = \begin{pmatrix}0 & 0\\ 0 & \sqrt{\kappa}\end{pmatrix}, \tag{12.6}$$

which reproduce (12.3), and $\hat K_0^\dagger\hat K_0 + \hat K_1^\dagger\hat K_1 = \hat I$.

## 12.3 Six error models

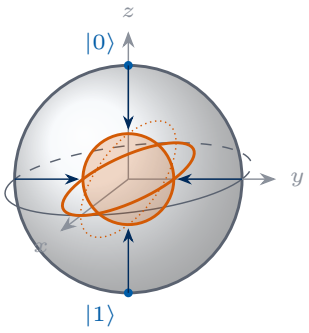
Six standard error models describe the common imperfections of qubits. For each model we give the Kraus operators, the action on the density matrix and on the Bloch vector $\mathbf{r}$ of equation (10.8), and an example. Every Kraus set below satisfies (12.5). Table 14 summarises the six models, and Figure 39 shows them as deformations of the Bloch ball.

### Bit flip

With probability $p$ the gate $\hat X$ acts by accident, and otherwise nothing happens,

$$\hat K_0 = \sqrt{1-p}\,\hat I, \quad \hat K_1 = \sqrt{p}\,\hat X, \qquad \mathcal{E}(\hat\rho) = (1-p)\hat\rho + p\hat X\hat\rho\hat X. \tag{12.7}$$

Since the conjugation $\hat X\hat\rho\hat X$ keeps $r_x$ and flips the signs of $r_y$ and $r_z$, the Bloch vector maps to $(r_x, (1-2p)r_y, (1-2p)r_z)$. The state $|1\rangle$ becomes $\mathrm{diag}(p, 1-p)$, a qubit that reads 0 with probability $p$.




bit flip, $p = 0.3$
$(r_x, 0.4r_y, 0.4r_z)$

### Phase flip (dephasing)

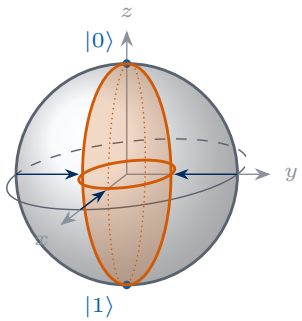
With probability $p$ the gate $\hat{Z}$ acts,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_1 = \sqrt{p}\,\hat{Z}, \qquad \mathcal{E}(\hat{\rho}) = (1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}, \tag{12.8}$$

and $\mathbf{r} \to ((1-2p)r_x, (1-2p)r_y, r_z)$. The populations in the $\hat{Z}$ basis survive and the coherences shrink by $1-2p$. The state $|+\rangle$ becomes

$$\mathcal{E}(|+\rangle\langle+|) = \frac{1}{2}\begin{pmatrix} 1 & 1-2p \\ 1-2p & 1 \end{pmatrix}, \tag{12.9}$$

and at $p = 1/2$ the channel is the unread $\hat{Z}$ measurement (11.6). Dephasing models a qubit whose energy splitting fluctuates, as for a spin in a fluctuating magnetic field, and the fluctuations randomise the relative phase.



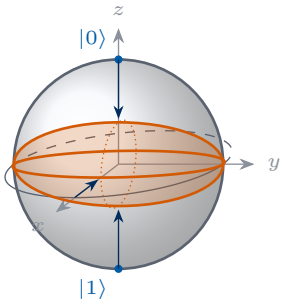

phase flip, $p = 0.3$
$(0.4r_x, 0.4r_y, r_z)$

### Bit-phase flip

With probability $p$ the gate $\hat{Y} = i\hat{X}\hat{Z}$ acts, a bit flip and a phase flip together,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_1 = \sqrt{p}\,\hat{Y}, \qquad \mathbf{r} \to ((1-2p)r_x,\ r_y,\ (1-2p)r_z). \tag{12.10}$$

The state $|1\rangle$ becomes $\mathrm{diag}(p, 1-p)$, as under the bit flip, and $|+\rangle$ loses its coherence as under the phase flip.



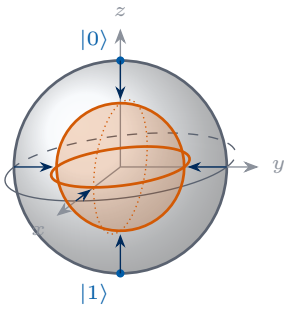

bit-phase flip, $p = 0.3$
$(0.4r_x, r_y, 0.4r_z)$

### Depolarising noise

With probability $p$ one of the three Pauli errors acts, each with probability $p/3$,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_{1,2,3} = \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}, \qquad \mathbf{r} \to \Big(1 - \frac{4}{3}p\Big)\mathbf{r}. \tag{12.11}$$

Each component keeps its sign under one of the three errors and flips it under the other two, and the factor becomes $1 - p + \frac{p}{3} - \frac{2p}{3} = 1 - \frac{4}{3}p$. Written as $\mathcal{E}(\hat{\rho}) = (1 - \frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\frac{\hat{I}}{2}$, the channel mixes the state with the maximally mixed state and shrinks the Bloch ball uniformly towards its centre. Depolarising noise is the basic model of errors without a preferred axis.



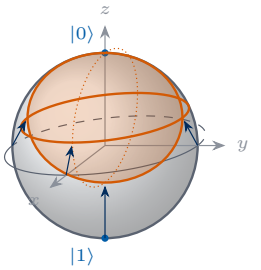

depolarising, $p = 0.3$
$0.6\,\mathbf{r}$

### Amplitude damping

A qubit encoded in two energy levels loses energy to its surroundings, for example by spontaneous emission from the excited state $|e\rangle = |1\rangle$ to the ground state $|g\rangle = |0\rangle$. The joint evolution of the qubit and an environment initially in $|0\rangle_E$ moves the excitation to the environment with probability $\gamma$, $|1\rangle|0\rangle_E \to \sqrt{1-\gamma}|1\rangle|0\rangle_E + \sqrt{\gamma}|0\rangle|1\rangle_E$, and leaves $|0\rangle|0\rangle_E$ unchanged. The Kraus operators ${}_E\langle k|\hat{U}|0\rangle_E$ are

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{r} \to (\sqrt{1-\gamma}\,r_x,\ \sqrt{1-\gamma}\,r_y,\ \gamma + (1-\gamma)r_z). \quad (12.12)$$

The operator $\hat{K}_1 = \sqrt{\gamma}\,|g\rangle\langle e| = \sqrt{\gamma}\,\hat{S}^+$, the decay operator (2.19), is the jump from $|1\rangle$ to $|0\rangle$, and $\hat{K}_0$ is the branch without a jump, in which the amplitude of $|1\rangle$ shrinks. Under the channel $|1\rangle$ becomes $\mathrm{diag}(\gamma, 1-\gamma)$ and the ball is pushed towards the north pole $|0\rangle$, which stays fixed. For $\gamma \to 1$ every state relaxes to $|0\rangle$. In spontaneous decay with the energy relaxation time $T_1$ the parameter grows as $\gamma = 1 - e^{-t/T_1}$.




amplitude damping, $\gamma = 0.3$
$(0.84r_x, 0.84r_y, 0.3 + 0.7r_z)$

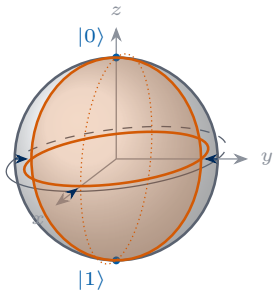
### Phase damping

The environment of Section 12.1 learns whether the qubit is excited without taking its energy. The channel has the parameter $\kappa$, the probability that the environment registers the excitation,

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\kappa} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\kappa} \end{pmatrix}, \qquad \mathbf{r} \to (\sqrt{1-\kappa}\,r_x,\ \sqrt{1-\kappa}\,r_y,\ r_z), \quad (12.13)$$

with $\kappa = \sin^2\frac{g}{2}$ for the controlled rotation. The populations stay fixed and the coherences shrink. For $1 - 2p = \sqrt{1-\kappa}$, phase damping and the phase flip are one channel written with two different sets of

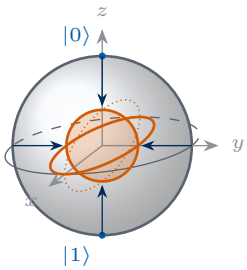
Kraus operators. All three descriptions of dephasing in these notes, the unread measurement, the environment record and the phase flip, are therefore one channel with different parameters.



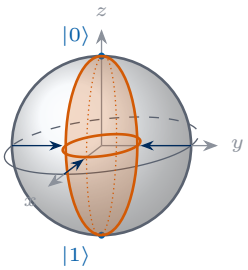

phase damping, $\kappa = 0.3$
$(0.84 r_x, 0.84 r_y, r_z)$

| channel | Kraus operators | Bloch vector $(r_x, r_y, r_z) \to$ |
|---|---|---|
| bit flip | $\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{X}$ | $(r_x,\ (1-2p)r_y,\ (1-2p)r_z)$ |
| phase flip (dephasing) | $\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Z}$ | $((1-2p)r_x,\ (1-2p)r_y,\ r_z)$ |
| bit-phase flip | $\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Y}$ | $((1-2p)r_x,\ r_y,\ (1-2p)r_z)$ |
| depolarising | $\sqrt{1-p}\,\hat{I},\ \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}$ | $(1-\frac{4}{3}p)\,(r_x, r_y, r_z)$ |
| amplitude damping | $\begin{pmatrix}1 & 0\\ 0 & \sqrt{1-\gamma}\end{pmatrix},\ \begin{pmatrix}0 & \sqrt{\gamma}\\ 0 & 0\end{pmatrix}$ | $(\sqrt{1-\gamma}\,r_x,\ \sqrt{1-\gamma}\,r_y,\ \gamma+(1-\gamma)r_z)$ |
| phase damping | $\begin{pmatrix}1 & 0\\ 0 & \sqrt{1-\kappa}\end{pmatrix},\ \begin{pmatrix}0 & 0\\ 0 & \sqrt{\kappa}\end{pmatrix}$ | $(\sqrt{1-\kappa}\,r_x,\ \sqrt{1-\kappa}\,r_y,\ r_z)$ |

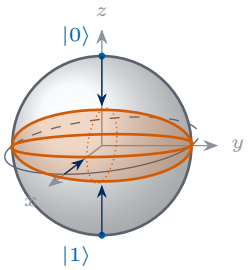
**Table 14:** Single-qubit error models. Each Kraus set satisfies $\sum_k \hat{K}_k^\dagger \hat{K}_k = \hat{I}$, and each Bloch-vector map was checked numerically on random states.



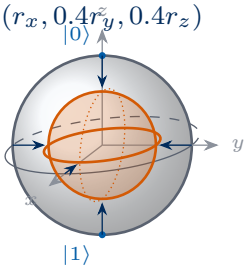

bit flip, $p = 0.3$
$(r_x, 0.4 r_y, 0.4 r_z)$



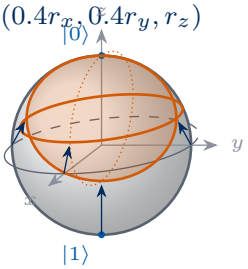

phase flip, $p = 0.3$
$(0.4 r_x, 0.4 r_y, r_z)$



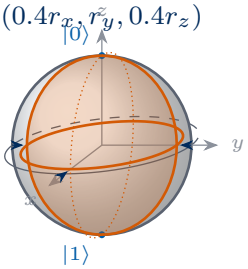

bit-phase flip, $p = 0.3$
$(0.4 r_x, r_y, 0.4 r_z)$




depolarising, $p = 0.3$
$0.6\,\mathbf{r}$




amplitude damping, $\gamma = 0.3$
$(0.84 r_x, 0.84 r_y, 0.3 + 0.7 r_z)$




phase damping, $\kappa = 0.3$
$(0.84 r_x, 0.84 r_y, r_z)$

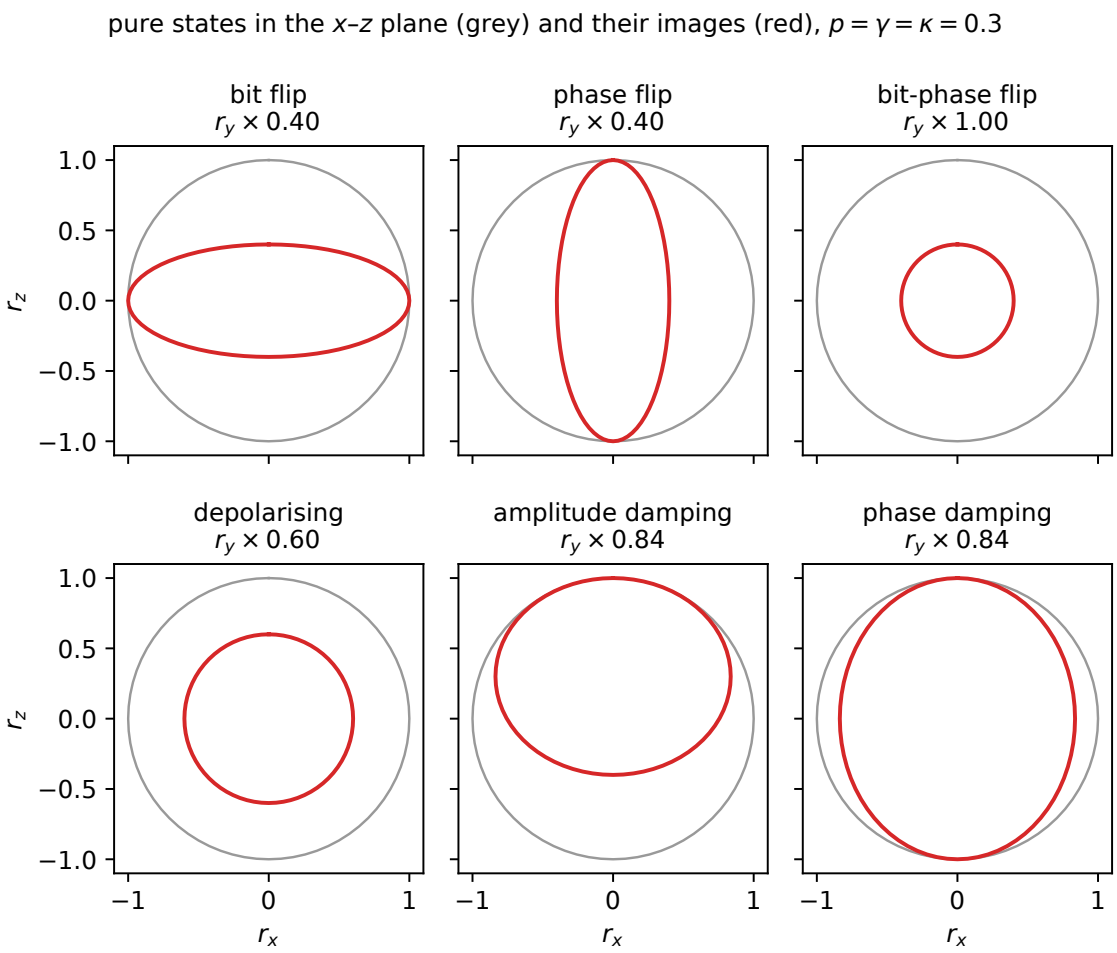



**Figure 39:** Images of the circle of pure states in the $x$–$z$ plane (grey) under the six error models (red), with strength $p = \gamma = \kappa = 0.3$. The bit flip keeps the $x$ axis and shrinks $z$ by $1 - 2p = 0.4$. The phase flip and phase damping keep the poles and shrink the coherences, and the bit-phase flip shrinks both axes. Depolarising noise shrinks the whole circle by $1 - \frac{4}{3}p = 0.6$, and amplitude damping shrinks the horizontal axis by $\sqrt{1-\gamma} \approx 0.84$ and shifts the ellipse towards $|0\rangle$.

## 12.4 Pauli channels, unital and non-unital noise

Each of the first four error models applies a Pauli matrix $\hat{P}$ with a probability $p_P$,

$$\mathcal{E}(\hat{\rho}) = \sum_{\hat{P}\in\{\hat{I},\hat{X},\hat{Y},\hat{Z}\}} p_P\, \hat{P}\hat{\rho}\hat{P}, \qquad \sum_{\hat{P}} p_P = 1, \tag{12.14}$$

which defines a *Pauli channel*. On $N$ qubits Pauli strings take the place of the Pauli matrices. A Pauli channel multiplies each Bloch-vector component by a factor, and the Bloch ball shrinks along the three axes with its centre fixed, $\mathcal{E}(\hat{I}/2) = \hat{I}/2$. A channel with $\mathcal{E}(\hat{I}) = \hat{I}$ is called *unital*. Every mixture of unitaries $\sum_j p_j \hat{U}_j \hat{\rho} \hat{U}_j^\dagger$ is unital, and so is phase damping. Amplitude damping is *non-unital*. It maps $\hat{I}/2$ to $\mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2})$, moves the centre of the ball towards the north pole and has the ground state $|0\rangle$ as its fixed point. Noise that releases energy into the environment always has a preferred final state and is non-unital.

## 12.5 The register under local noise

Let each qubit of the Bell register suffer the same noise independently. The channel on the register has the Kraus operators $\hat{K}_k \otimes \hat{K}_l$,

$$\hat{\rho} \;\longrightarrow\; \sum_{k,l} (\hat{K}_k \otimes \hat{K}_l)\, \hat{\rho}\, (\hat{K}_k \otimes \hat{K}_l)^\dagger. \tag{12.15}$$

On the density tensor $\hat{\rho}[a, b, a', b']$ a one-qubit channel on qubit $q$ contracts its Kraus operators with the ket and bra legs of that qubit only. As for the gates, we apply noise to a large register without building $2^N \times 2^N$ matrices.

Local dephasing does not touch the populations in the $\hat{Z}$ basis and leaves $\langle \hat{Z}\hat{Z} \rangle = 1$, whereas it multiplies $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ by $(1 - 2p)^2$, one factor per qubit (Figure 40, left). At $p = 0.3$ the correlator reaches $\langle \hat{X}\hat{X} \rangle = 0.16$ and the register purity 0.51. At $p = 1/2$ the coherences are gone and the register has become the correlated coin. Depolarising noise reduces all three correlators by the common factor $(1 - \frac{4}{3}p)^2$, to $\langle \hat{X}\hat{X} \rangle = 0.36$ at $p = 0.3$ (Figure 40, right). Under both channels the register purity falls.

The entropy of qubit 0 stays at one bit throughout, since $\hat{\rho}_0 = \hat{I}/2$ for every $p$ under both channels. As in Section 10.8, it misses the transition from the Bell state to the classical coin. Amplitude damping has a preferred final state and changes the reduced states as well. With a common $\gamma$ on both qubits it turns the reduced state of each qubit into $\mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2})$ and the correlator into $\langle \hat{Z}\hat{Z} \rangle = 1 - 2\gamma + 2\gamma^2$.

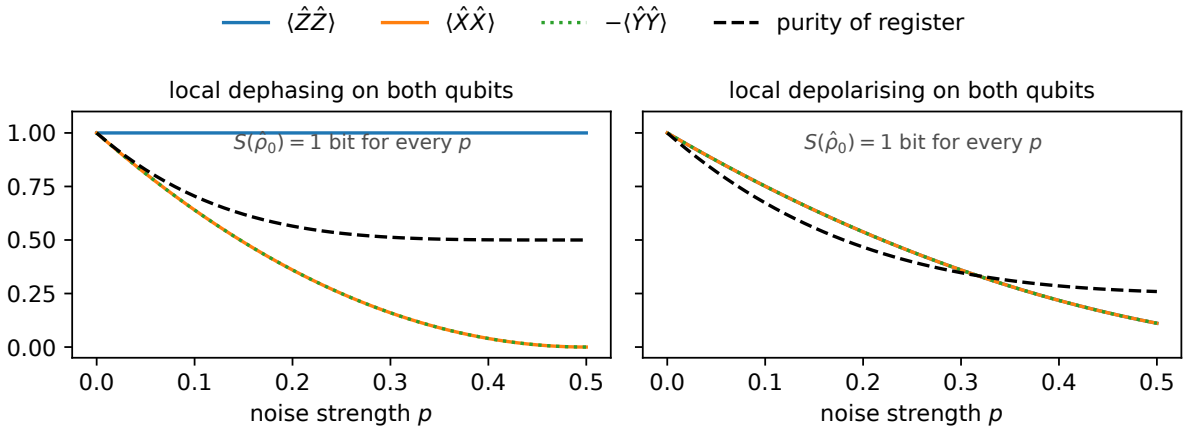



**Figure 40:** Correlators, register purity and entropy of qubit 0 for the Bell state under local dephasing (left) and local depolarising noise (right), against the noise strength $p$. Under dephasing $\langle \hat{Z}\hat{Z} \rangle$ stays at one while $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ decay as $\pm(1 - 2p)^2$, and at $p = 1/2$ only the classical correlation of the coin remains. Under depolarising noise all three correlators decay together as $(1 - \frac{4}{3}p)^2$. In both panels the register purity falls while the entropy of qubit 0 stays at one bit, so the reduced state of one qubit cannot see the loss of entanglement.

## 12.6 The Lindblad equation

A channel describes the noise accumulated over a fixed time. In a laboratory the noise acts continuously, and the density matrix obeys a differential equation, the open-system counterpart of the Schrödinger equation (3.1). For a closed system $\hat{\rho}(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}(t)^\dagger$ with $\hat{U} = e^{-it\hat{\mathcal{H}}}$, and differentiation yields the von Neumann equation $\frac{d\hat{\rho}}{dt} = -i[\hat{\mathcal{H}}, \hat{\rho}]$, with the commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. An environment that carries information away without returning it, the memoryless or Markovian case, adds extra terms. The equation then takes the *Lindblad form* [13,14],

$$\frac{d\hat{\rho}}{dt} = \mathcal{L}(\hat{\rho}) = -i[\hat{\mathcal{H}}, \hat{\rho}] + \sum_k \Big( \hat{L}_k \hat{\rho} \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \hat{\rho}\} \Big), \tag{12.16}$$

with the anticommutator $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ and the *jump operators* $\hat{L}_k$, one for each noise process. The first term in the bracket describes the jumps. The anticommutator removes the corresponding probability from the no-jump evolution and keeps the trace at one. The solution $\hat{\rho}(t) = \mathcal{E}_t(\hat{\rho}(0))$ with $\mathcal{E}_t = e^{t\mathcal{L}}$ is the open-system analogue of $\hat{U}(t) = e^{-it\hat{\mathcal{H}}}$. Every $\mathcal{E}_t$ is a channel, and the family obeys $\mathcal{E}_{t+s} = \mathcal{E}_t \mathcal{E}_s$. Unlike $\hat{U}(t)$, it has in general no physical inverse, since the information lost to the environment cannot be recovered, and such a family is called a *semigroup*. On a computer we flatten $\hat{\rho}$ into a vector of $d^2$ entries. The generator $\mathcal{L}$ is then a $d^2 \times d^2$ matrix and $e^{t\mathcal{L}}$ a matrix exponential.

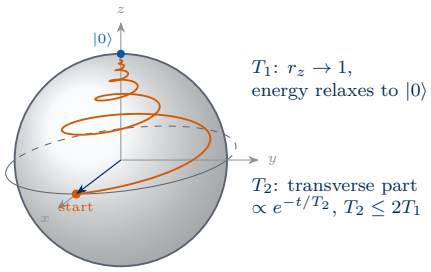
Two jump operators describe the decoherence of a qubit. *Energy relaxation* with the rate $\Gamma$ has the jump operator $\hat{L} = \sqrt{\Gamma}\,|g\rangle\langle e| = \sqrt{\Gamma}\,\hat{S}^{+}$, the decay operator (2.19), which takes the excited state $|1\rangle$ to the ground state $|0\rangle$. In the atomic-physics convention of the box in Section 2.6 the same operator is written $\sqrt{\Gamma}\,\hat{\sigma}^{-}_{\text{atom}}$. With $\hat{L}\hat{\rho}\hat{L}^{\dagger} = \Gamma\rho_{11}|0\rangle\langle 0|$ and $\hat{L}^{\dagger}\hat{L} = \Gamma|1\rangle\langle 1|$, the elements of (12.16) with $\hat{\mathcal{H}} = 0$ read

$$\frac{d\rho_{11}}{dt} = -\Gamma\rho_{11}, \qquad \frac{d\rho_{00}}{dt} = +\Gamma\rho_{11}, \qquad \frac{d\rho_{01}}{dt} = -\frac{\Gamma}{2}\,\rho_{01}, \tag{12.17}$$

so $\rho_{11}(t) = e^{-\Gamma t}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-\Gamma t/2}\rho_{01}(0)$. The decay is amplitude damping with $\gamma = 1 - e^{-\Gamma t}$, whose coherence factor $\sqrt{1-\gamma}$ equals $e^{-\Gamma t/2}$, and the *relaxation time* is $T_1 = 1/\Gamma$. *Pure dephasing* with the rate $\gamma_\phi$ has $\hat{L} = \sqrt{\gamma_\phi/2}\,\hat{Z}$. With $\hat{L}^{\dagger}\hat{L} = \frac{\gamma_\phi}{2}\hat{I}$ the equation becomes $\frac{d\hat{\rho}}{dt} = \frac{\gamma_\phi}{2}(\hat{Z}\hat{\rho}\hat{Z} - \hat{\rho})$, and the populations stay constant. Since $\hat{Z}\hat{\rho}\hat{Z}$ flips the sign of the coherences, they decay as $\rho_{01}(t) = e^{-\gamma_\phi t}\rho_{01}(0)$, and the channel is phase damping with $\sqrt{1-\kappa} = e^{-\gamma_\phi t}$. With both processes the populations relax with $T_1$ and the coherences decay with the *coherence time* $T_2$,

$$\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi \qquad \Longrightarrow \qquad T_2 \le 2T_1. \tag{12.18}$$

The Bloch vector spirals towards the north pole $|0\rangle$, its $z$ component relaxing with $T_1$ and its length in the $x$–$y$ plane shrinking with $T_2$.




## 12.7 Noise in quantum circuits

We describe a noisy quantum computer by attaching channels to the circuit. The standard model has four ingredients, all built from the channels above.

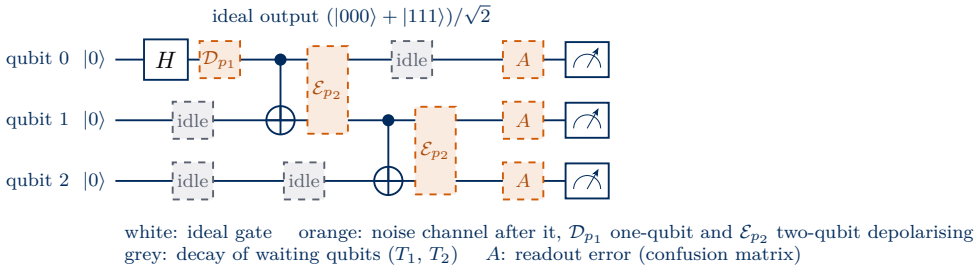



**Figure 41:** The circuit noise model for the GHZ circuit. Each gate is followed by a depolarising channel on the qubits it touches, and waiting qubits decay with $T_1$ and $T_2$. The readout reports wrong bits with the probabilities of the confusion matrix.

### Gate errors

Each gate is applied ideally and followed by a channel on the qubits it touches. After a one-qubit gate this is the depolarising channel with the probability $p_1$, which equals $(1 - \frac{4}{3}p_1)\hat{\rho} + \frac{4}{3}p_1\frac{\hat{I}}{2}$. After a two-qubit gate it is the two-qubit depolarising channel with the error probability $p_2$, uniform over the 15 Pauli strings of the pair other than $\hat{I}\otimes\hat{I}$,

$$\mathcal{E}_2(\hat{\rho}) = (1-p_2)\hat{\rho} + \frac{p_2}{15}\sum_{\hat{P}\neq\hat{I}\otimes\hat{I}} \hat{P}\hat{\rho}\hat{P} = \Big(1 - \frac{16}{15}p_2\Big)\hat{\rho} + \frac{16}{15}p_2\,\frac{\hat{I}}{4}\otimes \mathrm{Tr}_{\text{pair}}\hat{\rho}. \tag{12.19}$$

The second form uses the identity $\sum_{\hat{P}} \hat{P}\hat{\rho}\hat{P} = 4\,\hat{I} \otimes \mathrm{Tr}_{\mathrm{pair}}\hat{\rho}$ for the sum over all 16 strings of the pair. The identity holds because the average over all Pauli strings removes every coherence and population difference of the pair. In this form the channel mixes the pair with its maximally mixed state with the weight $\frac{16}{15}p_2$. It multiplies the expectation value of every non-identity Pauli string on the pair by $1 - \frac{16}{15}p_2$. Some texts quote the mixing weight $\frac{16}{15}p_2$ as the error probability, and error rates from different sources must be compared with this convention in mind. Two-qubit gates are usually the noisiest operations of a device, with $p_2$ often about ten times larger than $p_1$.

### Idle decoherence

A layer of the circuit lasts $t_{\mathrm{layer}}$. During each layer the qubits that wait while gates act on others undergo amplitude damping with $\gamma = 1 - e^{-t_{\mathrm{layer}}/T_1}$ and phase damping with $\sqrt{1-\kappa} = e^{-t_{\mathrm{layer}}/T_\phi}$, where $1/T_\phi = 1/T_2 - 1/(2T_1)$ by (12.18). Both are finite-time channels of the Lindblad equation of Section 12.6. In our model a qubit acted on by a gate receives only the gate channel, which is taken to include its decoherence during the gate. A deep circuit is limited by the ratio of its total duration to $T_1$ and $T_2$. A gate lasts as long as its angle divided by the coupling that produces it, $\theta/\Omega$ for a single-qubit rotation and $\theta/J$ for a two-qubit rotation (3.26), (7.18). Larger angles and weaker couplings expose the qubits longer to these channels. Including decoherence during the gate, a rotation by $\theta$ held for the time $\theta/\Omega$ suffers amplitude damping with $\gamma = 1 - e^{-\theta/(\Omega T_1)} \approx \theta/(\Omega T_1)$.

### Readout errors

The detector reports the wrong bit with a probability that may depend on the true bit. A *confusion matrix* $A_{mb} = p(\text{read } m \,|\, \text{true } b)$, with columns that sum to one, maps the true probabilities of the bits to the recorded ones,

$$\mathbf{p}_{\mathrm{meas}} = A\,\mathbf{p}_{\mathrm{true}}, \qquad A = \begin{pmatrix} 1-\epsilon_{0\to1} & \epsilon_{1\to0} \\ \epsilon_{0\to1} & 1-\epsilon_{1\to0} \end{pmatrix}. \tag{12.20}$$

The confusion matrix corresponds to the POVM $\hat{\pi}_m = \sum_b A_{mb}|b\rangle\langle b|$, which for $\epsilon_{0\to1} = \epsilon_{1\to0} = \epsilon$ is the noisy detector (11.7). With independent readout errors on $N$ qubits the register has the confusion matrix $A_0 \otimes \cdots \otimes A_{N-1}$, acting on the vector of $2^N$ bit-string probabilities in the order of Table 5. *Readout mitigation* measures $A$ with known basis states as inputs and estimates $\mathbf{p}_{\mathrm{true}} \approx A^{-1}\mathbf{p}_{\mathrm{meas}}$. From a finite record the inverted vector can have small negative entries, which must be handled before the vector serves as a probability distribution.

### Coherent and stochastic errors

A *coherent* error is a small unwanted unitary, for example an over-rotation $\hat{R}_x(\epsilon)$ in place of the identity. Over $n$ repetitions the angles add, $\hat{R}_x(\epsilon)^n = \hat{R}_x(n\epsilon)$, and a qubit started in $|0\rangle$ is found in $|1\rangle$ with the probability $\sin^2(n\epsilon/2) \approx (n\epsilon)^2/4$. Because amplitudes add, the error grows quadratically with the number of gates. A *stochastic* error is a channel, for example a bit flip with probability $p$ at each gate. After $n$ gates the flip probability is $\frac{1}{2}(1-(1-2p)^n) \approx np$, so probabilities add and the error grows linearly. At equal single-gate error, $p = \epsilon^2/4$, the coherent error after $n$ gates is larger by the factor $n$. Depolarising models describe stochastic errors only, and coherent errors are usually the first to be calibrated away.

### Simulating a noisy circuit

We simulate a noisy circuit with the density matrix. Each gate acts as $\hat{\rho} \to \hat{U}\hat{\rho}\hat{U}^\dagger$, with $\hat{U}$ a Kronecker product with identities on the untouched qubits as in Section 7.5. Each channel acts as $\hat{\rho} \to \sum_k \hat{K}_k\hat{\rho}\hat{K}_k^\dagger$, where a Kraus operator on qubit $q$ is again $\hat{I} \otimes \cdots \otimes \hat{K}_k \otimes \cdots \otimes \hat{I}$. On a computer these operations are contractions with the ket and bra legs of the touched qubits. With $4^N$ entries, $16 \cdot 4^N$ bytes, the

density matrix limits this simulation to about half the number of qubits of a state-vector simulation. The *quantum trajectory* method (optional) keeps a state vector, chooses at each channel the branch $k$ with the probability $\|\hat{K}_k|\psi\rangle\|^2$ and continues with $\hat{K}_k|\psi\rangle/\|\hat{K}_k|\psi\rangle\|$. Averaging $|\psi\rangle\langle\psi|$ over many trajectories reproduces $\hat{\rho}$, with a statistical error that falls as one over the square root of the number of trajectories.

### A noisy GHZ circuit

The GHZ circuit (7.27) with one Hadamard gate and $N-1$ CNOT gates prepares a density matrix $\hat{\rho}$. We measure its quality by the fidelity with the target, $F = \langle \mathrm{GHZ}_N|\hat{\rho}|\mathrm{GHZ}_N\rangle$, the probability that $\hat{\rho}$ passes a test projecting on the target state. We add depolarising noise $p_1$ after the Hadamard gate and $p_2$ after each CNOT. Each channel does nothing with the weight $1-\frac{4}{3}p_1$ or $1-\frac{16}{15}p_2$ and replaces the touched qubits by a maximally mixed state with the remaining weight. These weights are non-negative for $p_1 \le 3/4$ and $p_2 \le 15/16$. In the expanded product of all channels the branch without replacement prepares the ideal GHZ state. Every other branch adds a positive matrix, whose overlap with the GHZ state is non-negative. The fidelity therefore obeys the lower bound

$$F \;\ge\; \Big(1-\frac{4}{3}p_1\Big)\Big(1-\frac{16}{15}p_2\Big)^{N-1}, \tag{12.21}$$

which falls exponentially with the number of CNOT gates. The bound overestimates the loss, since some Pauli errors leave the GHZ state unchanged. To first order in the error probabilities, an $\hat{X}$ error after the Hadamard gate leaves $|+\rangle$ as it is. After the first CNOT, three of the 15 errors are harmless, because $\hat{X}\otimes\hat{X}$, $\hat{Y}\otimes\hat{Y}$ and $\hat{Z}\otimes\hat{Z}$ leave the pair state $(|00\rangle+|11\rangle)/\sqrt{2}$ unchanged up to a global phase. After each later CNOT only $\hat{Z}\otimes\hat{Z}$ is harmless, and counting the harmful errors yields

$$1-F = \frac{2}{3}p_1 + \frac{12}{15}p_2 + (N-2)\,\frac{14}{15}\,p_2 + O(p^2). \tag{12.22}$$

The full model adds idle noise on the waiting qubits of each layer and readout errors. We take parameters of the order found on current devices, the gate errors $p_1 = 0.003$ and $p_2 = 0.03$ and the idle noise $\gamma = 0.002$ and $\kappa = 0.004$ per layer on the waiting qubits. The readout errors are $\epsilon_{0\to1} = 0.01$ and $\epsilon_{1\to0} = 0.03$. With them the simulated four-qubit circuit reaches $F = 0.915$, close to the cruder estimate $(1-p_1)(1-p_2)^{N-1} = 0.910$ in which every gate fails with its error probability (Figure 42). After the readout errors a fraction 0.136 of the recorded bit strings differs from 0000 and 1111. The parity of the outer qubits measured from the bit strings is $\langle \hat{Z}_0\hat{Z}_3\rangle = 0.833$, against one for the ideal state.

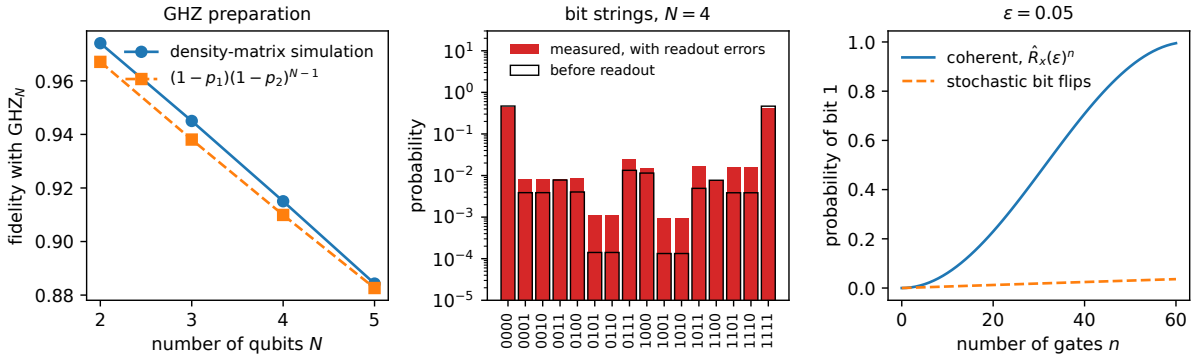



**Figure 42:** Noisy GHZ preparation with gate noise, idle noise on the waiting qubits and readout errors. Left, the fidelity against the number of qubits, with the estimate in which every gate fails with its error probability. Middle, the bit-string probabilities for four qubits before and after the readout errors, on a logarithmic scale. Right, the error after $n$ coherent over-rotations, growing quadratically, and after $n$ stochastic bit flips of the same single-gate probability, growing linearly.

# 13 Comparing states

The *fidelity* between two pure states is their overlap probability

$$F = |\langle\varphi|\psi\rangle|^2, \tag{13.1}$$

which is one for equal states, up to a global phase, and zero for orthogonal ones. It is the probability that $|\psi\rangle$ passes a test that projects on $|\varphi\rangle$. When the target $|\psi\rangle$ is pure and the prepared state is a density matrix $\hat{\sigma}$, the same test gives

$$F = \langle\psi|\hat{\sigma}|\psi\rangle, \tag{13.2}$$

the quantity used for the noisy GHZ circuit of Section 12.7.

For two mixed states (optional) the Uhlmann fidelity

$$F(\hat{\rho},\hat{\sigma}) = \Big(\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\hat{\sigma}\sqrt{\hat{\rho}}}\Big)^2 \tag{13.3}$$

generalises it. The square root of a positive matrix is computed in its eigenbasis. For $\hat{\rho} = \sum_k \lambda_k |k\rangle\langle k|$ it is $\sqrt{\hat{\rho}} = \sum_k \sqrt{\lambda_k}|k\rangle\langle k|$, the matrix whose square is $\hat{\rho}$. When $\hat{\rho} = |\psi\rangle\langle\psi|$ is pure, $\sqrt{\hat{\rho}} = \hat{\rho}$ and the formula reduces to (13.2).

The *trace distance*

$$D(\hat{\rho},\hat{\sigma}) = \tfrac{1}{2}\|\hat{\rho}-\hat{\sigma}\|_1 = \tfrac{1}{2}\sum_k |\mu_k|, \tag{13.4}$$

with $\mu_k$ the eigenvalues of the Hermitian matrix $\hat{\rho}-\hat{\sigma}$, uses the *trace norm* $\|\hat{A}\|_1$ of a Hermitian matrix, the sum of the moduli of its eigenvalues. Since the difference of two density matrices is traceless, its positive and negative eigenvalues cancel in the sum, and $D$ equals the sum of the positive ones. Given one copy that is $\hat{\rho}$ or $\hat{\sigma}$ with equal probability, the best possible measurement identifies it correctly with probability $\frac{1}{2}(1+D)$. For one qubit $D$ is half the Euclidean distance between the two Bloch vectors, since $\hat{\rho}-\hat{\sigma} = \frac{1}{2}(\mathbf{r}-\mathbf{s})\cdot\hat{\boldsymbol{\sigma}}$ has the eigenvalues $\pm\frac{1}{2}|\mathbf{r}-\mathbf{s}|$.

The state $|+\rangle$ and the mixture $\hat{\rho}_{\mathrm{mix}}$ have fidelity and trace distance one half, since their Bloch vectors are $\hat{x}$ and 0. The Bell state and the correlated coin also have $F = 0.50$ and $D = 1/2$, so one copy is identified correctly with probability 3/4 at best, for example by measuring $\hat{X}\hat{X}$. For the Bell state after local dephasing with $p = 0.3$, the decomposition $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(\hat{I}+\hat{X}\hat{X}-\hat{Y}\hat{Y}+\hat{Z}\hat{Z})$ gives

$$F = \langle\Phi^+|\hat{\rho}|\Phi^+\rangle = \tfrac{1}{4}(1+\langle\hat{X}\hat{X}\rangle-\langle\hat{Y}\hat{Y}\rangle+\langle\hat{Z}\hat{Z}\rangle) = \tfrac{1}{2}(1+(1-2p)^2) = 0.58, \tag{13.5}$$

with the trace distance $D = 0.42$. Fidelities with a target state and distances between estimated and true states measure success in variational state preparation and in quantum state tomography.

# 14 Many qubits and outlook

## 14.1 The same rules for $N$ qubits

Everything introduced on the two-spin register holds for $N$ qubits without change. The state is a vector of $2^N$ complex amplitudes in the computational basis, ordered by equation (5.5) with qubit 0 as the most significant bit. Reshaping it yields a state tensor with $N$ indices. Gates on one qubit or on a pair of qubits are Kronecker products with identities on the other qubits, as Section 7.5 works out. Observables are combinations of Pauli strings, whose expectation values are estimated from bit strings in measurement settings. The reduced state of any subset of qubits is the contraction of $\psi$ with $\psi^*$ over the indices of the other qubits. The state alone occupies $16\cdot 2^N$ bytes of memory (Section 5.2), which restricts exact simulation to a few tens of qubits.

### 14.2 GHZ and W states

Two states of three or more qubits show that multipartite entanglement comes in different kinds. The GHZ state (7.27) of Section 7.5 extends the Bell state. It has only two nonzero amplitudes, at the indices 0 and $2^N - 1$, but a state-vector simulation still stores all $2^N$ of them. The *W state* of three qubits

$$|W_3\rangle = \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{3}} \tag{14.1}$$

shares one excitation among the three qubits. The reduced state of one qubit is $\hat{I}/2$ for the GHZ state and $\mathrm{diag}(2/3, 1/3)$ for the W state, as the partial trace with a three-qubit state tensor shows. A measurement of one qubit of the GHZ state in the $\hat{Z}$ basis leaves the others in a product state. In the W state, the outcome 0 on one qubit leaves the other two in a Bell-type state $(|01\rangle + |10\rangle)/\sqrt{2}$. The two kinds of entanglement respond differently to the loss of a qubit and cannot be converted into each other by local operations [15].

### 14.3 Circuits and universality

A circuit on $N$ qubits is a list of gates, each acting on one or two qubits. The gates are applied in order to $|0 \ldots 0\rangle$, and a measurement in a chosen setting follows. Its *depth* is the number of layers of gates that act on disjoint qubits. Its gate counts, in particular the number of two-qubit gates, measure its cost on a device. Among the gates on three qubits, the *Toffoli* gate flips qubit 2 when qubits 0 and 1 are both 1. It is the evolution generated by the projector on $|1\rangle|1\rangle|-\rangle$,

$$\widehat{\mathrm{Toffoli}} = \exp\Big[-\frac{i\pi}{8}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})\Big], \tag{14.2}$$

an $8 \times 8$ matrix in the binary order of Table 5. By the same projector argument as for CNOT in Section 7.2, it exchanges the amplitudes of 110 and 111. On basis states it maps $|b_0 b_1 b_2\rangle$ to $|b_0,\, b_1,\, b_2 \oplus (b_0 \wedge b_1)\rangle$, with the AND operation $\wedge$. For $b_2 = 0$ it computes the AND of the first two bits and keeps them, so the Toffoli gate is a reversible AND. Since AND and NOT suffice for every classical computation, a quantum computer can run any classical program reversibly. Single-qubit rotations together with CNOT are universal (Section 7.3), and an arbitrary unitary on $N$ qubits built from them generally needs a number of gates that grows exponentially with $N$ [1]. Parametrised circuits use these building blocks, rotations $\hat{R}_\alpha(\theta)$ with trainable angles and fixed entangling gates or interactions $\hat{R}_{\alpha\alpha}(\theta)$.

### 14.4 Methods built on these foundations

Quantum machine-learning methods combine the objects defined here. Variational quantum algorithms prepare states with parametrised rotations and entangling gates. They estimate the energy of a Hamiltonian written as Pauli strings from bit strings in a few measurement settings, as in Section 9.3, and budget the shot noise of Section 4.2. Quantum state tomography reconstructs density matrices from records in many settings and uses POVMs and randomised measurements. Hamiltonian learning infers the generator of the dynamics from time traces such as those of the hopping excitation. Ground states of spin models can be classified by their phases, and classical data enter quantum states through rotation gates. Tensor networks turn the reshape-and-SVD test of Section 8.2 into matrix product states and the contractions of these notes into algorithms for many qubits.

## 15 Summary

The results are collected here in the order of their derivation, each under the number of the section that derives it.

## 1 The four rules

1. *State.* $N$ qubits are described by a unit vector $|\psi\rangle \in \mathbb{C}^{2^N}$, up to a global phase, or by a density matrix $\hat{\rho}$ when the preparation is uncertain.
2. *Evolution.* $|\psi\rangle \to \hat{U}|\psi\rangle$ with a unitary $\hat{U} = e^{-it\hat{\mathcal{H}}}$ for a constant Hamiltonian.
3. *Measurement.* The bit string $b$ appears with the probability $|\langle b|\psi\rangle|^2 = \mathrm{Tr}(|b\rangle\langle b|\hat{\rho})$.
4. *Composition.* Systems combine by the tensor product, computed as the Kronecker product.

## 2 One qubit and the Pauli algebra

A qubit is a two-level system, with the ground state $|g\rangle = |0\rangle = |\uparrow\rangle$ and the excited state $|e\rangle = |1\rangle = |\downarrow\rangle$. Its state $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ is a point of the Bloch sphere with the Bloch vector $\mathbf{r} = (\langle\hat{X}\rangle, \langle\hat{Y}\rangle, \langle\hat{Z}\rangle)$, and $p(0) = |\alpha|^2 = (1 + r_z)/2$. The Pauli matrices obey (2.12),

$$\hat{\sigma}_a\hat{\sigma}_b = \delta_{ab}\hat{I} + i\sum_c \varepsilon_{abc}\hat{\sigma}_c, \qquad \mathrm{Tr}(\hat{\sigma}_a\hat{\sigma}_b) = 2\delta_{ab}, \qquad \hat{M} = c_0\hat{I} + \mathbf{c}\cdot\hat{\boldsymbol{\sigma}}.$$

The ladder operators are $\hat{\sigma}^{\pm} = \hat{\sigma}_x \pm i\hat{\sigma}_y$, with $\hat{\sigma}^+ = 2|0\rangle\langle 1|$ and $\hat{S}^{\pm} = \hat{\sigma}^{\pm}/2$. The energy jumps are the excitation $|e\rangle\langle g| = \hat{S}^-$ and the decay $|g\rangle\langle e| = \hat{S}^+$ (2.19). Since spin up is the ground state here, $\hat{\sigma}^+$ raises the spin but lowers the energy (box in Section 2.6).

## 3 Evolution, the laser-driven atom and gates as rotations

The Schrödinger equation $i\frac{d}{dt}|\psi\rangle = \hat{\mathcal{H}}|\psi\rangle$ with a time-independent $\hat{\mathcal{H}}$ is solved by

$$\hat{U}(t) = e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 0}\frac{(-it\hat{\mathcal{H}})^k}{k!} = \sum_n e^{-iE_nt}|E_n\rangle\langle E_n|, \qquad \hat{U}(t_1+t_2) = \hat{U}(t_1)\hat{U}(t_2),$$

where the last form uses the eigenvalues $E_n$ and eigenvectors $|E_n\rangle$ of $\hat{\mathcal{H}}$ (3.5). An atom with the gap $\omega_0$, $\hat{\mathcal{H}}_{\mathrm{atom}} = -\frac{\omega_0}{2}\hat{Z}$, is driven by a laser of frequency $\omega_L$ and Rabi frequency $\Omega$. In the rotating frame it evolves with (3.15), $\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}$ with $\Delta = \omega_L - \omega_0$. On resonance the excited-state probability oscillates as $\sin^2(\Omega t/2)$, a $\pi$ pulse is an $\hat{X}$ gate up to a phase, and a $\pi/2$ pulse creates an equal superposition. For any matrix with $\hat{P}^2 = \hat{I}$ (3.28),

$$e^{-ia\hat{P}} = \cos a\,\hat{I} - i\sin a\,\hat{P}, \qquad \hat{R}_{\mathbf{n}}(\theta) = e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2},$$

and every single-qubit gate is a rotation of the Bloch sphere up to a global phase, $\hat{X} = i\hat{R}_x(\pi)$, $\hat{Z} = i\hat{R}_z(\pi)$, $\hat{H} = i\hat{R}_{\mathbf{n}_H}(\pi)$, $\hat{S} = e^{i\pi/4}\hat{R}_z(\pi/2)$, $\hat{T} = e^{i\pi/8}\hat{R}_z(\pi/4)$ (Table 3).

## 4 Other bases and shots

A measurement in the $\hat{X}$ basis is $\hat{H}$ followed by a $\hat{Z}$ measurement, and in the $\hat{Y}$ basis $\hat{S}^\dagger$, then $\hat{H}$, then $\hat{Z}$. An expectation value estimated from $M$ shots has the standard error $\sqrt{(1-\langle\hat{O}\rangle^2)/M}$ (4.3), which falls as $1/\sqrt{M}$.

## 5 The register in a computer

The state of $N$ qubits is stored as $2^N$ complex amplitudes in the computational basis, the common eigenbasis of $\hat{Z}_0, \ldots, \hat{Z}_{N-1}$. The Kronecker product fixes the order,

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}, \qquad i = \sum_{q=0}^{N-1} b_q\, 2^{N-1-q},$$

so qubit 0 is the most significant bit (Table 5). Operators follow the same rule (5.9),

$$(\hat{A}\otimes\hat{B})_{(b_0b_1),(b_0'b_1')} = A_{b_0b_0'}B_{b_1b_1'}, \qquad (\hat{A}\otimes\hat{B})(|u\rangle\otimes|v\rangle) = \hat{A}|u\rangle\otimes\hat{B}|v\rangle, \qquad (\hat{A}\otimes\hat{B})(\hat{C}\otimes\hat{D}) = \hat{A}\hat{C}\otimes\hat{B}\hat{D}.$$

A gate $\hat{G}$ on qubit $q$ of $N$ is $\hat{I}\otimes\cdots\otimes\hat{G}\otimes\cdots\otimes\hat{I}$, and $\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i\otimes\hat{I} + |1\rangle\langle 1|_i\otimes\hat{X}_j$ with identities on all other qubits (Section 7.5). These matrices are sparse, with $2\cdot 2^N$ or $2^N$ nonzero entries out of $4^N$. Pauli strings $\hat{P}_0\otimes\cdots\otimes\hat{P}_{N-1}$ square to the identity and have the eigenvalues $\pm 1$. They form a basis in which every Hamiltonian is a sum $\hat{\mathcal{H}} = \sum_{\hat{P}} c_P\hat{P}$ with $c_P = \mathrm{Tr}(\hat{P}\hat{\mathcal{H}})/2^N$. On a computer every one of these operations is also a tensor contraction of the state tensor $\psi[b_0,\ldots,b_{N-1}]$ (Table 6).

## 7 Two-qubit gates as rotations

An Ising coupling switched on for a time gives $\hat{R}_{\alpha\alpha}(\theta) = \cos\frac{\theta}{2}\hat{I} - i\sin\frac{\theta}{2}\,\hat{\sigma}_\alpha\otimes\hat{\sigma}_\alpha$ (7.2). $\widehat{\mathrm{CNOT}}$ is the conditional half turn about $x$, the reversible XOR $(b_0,b_1)\to(b_0,b_0\oplus b_1)$, and $\widehat{\mathrm{CZ}}$, $\widehat{\mathrm{SWAP}}$ and Toffoli are exponentials of Pauli products in the same way.

## 8 Entanglement

Reshaping the four amplitudes into the $2\times 2$ matrix $\Psi_{b_0b_1} = \psi_{b_0b_1}$ and decomposing $\Psi$ by the SVD gives the Schmidt decomposition (8.6),

$$|\psi\rangle = \sum_k s_k|u_k\rangle\otimes|v_k\rangle, \qquad s_1\geq s_2\geq 0, \qquad s_1^2+s_2^2 = 1.$$

Schmidt rank one means a product state. The entanglement entropy is $S = -\sum_k \lambda_k\log_2\lambda_k$ with $\lambda_k = s_k^2$, zero for a product state and one bit for the Bell state $(|00\rangle+|11\rangle)/\sqrt{2} = \widehat{\mathrm{CNOT}}(\hat{H}\otimes\hat{I})|00\rangle$. A gate on one qubit alone leaves the Schmidt coefficients unchanged and cannot create entanglement.

## 10 Density matrices and the partial trace

A pure state has $\hat{\rho} = |\psi\rangle\langle\psi|$, a rank-one projector with $\hat{\rho}^2 = \hat{\rho}$ and $\mathrm{Tr}\hat{\rho}^2 = 1$. A mixture $\hat{\rho} = \sum_i p_i|\psi_i\rangle\langle\psi_i|$ has rank above one and purity $\mathrm{Tr}\hat{\rho}^2 < 1$. For one qubit $\hat{\rho} = \frac{1}{2}(\hat{I} + \mathbf{r}\cdot\hat{\boldsymbol{\sigma}})$ with $|\mathbf{r}|\leq 1$ and $\mathrm{Tr}\hat{\rho}^2 = (1+|\mathbf{r}|^2)/2$ (10.8). The reduced state of qubit 0 is

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\psi^*_{a'b}, \qquad \hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^T,$$

equivalently the traces of the four $2\times 2$ blocks of the $4\times 4$ matrix $\hat{\rho}$ (10.13). The eigenvalues of $\hat{\rho}_0$ are $\lambda_k = s_k^2$. The entropy of a reduced state measures entanglement only when the whole register is pure (Table 13). The classical correlated coin and one qubit of a Bell pair have the same reduced state.

## 11 Measurement

A projective measurement has $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$ and leaves $\hat{\rho}_k = \hat{P}_k\hat{\rho}\hat{P}_k/p_k$ (11.1). An unread measurement removes the coherences between different outcome sectors, which is dephasing. The outcome on one qubit of a Bell pair fixes the state of the other, and no signal is transmitted. A Pauli string is measured as a parity, either by rotating each qubit to the $\hat{Z}$ basis or with an ancilla and CNOT gates (Section 11.2). A general measurement is a POVM, positive $\hat{\pi}_k$ with $\sum_k\hat{\pi}_k = \hat{I}$ and $p_k = \mathrm{Tr}(\hat{\pi}_k\hat{\rho})$, for example a detector with readout error (11.7). Informationally complete measurements, the $3^N$ Pauli settings or the tetrahedral POVM, determine $\hat{\rho}$ (Section 11.8).

## 12 Qubit errors and noise channels

A channel is $\mathcal{E}(\hat\rho)=\sum_k \hat K_k\hat\rho\hat K_k^\dagger$ with $\sum_k \hat K_k^\dagger\hat K_k=\hat I$ (12.4), obtained from a unitary coupling to an environment that is then traced out. The six standard error models of Table 14 deform the Bloch ball,

| | |
|---|---|
| bit flip, phase flip, bit-phase flip | two axes shrink by $1-2p$, one is kept |
| depolarising | $\mathbf{r}\to(1-\frac{4}{3}p)\mathbf{r}$ |
| amplitude damping | $\mathbf{r}\to(\sqrt{1-\gamma}\,r_x,\sqrt{1-\gamma}\,r_y,\gamma+(1-\gamma)r_z)$, towards $\lvert g\rangle$ |
| phase damping | $\mathbf{r}\to(\sqrt{1-\kappa}\,r_x,\sqrt{1-\kappa}\,r_y,r_z)$ |

In continuous time the Lindblad equation (12.16) with the decay jump $\sqrt{\Gamma}\lvert g\rangle\langle e\rvert$ gives amplitude damping with $\gamma=1-e^{-t/T_1}$, and pure dephasing gives phase damping, with $1/T_2=1/(2T_1)+\gamma_\phi$ (12.18).

### 12.7 Noise in quantum circuits

In the standard model every gate is followed by depolarising noise on the qubits it touches, $p_1$ after one-qubit gates and $p_2$ after two-qubit gates (12.19). Waiting qubits decay with $T_1$ and $T_2$ in every layer, and readout errors act as a confusion matrix on the bit-string probabilities. Coherent errors add in amplitude and grow as $(n\epsilon)^2$, while stochastic errors add in probability and grow as $np$. The fidelity of a GHZ state prepared with $N-1$ CNOT gates obeys the exponentially falling bound $F\ge(1-\frac{4}{3}p_1)(1-\frac{16}{15}p_2)^{N-1}$ (12.21), and the full model gives $F=0.915$ for $N=4$. The distance to a target is measured by the fidelity $F=\langle\psi\rvert\hat\sigma\lvert\psi\rangle$ and the trace distance (Section 13).

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